# Prevention, treatment and lifecycle health types

Streamlined outline: figures, equations and essential points

November 2026

Boiled down from `paper_skeleton.tex` (7 Nov), which keeps the full text, numbers and caveats. Every figure, equation and table is kept here; the text is cut to the argument. Open questions from the skeleton are collected in red at the end of each section.

## 1 Introduction

- Question: what are the returns to prevention over the life course, for whom, and when?
- Prevention acts on how health *changes*; treatment pays for the level once realised. So the return to prevention depends on each lifecycle health type's mean path, not on the cross-section of levels.
- We estimate those types on UKHLS 2009–24 (ages 20–90), price them with an NHS cost proxy, and find a worst type of 15–18% that carries most of the targetable return.

### 1.1 Related literature

- Earnings dynamics and its use of the covariance matrix.
- Health measurement in economics (factor and item-response models).
- Stochastic specification closest to Hosseini et al. (2022): log frailty as fixed effect plus AR(1) plus transitory shock.

[**Open:** Introduction still to be written.]

## 2 Conceptual framework

### 2.1 Setup: deterministic type and stochastic noise

- Health of person $i$ of type $j$ at age $a$:

$$h_{i,a} = \bar{h}_{j,a} + u_{i,a}, \qquad \bar{h}_{j,a} \equiv H_j + \sum_{a'=1}^{a} d_{j,a'}. \tag{1}$$

- Three kinds of information: what *stays* (level), what is *in the moment* (transitory, measurement error), how one is *evolving* ($d_{j,a}$). Prevention targets the third; the first two are paid for as treatment.
- Each leaves a different trace on the covariance matrix: the basis for identification.
- Types are the object of interest because the return to prevention for a type depends only on its mean path (Result 1).

- Linear illustration ($d_{j,a} = d_j$):

$$\begin{aligned} \bar{h}_a &= \bar{H} + a\bar{d}, \\ \mathrm{Var}(h_a) &= V_H + 2aC_{Hd} + a^2 V_d + v_a, \\ \mathrm{Cov}(h_s, h_t) &= V_H + (s+t)C_{Hd} + st\, V_d + \mathrm{Cov}(u_s, u_t). \end{aligned} \tag{2}$$

- Units matter: the covariance matrix treats health as cardinal, so we need an anchor (costs, Section 3).

### 2.2 Policy: the treatment bill, and prevention as avoiding it

- Prevention from $r$ at intensity $\mu$ scales every future mean change by $(1-\mu)$:

$$\bar{h}^{\mu}_{j,a} = \bar{h}_{j,a} + \mu\,(\bar{h}_{j,r-1} - \bar{h}_{j,a}) \quad \text{for } a \geq r.$$

- We are assuming that prevention acts on health dispersion that is (i) deterministic and (ii) actionable.
- Treatment bill still to come from age $r$:

$$C(r) = \sum_{a\geq r} \mathbb{E}[c(h_{i,a})] = \sum_{a\geq r} \Big[ \underbrace{c(\bar{h}_a)}_{\text{mean path}} + \underbrace{\textstyle\sum_j \pi_{j,a} c(\bar{h}_{j,a}) - c(\bar{h}_a)}_{\text{type differences}} + \underbrace{\mathbb{E}[c(h_{i,a})] - \textstyle\sum_j \pi_{j,a} c(\bar{h}_{j,a})}_{\text{realised deviations}} \Big]. \tag{3}$$

  The third part is transitory and never prevented.
- Illustrative cost $c(h) = (1-h)^2$. Population cost at age $a$:

$$\mathbb{E}[c(h_{i,a})] = (1-\bar{h}_a)^2 + \mathrm{Var}_j(\bar{h}_{j,a}) + \textstyle\sum_j \pi_j \mathrm{Var}(u_{i,a} \mid j). \tag{4}$$

- In which case the total bill from age $r$ onwards is

$$C(r) = \sum_{a\geq r} \big[(1-\bar{h}_a)^2 + \mathrm{Var}(h_{i,a})\big] \tag{5}$$

- Returns to prevention are priced as the treatment costs that the prevention avoids.

**Result 1** (Return to prevention by type)**.**

$$P_j(r) = 2\sum_{a\geq r} \underbrace{(1-\bar{h}_{j,a})}_{\textit{gap at } a} \quad \underbrace{(\bar{h}_{j,r-1} - \bar{h}_{j,a})}_{\textit{preventable decline by } a} \quad \equiv \sum_{a\geq r} s_j(a \mid r). \tag{6}$$

*A function of the type's mean path alone. A type pays back most where it is already in poor health and still declining.*

**Corollary 1** (The uniform policy)**.**

$$P(r) = 2\sum_{a\geq r} \Big[ \underbrace{(1-\bar{h}_a)(\bar{h}_{r-1} - \bar{h}_a)}_{\textit{mean channel}} + \underbrace{\mathrm{Cov}_j\big(1-\bar{h}_{j,a},\, \bar{h}_{j,r-1} - \bar{h}_{j,a}\big)}_{\textit{between-type channel}} \Big]. \tag{7}$$

*The between-type channel splits into pathway, persistence and accumulation margins. The mean channel is most of the uniform return; the types add the dispersion of $P_j(r)$, which is what targeting works through.*

- General convex cost: marginal cost times decline,

$$P_j(r) = \sum_{a \geq r} G_{j,a} \, (\bar{h}_{j,r-1} - \bar{h}_{j,a}), \qquad G_{j,a} \equiv \mathbb{E}[-c'(h_{i,a}) \mid j]. \tag{8}$$

- Prevention acts on change, so the policy needs $(\pi_j, \bar{h}_{j,\cdot})$; the covariance structure is needed only to separate path from noise and to recover a person's type.
- $P(r)$ is first order and can exceed the bill. For the cubic cost we estimate, the saving at any $\mu$ is give by a third order approximation, exact.

**Result 2** (Targeting)**.**

- *Any allocation $\mu(z)$ with the same mean saves* $\mathrm{Cov}(\mu(z), \mathbb{E}[P_i \mid z])$ *over uniform: a signal is worth the return dispersion it reveals.*
- *Targeting on initial health works through the first row of the covariance matrix, attenuated by the signal share; targeting on type gains* $\max_j P_j(r) - \sum_j \pi_j P_j(r)$.

## 2.3 Identification

- Observed covariance = between-type part + noise part; the returns need the first, so we need to separate it from the second.
- Assumptions: A1 window linearity, A2 noise orthogonal to $(H, d)$, A3 a noise class. In the linear case

$$\mathrm{Cov}(h_s, h_t) = V_H + (s+t)C_{Hd} + st\, V_d + \mathrm{Cov}(u_s, u_t). \tag{9}$$

- Unrestricted noise variances can never rule out pure noise; stationary noise identifies $(C_{Hd}, V_d)$ from the diagonal but not $V_H$ from $\sigma_u^2$.

**Terminology.** The noise class below, a persistent AR(1) component plus a one-period shock, is the *persistent–transitory* specification, and the case with no persistent component is *transitory only*: the standard specification and name in earnings and income dynamics (MaCurdy, 1982; Abowd and Card, 1989; Meghir and Pistaferri, 2004; Blundell et al., 2008), also used for medical spending (French and Jones, 2004; De Nardi et al., 2010) and frailty (Hosseini et al., 2022). The transitory shock includes both pure noise (measurement error) and an *episodic* shock, a genuine change in health that is gone by the next period; the covariance matrix cannot tell them apart.

**Proposition 1** (Identification within the persistent–transitory class)**.** *Under A1–A2 and* $\mathrm{Cov}(u_s, u_t) = \rho^{t-s} v_s + \mathbf{1}\{s = t\}\sigma^2_{\varepsilon,s}$*: five waves identify $V_H, C_{Hd}, V_d, \rho$ and both noise variances; four suffice without the transitory shock or with stationary noise; three without it and with stationary variance; two if also $C_{Hd} = 0$ and memoryless noise.*

- One identifying force per region of the matrix: **diagonal** ($C_{Hd}$, $V_d$ from differences); **first row** (a line cannot mimic a decay: splits level from noise, gives $\rho$; the transitory shock is the anchor's excess); **interior** ($V_d$, overidentification).
- Linearity is local, so $C_{Hd}(a)$ can change sign: a flattening variance profile requires it.
- The uniform return is priced by the surface's moments alone:

$$P(r) = 2 \sum_{a \geq r} \big[ (1 - \bar{h}_a)(\bar{h}_{r-1} - \bar{h}_a) + (a - r + 1)\, C_{Hd} + a(a - r + 1)\, V_d \big]. \tag{10}$$

| Policy object | Moments it needs |
|---|---|
| $C(r)$, and the mean channel of $P(r)$ | mean path and variance: Figure 2(a), (b) |
| accumulation margin; timing $r \mapsto P(r)$ | $V_d$ (with $C_{Hd}$): curvature of the surface |
| pathway margin; targeting on a level | $C_{Hd}(a)$, its sign: Figure 2(c) |
| targeting on a type | $P_j(r)$ across types: Section 4 |

[**Open:** which noise class to lead with (data point to a transitory shock and $\rho$ near 1); is the AR(1) "noise within class" or part of "what stays"; mention selection here; how many identification cases to show.]

# 3 Data and motivating descriptives

## 3.1 Data and the health measure

- UKHLS waves 1–15 (2009–24), ages 20–90. Lifecycle sample: at least three observed ages, 334,194 person-ages on 38,963 people.
- Eight physical-health items (Table 1), none tied to employment.

Table 1: Underlying health variables

| Item | What it records | Categories |
|---|---|---|
| GH | self-rated general health | 5 |
| PF | moderate activities and climbing stairs | 5 |
| RP | accomplished less, limited in kind of work or other activity | 9 |
| BP | pain interference with normal activity | 5 |
| LIM_FL | functional limitations: mobility, lifting, dexterity, co-ordination | 5 |
| LIM_SELF | self-care: personal care, continence | 3 |
| LIM_SENS | sensory: hearing, sight | 3 |
| COND | ever-diagnosed physical conditions, 0–5 and 6 or more | 7 |

- Graded-response model (Samejima, 1969):

$$P(Y_j \geq c \mid \theta) \;=\; \frac{1}{1+\exp\{-a_j(\theta - b_{j,c-1})\}} \tag{11}$$

- Bounded expected-score twin:

$$h(\theta) \;=\; \frac{\sum_{j=1}^{J}\sum_{c=2}^{k_j} P(Y_j \geq c \mid \theta)}{\sum_{j=1}^{J}(k_j - 1)}, \qquad h_i = h(\hat{\theta}_i), \quad \hat{\theta}_i = E[\theta \mid \mathbf{y}_i]. \tag{12}$$

- Plus a 31-deficit frailty index and $\log(\text{frailty} + 1/31)$; $h$ behaves like frailty, $\theta$ like log frailty.
- No measure is uniquely cardinal, but covariances are; costs give the anchor. Main text uses $\theta$ (unbounded, convex cost curve); the others are in Appendix I.
- Cost proxy: annual NHS cost from in-patient, out-patient and GP use, waves 7–15, 2024/25 prices:

$$\widehat{C}_{it} \;=\; \underbrace{\text{adm}_{it}\big[\,c_{\text{short}}\,\mathbf{1}\{n_{it}\leq 1\} + c_{\text{bed}}\,n_{it}\,\mathbf{1}\{n_{it}\geq 2\}\big]}_{\text{in-patient}} \;+\; \underbrace{c_{\text{op}}\cdot\widehat{n}^{\text{op}}_{it}}_{\text{out-patient}} \;+\; \underbrace{c_{\text{gp}}\cdot\widehat{n}^{\text{gp}}_{it}}_{\text{GP}}. \tag{13}$$

Wrong in level (£900 vs £3,000 a head), right in shape.

- Figure 1: cost is convex in poor health ($t = 17.4$ on the quadratic) and close to age-invariant above $\theta \approx -2$; mortality is roughly log-linear in $\theta$, but age carries extra mortality risk.

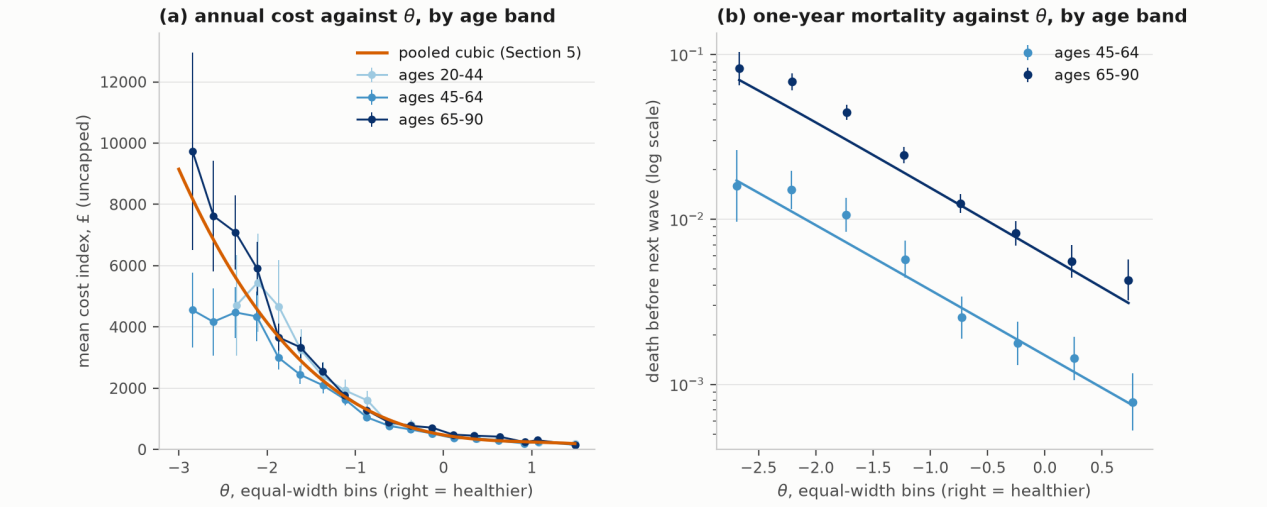



Figure 1: Cost (a) and one-year mortality (b) against $\theta$, by age band; orange line in (a) is the pooled cubic used for pricing.

### 3.2 The moments of health over the life course

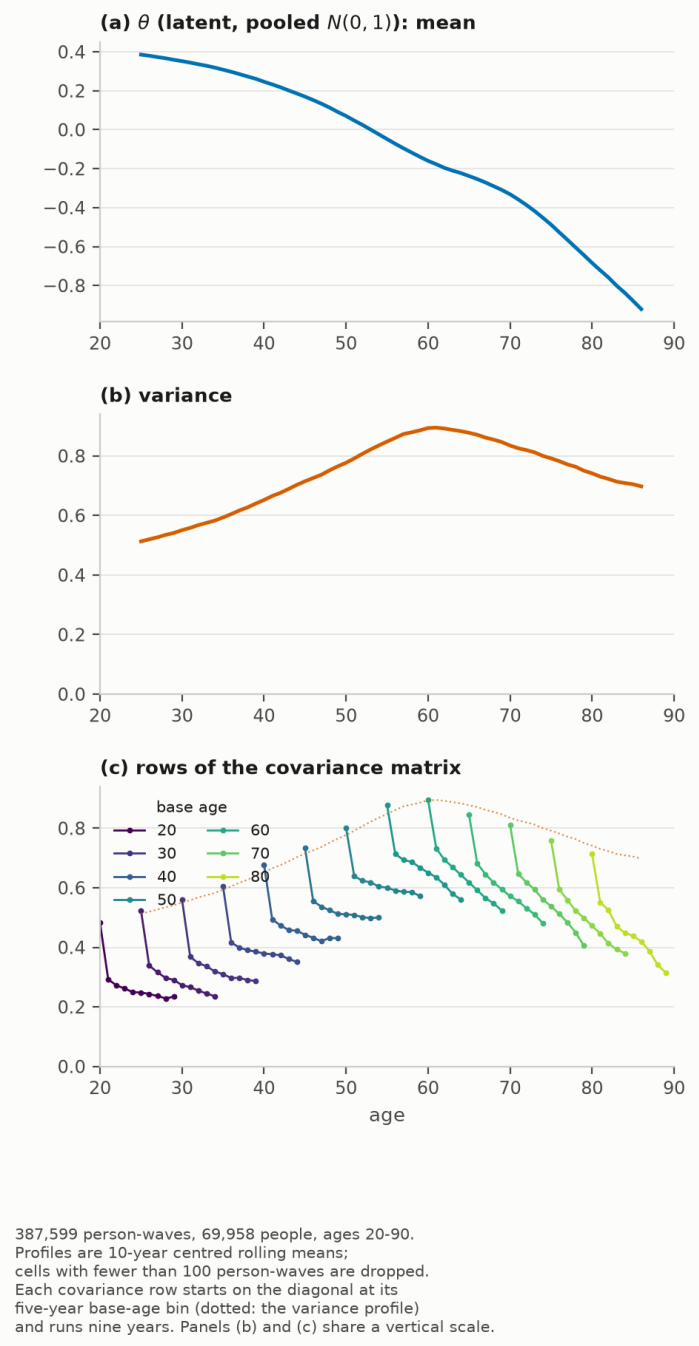



Figure 2: Mean path, variance profile and covariance rows of $\theta$, model-free.

- (a) The mean decline accelerates with age: on its own this says intervene late.
- (b) Variance rises 60% from 30 to 60, peaks at 62, then falls: $C_{Hd}(a)$ turns negative after 60.
- (c) A one-year transitory drop in every row, then divergence among the young and convergence among the old.
- One stationary noise process cannot produce both, so either noise moves with age or $C_{Hd}$ changes sign. Implication: target the low early; later, targeting on levels pays little.

- Same on log frailty; bounded measures ($h$, frailty) keep variance growing after 60.

### 3.3 The linear approximation on real data

- Fit $V_H$, $C_{Hd}$, $V_d$ and noise by constrained least squares on the ten moments of four ages four years apart, within four age bands (Case 4: persistent–transitory).
- Corr$(H, d)$ goes from positive to negative with age (+0.45, +0.78, −0.56, −0.53); the sign path is robust across cases and measures.

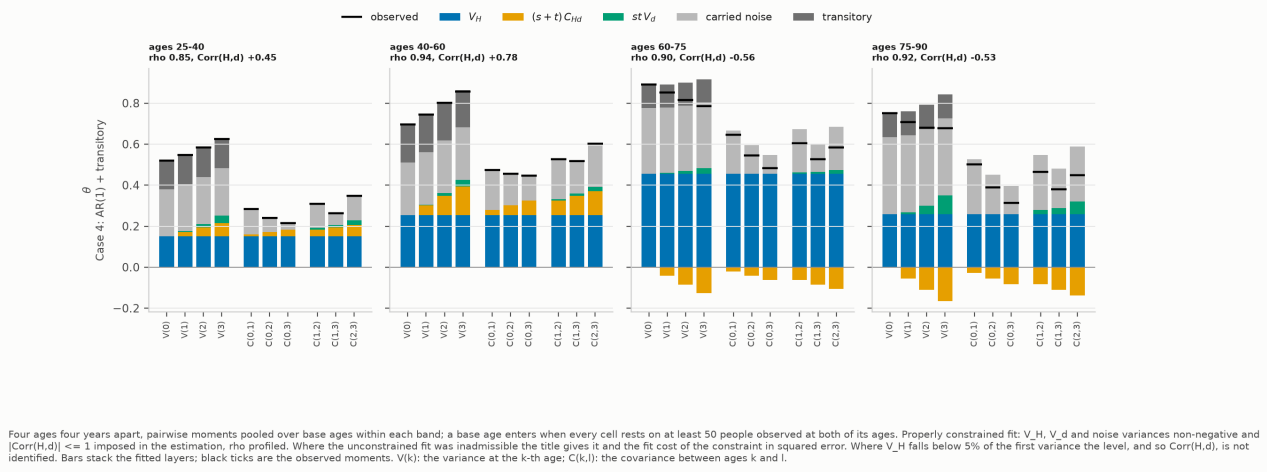



Figure 3: The linear approximation on $\theta$, Case 4: fitted layers against observed moments by age band.

[**Open:** which identification cases to present here, and pairwise vs balanced moments; PSID replication to bring in; take posterior uncertainty in $\theta$ into the clustering.]

## 4 Lifecycle types

### 4.1 Why and how

- Health at 55 is an outcome of earlier dynamics, and divergence happens before late middle age, so we need the full lifecycle (most work uses the old or short windows).
- Cohort-independent types with cohort-dependent shares, estimated three ways: partial K-means, Bayesian mixtures, then against observables.
- Challenges: power and type definition; mortality and attrition; cohort effects; units and bounds.

### 4.2 (A) Non-parametric types: partial K-means

- Minimise squared deviations over observed entries only, $K = 3$, 50 starts; every person-wave weighted equally.
- Three ordered paths, 18/43/40%. The worst type falls fastest to 60, then the order reverses: divergence then convergence.
- Within-type deviations are persistent (lag-1 autocorrelation 0.37–0.64), so the final model needs persistence.

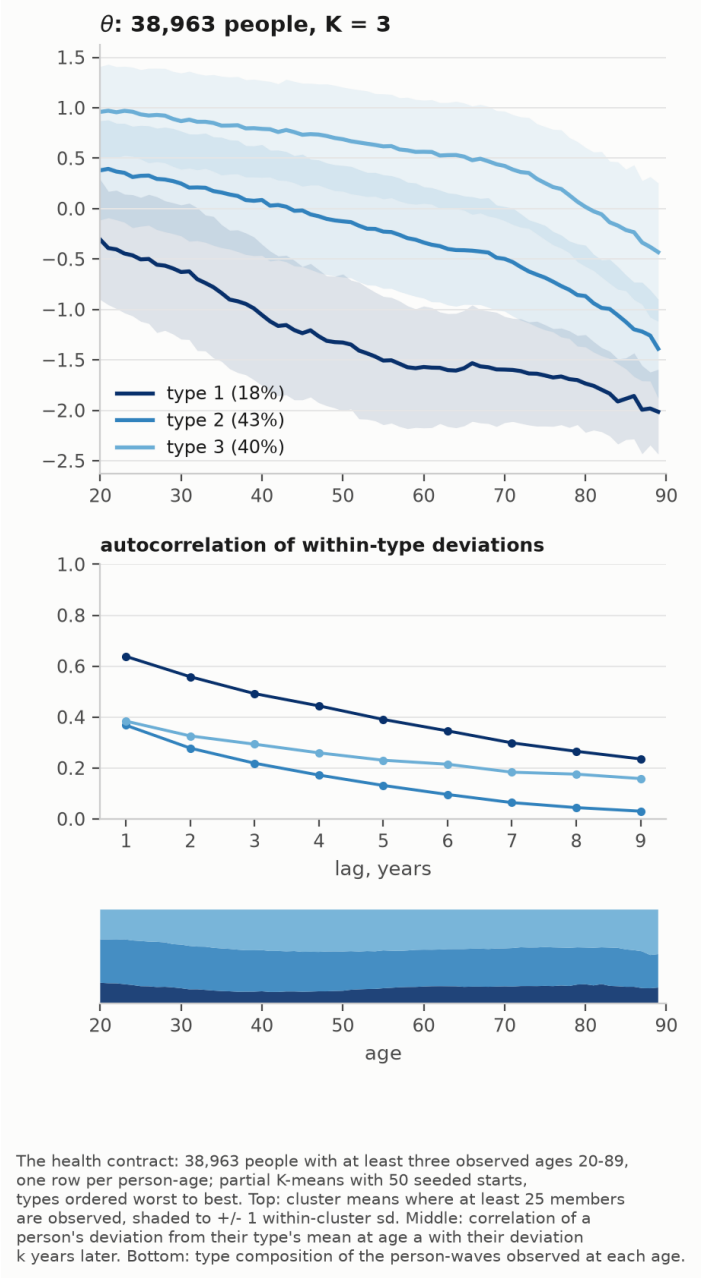



Figure 4: Partial K-means types ($K = 3$) on $\theta$: type means, within-type autocorrelation, composition by age.

- Between-type share of variance 48% at 30, 65% at 70; $\mathrm{Cov}_j(\bar{h}_{j,a}, d_{j,a})$ changes sign at 61.
- Cohort adjustment leaves the typology unchanged (ARI 0.90).

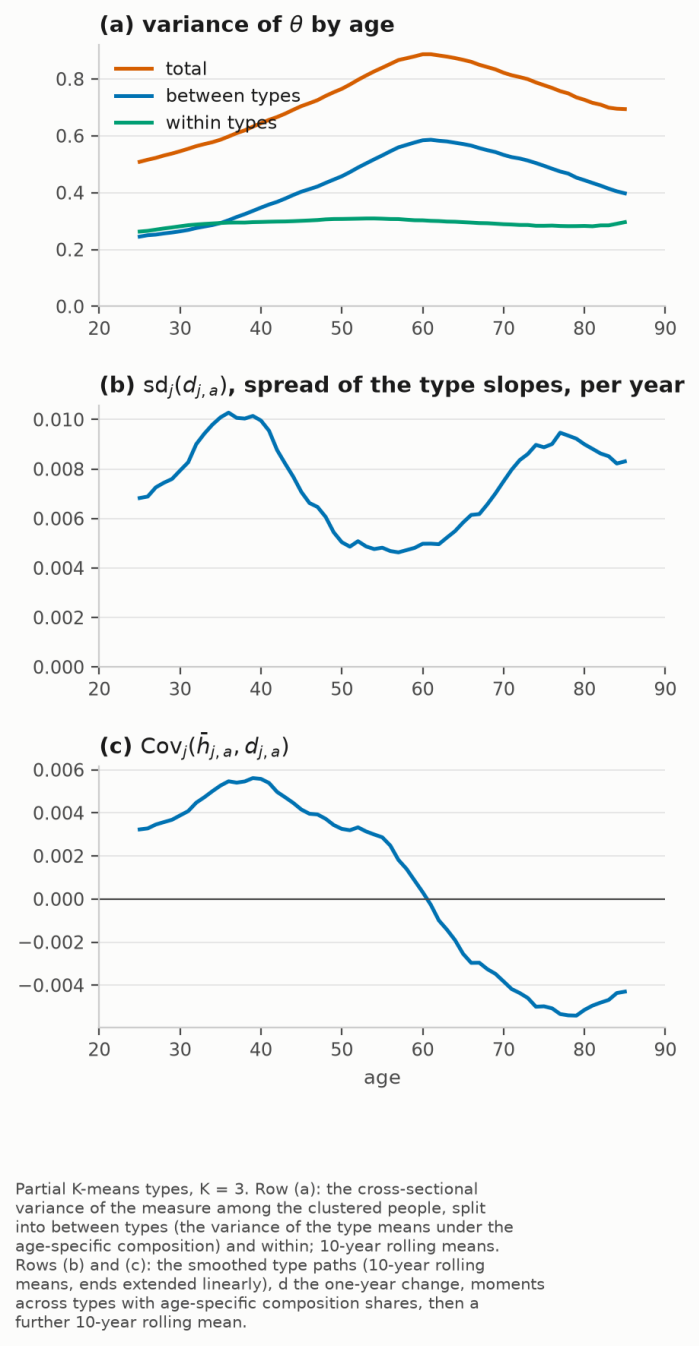



Figure 5: Between versus within by age for the K-means types: variance split, spread of type slopes, level–slope covariance.

### 4.3 (B) Parametric types: Bayesian mixtures

- Baseline: mixture of quadratic growth curves, $\tilde{a} = (a - 55)/10$, estimated in Stan:

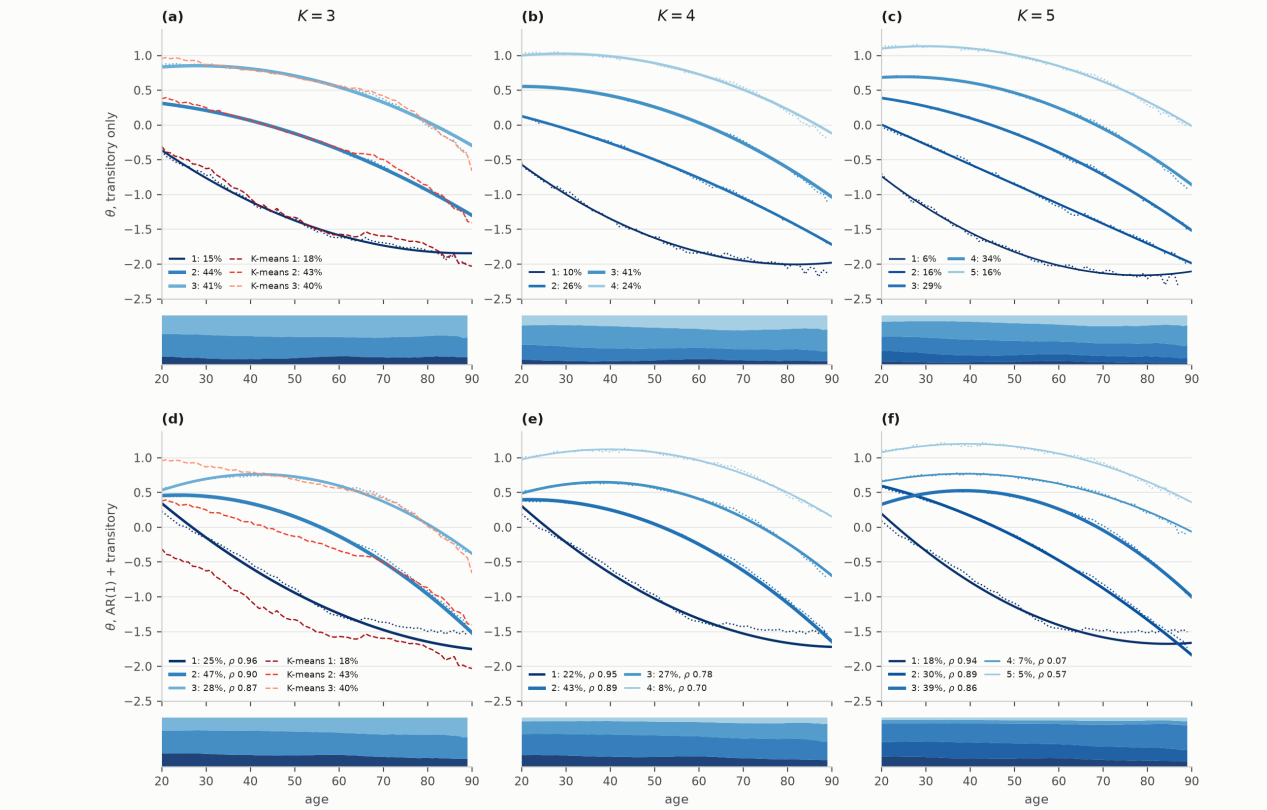
$$y_{ia} = \alpha_{k_i} + \beta_{k_i}\tilde{a} + \gamma_{k_i}\tilde{a}^2 + \varepsilon_{ia}, \qquad \varepsilon_{ia} \sim N(0, \sigma^2) \text{ i.i.d.} \tag{14}$$

The transitory-only specification; same typology as K-means (ARI 0.85, 95% same class).

- Main specification, persistent–transitory: an AR(1) state plus a transitory shock:

$$y_{ia} = \mu_{k_i}(a) + u_{ia} + \varepsilon_{ia}, \qquad u_{ia} = \rho^{g}_{k_i}\, u_{i,a-g} + \eta_{ia}, \quad \eta_{ia} \sim N\Big(0,\ \sigma^2 \frac{1-\rho^{2g}_{k_i}}{1-\rho^{2}_{k_i}}\Big), \quad \varepsilon_{ia} \sim N(0, \sigma^2_{\varepsilon}). \tag{15}$$

  Shares .25/.47/.29, persistence .87–.96. The worst class now starts near average and falls away.
- On $h$ and frailty the healthiest class collapses at the bound: one reason to use $\theta$.




Quadratic growth mixtures on the health contract (38,963 people, 334,194 person-ages), parameters at the posterior mean, in each variant's own units; class 1 is worst health. Line width is proportional to the class share; the legend gives the share and, under AR(1), the class persistence. Dotted: the observed class mean at each age, the measure averaged over the people observed at that age weighted by their posterior class probabilities (where the summed weight is at least 25). Dashed red ($K = 3$ only): the partial K-means type means on the same rows, with the type shares in the legend. The strip beneath each panel is the posterior class composition of the person-waves observed at each age. Rows: the stochastic specification, transitory only or AR(1) + transitory; columns: the number of classes. All six panels share one vertical scale.

Figure 6: Mixtures on $\theta$: transitory only (a)–(c) and persistent–transitory (d)–(f), at $K = 3, 4, 5$; red dashed is K-means.

**Is there always a bad group?**

- Yes, 15–18% under every method, and mostly the same people (74–85% overlap).
- The stochastic structure changes how they got there: level-based methods overstate early separation and understate decline, and decline is what prevention prices.
- This group drives the level–slope reversal: drop it and the reversal disappears.

**Extensions.**

- $K = 4, 5$: under the transitory-only specification each new class is a sicker group; under the persistent–transitory specification, small healthy groups.

- Multidimensional model (physical, mental, Gompertz–Makeham mortality, shared classes): the physical typology survives; mental health is faster and noisier; the worst class's hazard is 11.5× the best at 55 and 3.2× at 85.
- Windows: a window finds the bad type only if it contains the 40–60 divergence; about fifty years of ages are needed.
- Cohort shifts leave the typology unchanged but should not be read as cohort effects (age–cohort collinearity).
- Life expectancy from 25: 47.7/57.2/62.2 years; healthy LE separates the classes far more (17.5 vs 59.2).
- Death is hard to predict: class adds a little over age (AUC 0.85 to 0.88), less than current health (0.90).

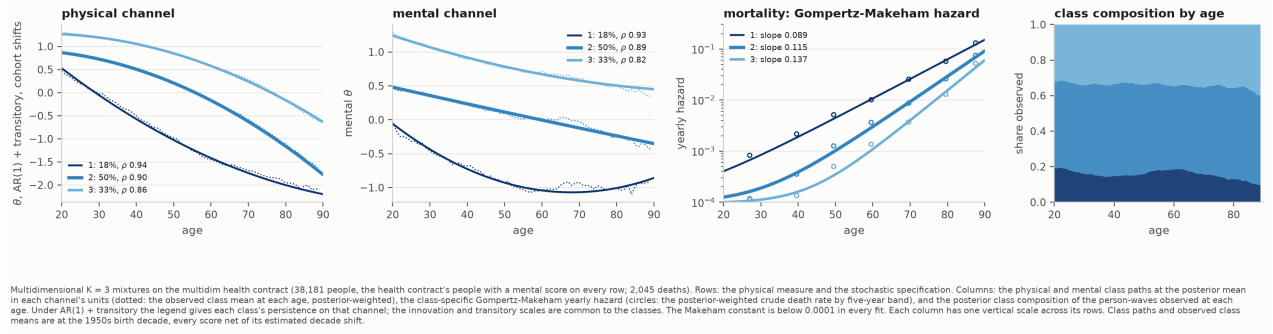



Figure 7: Multidimensional $K = 3$ mixture with birth-decade shifts: physical and mental class paths, class hazards, composition.

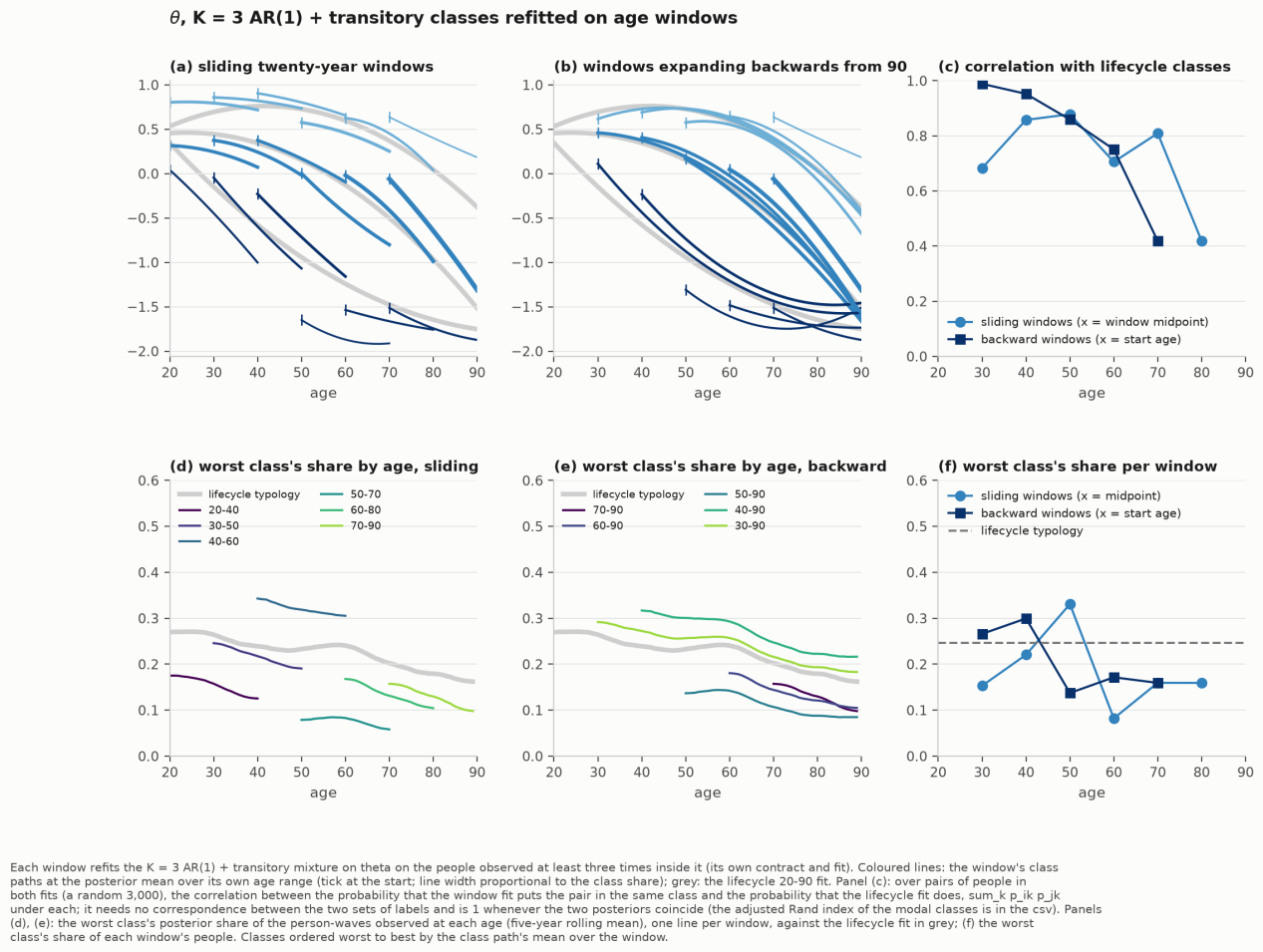
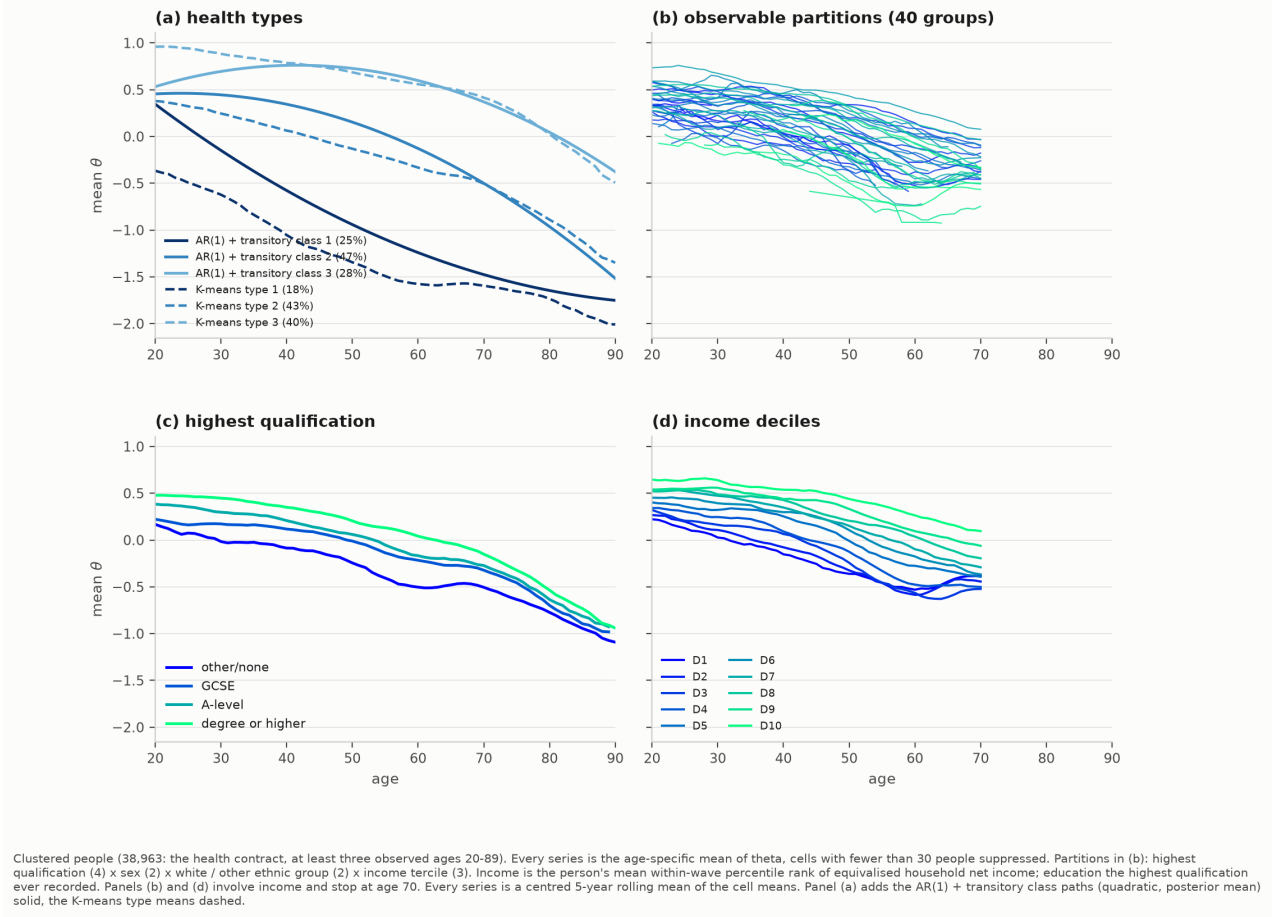



Figure 8: Persistent–transitory classes refitted on sliding and backward-expanding age windows, agreement with the lifecycle classes, and the worst class's share.

## 4.4 (C) Types against observables

- Gradients by education and income exist and run the expected way.
- But even 40 observable cells span half the range the types do; between-group variance shares 3–9% against 48–65% for types; ARI with types 0.01–0.03.




Clustered people (38,963: the health contract, at least three observed ages 20-89). Every series is the age-specific mean of theta, cells with fewer than 30 people suppressed. Partitions in (b): highest qualification (4) x sex (2) x white / other ethnic group (2) x income tercile (3). Income is the person's mean within-wave percentile rank of equivalised household net income; education the highest qualification ever recorded. Panels (b) and (d) involve income and stop at age 70. Every series is a centred 5-year rolling mean of the cell means. Panel (a) adds the AR(1) + transitory class paths (quadratic, posterior mean) solid, the K-means type means dashed.

Figure 9: Lifecycle trajectories of $\theta$ by (a) type, (b) 40 observable cells, (c) education, (d) income decile.

## 4.5 Predictive value

- Predicting ahead (last two rows held out): observables add nothing; current health and the class add about the same; the model forecast does a little better.
- Predicting type (first two rows held out): observables barely predict it; one health reading is most of the value; health is the weakest guide to type when young.
- Headline: health is what matters, the model helps a bit, observables are almost irrelevant.

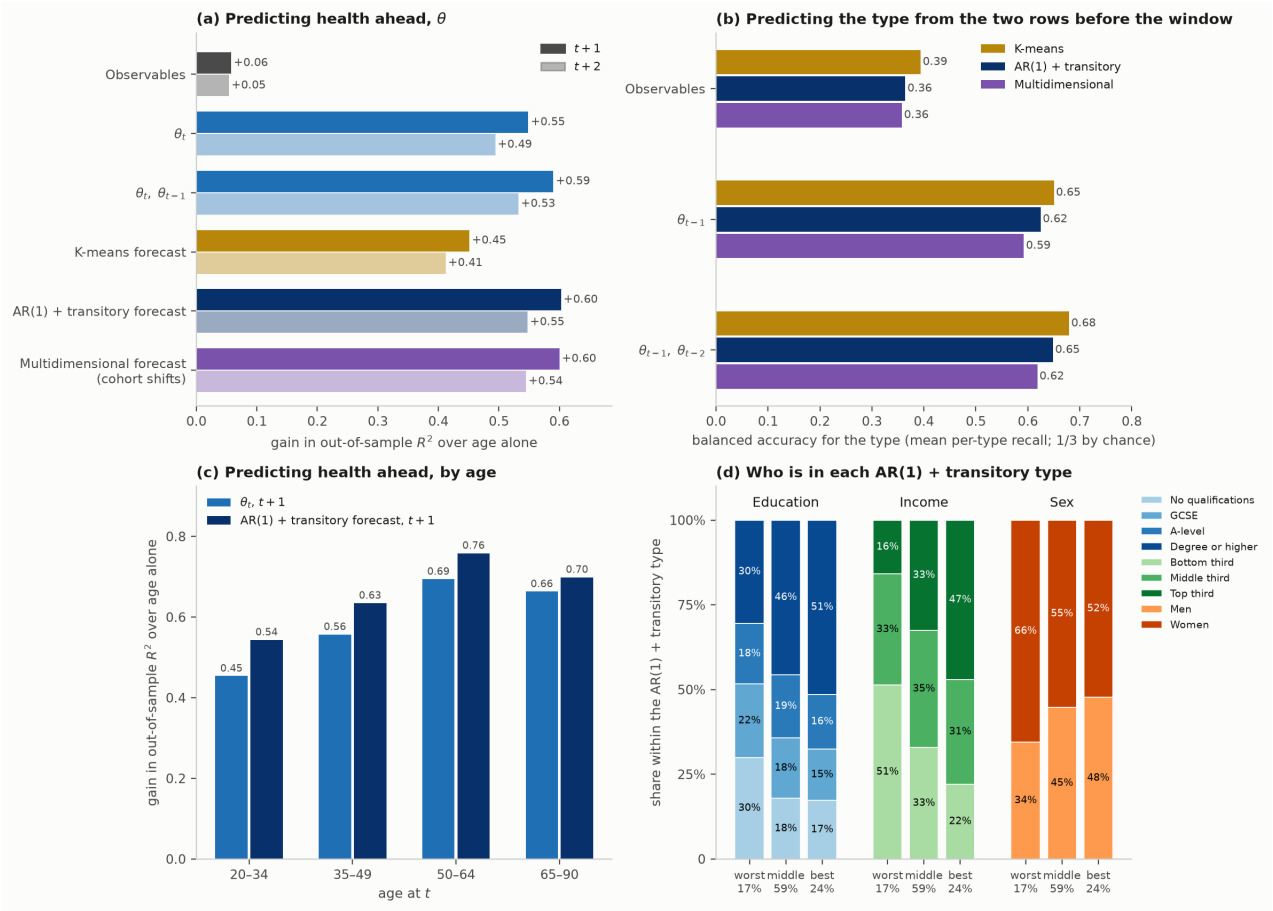
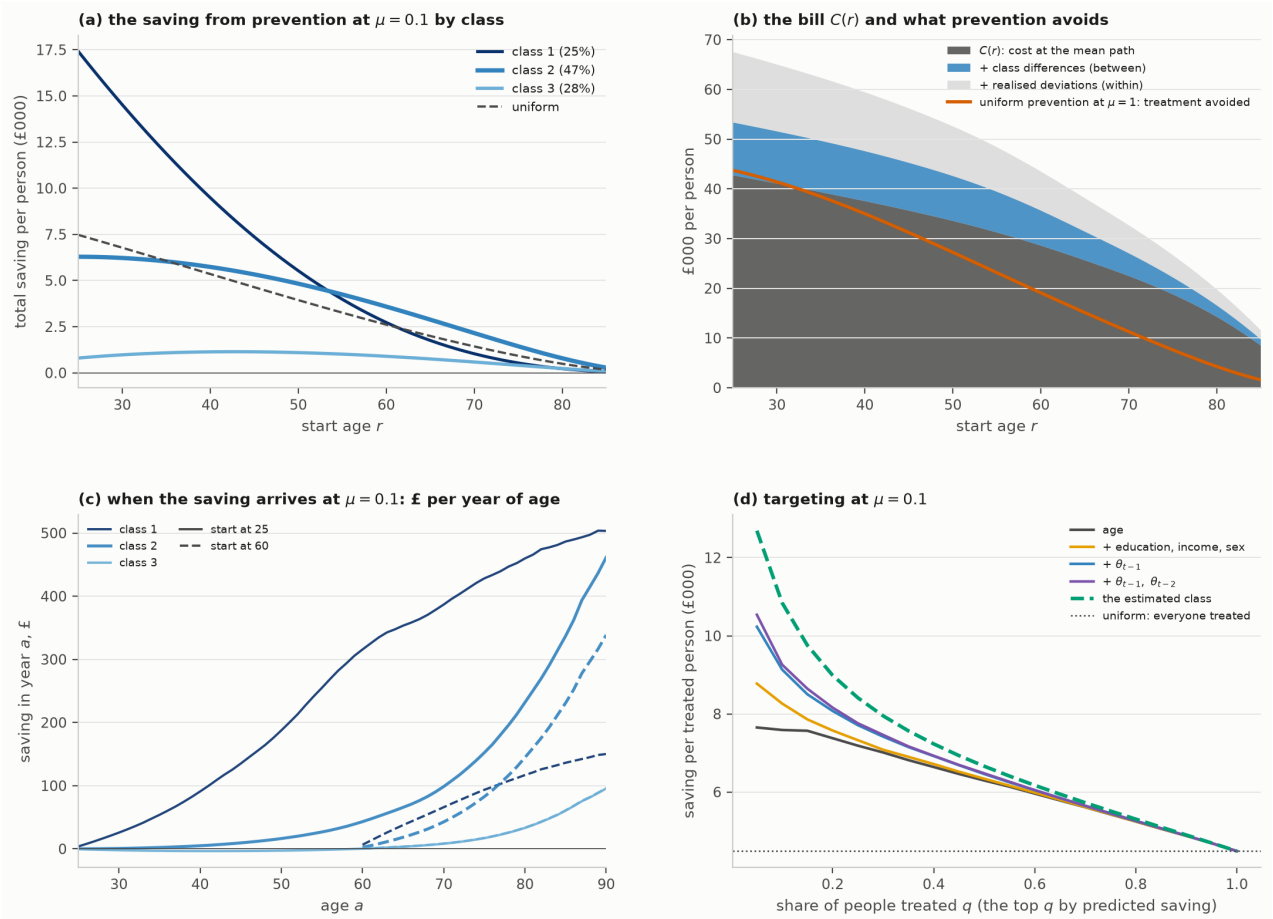



Figure 10: Out-of-sample gain over age in predicting $\theta$ ahead (a, c) and balanced accuracy in predicting the type (b), with the type composition by observables (d).

[**Open:** bounded observation model for $h$ and frailty; class composition in the persistent–transitory fits looks like mortality selection; calibrate healthy LE to ONS.]

# 5 Prevention exercises

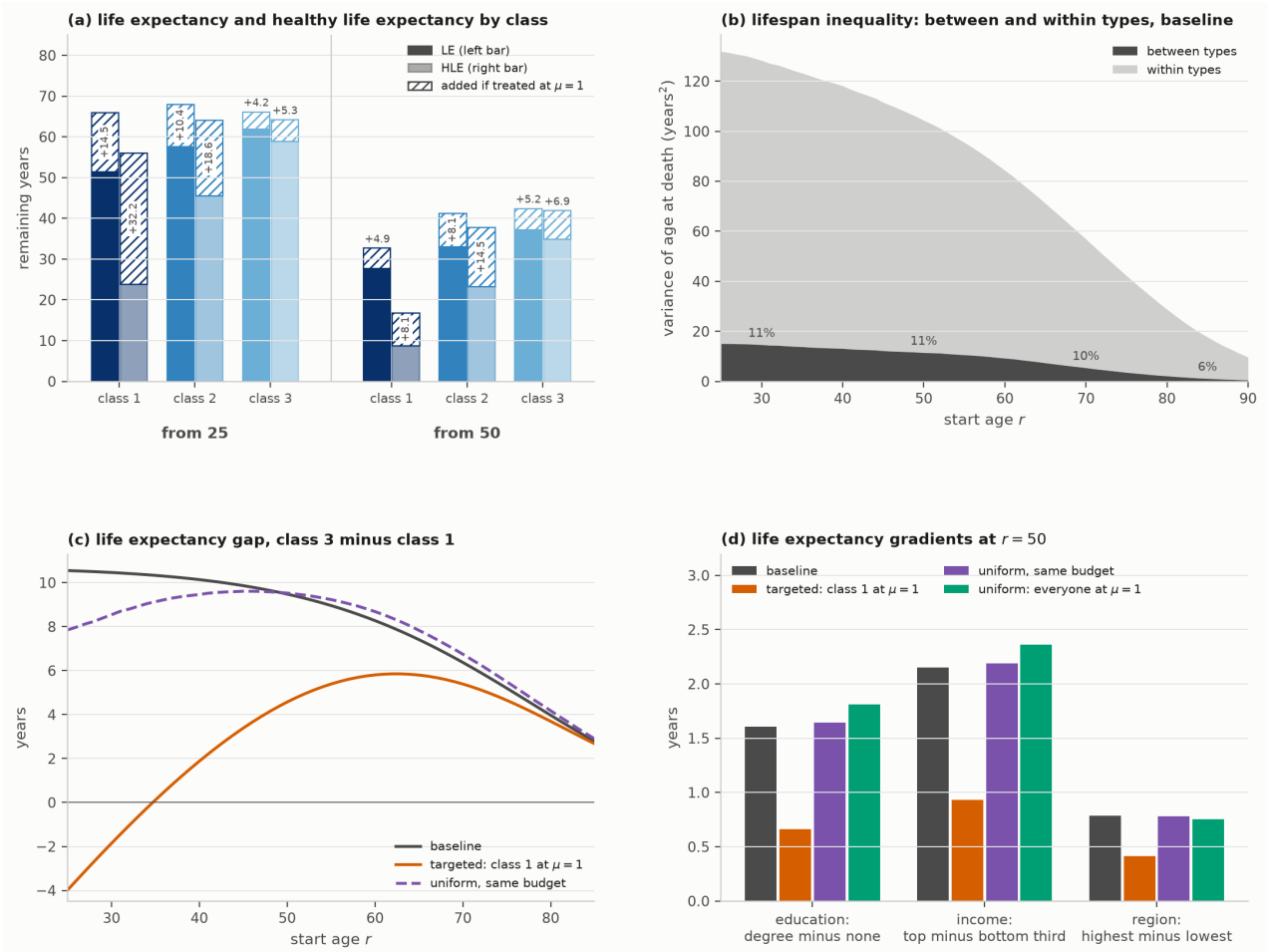
- Priced on the $K = 3$ persistent–transitory classes with the estimated cubic cost curve; pounds to 90, undiscounted.
- **The bill**: £67,500 per person from 25; 63% mean path, 16% type differences, 21% unpreventable deviations.
- **Uniform prevention**: $P(r)/C(r)$ is 1.17 at 25, 0.78 at 50, 0.45 at 70; the exact saving at $\mu = 1$ is 65%, 52%, 35% of the bill. The later the start, the more is already determined.
- **By class**: the worst class returns 2.3× the uniform amount from 25 because its gap and decline coincide; the best class a tenth.
- **Timing**: the between-class channel stays positive at every start age once the transitory shock is separated (it turns negative by 50 under the transitory-only specification).
- **Targeting on class**: gains of 133% from 25, 38% from 50, 51% from 70.




The K = 3 AR(1) + transitory classes on theta with the cost curve estimated in pounds on theta (a degree-3 polynomial on the cost index, waves 7-15). Savings are exact: the cost curve is a cubic, so the expansion in mu stops at the cube. (a) The treatment cost avoided by prevention at mu = 0.1 (a tenth of each class's future decline held back) from start age r, by class, line width proportional to the class share, dashed the uniform policy. (b) C(r), the expected treatment cost from r to 90 on the observed cross-section, split into the cost at the mean path, what the class differences add and what realised deviations add (the part prevention cannot reach), with the saving from uniform prevention at mu = 1 (the whole decline held back). (c) The saving at mu = 0.1 in each year of age a from prevention started at 25 (solid) and 60 (dashed), by class, the per-age terms whose sum is (a). Pounds to 90, no survival weighting, no discounting. (d) Targeting at mu = 0.1, on the same classes refitted with each person's first two rows held out (29,899 people): each rule predicts a person's saving from what is known at t-1, the later of the two held rows (t is the first fitted row) (a multinomial logit of the class on a quadratic in age plus the set shown, fitted on a 70% split), treats the top q of the other 30% by predicted saving and is scored by the realised saving, the fitted-window class posterior times S_j(r; 0.1) at the person's age at t-1. Dashed, targeting on the estimated class, the posterior from the whole fitted window; dotted, everyone treated. The other rules and the share-of-return view are in the appendix targeting figure.

Figure 11: Prevention in pounds: (a) saving at $\mu = 0.1$ by class and start age; (b) the bill $C(r)$ and the uniform saving at $\mu = 1$; (c) when the saving arrives; (d) targeting with what is known before the fitted window.

- **When**: savings are back-loaded (half by 72 for the worst class, 82 middle, 86 best), so discounting will bite.
- **What to target on**: age first; then a health reading (£9,100 per treated person at 10% coverage) beats observables (£8,300); the estimated class gets £10,800. Two readings close about half the gap from age to class.




The univariate K = 3 AR(1) + transitory classes on theta (38,963 people), class 1 worst health, with Gompertz-Makeham class hazards estimated by posterior-weighted maximum likelihood on the multidim contract's death intervals (2,045 deaths) and calibrated to the ONS life tables (one common age-varying factor on every class hazard, matching the population LE at 65 and 90). Prevention from r at intensity mu holds back a share mu of the class's mean change in theta after r - 1. Mortality responds through the within-age gradient of death before the next wave in theta (a logit on theta, a quadratic in age and theta x age over 363,033 person-waves at risk, 1,825 deaths: log-odds -0.99 per unit of theta at 55); the healthy share (general health good, very good or excellent) through the same design fitted to the healthy indicator. HLE by the Sullivan method: survival times each class's healthy share at each age. Policies: targeted, class 1 at mu = 1; uniform with the same budget, everyone at mu = 1 pi_1(r) (class 1's share at r, 0.23 at 50); uniform, everyone at mu = 1. (a) solid, the baseline; hatched, what the class gains if treated on its own from r at mu = 1. (b) the variance of age at death from r at baseline over the composition-weighted mixture, split by the law of total variance into the part between types, sum_k pi_k (m_k - m)^2 with m_k class k's mean age at death, and the part within them; the labels give the between share. (d) each group's LE at 50 is its class composition among the person-waves within five years of 50 times the class LE, so the gradients are the part the types carry; income thirds of the person's mean percentile rank of equivalised household income within wave, over all adults.

Figure 12: Targeted prevention in life years: (a) LE and HLE by class with gains if treated; (b) variance of age at death; (c) LE gap between best and worst class; (d) socioeconomic LE gradients.

- Prevention adds more healthy years than years of life.
- Most lifespan inequality is within type (types explain about 11%).
- Targeting the worst class from 50 halves the LE gap between types (9.5 to 4.6 years); the same budget spread evenly does not.
- It also more than halves the education and income LE gradients the types carry.

[**Open:** gap between first-order and exact savings in the text; survival weighting and discounting; cost of a unit of $\mu$; within-class gradients; additive-prevention robustness; posterior uncertainty.]

# 6 Conclusions

[**Open:** To be written.]

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# Appendices

## A Proofs and Additional Material for the Conceptual Framework

[**Open:** Proofs and illustrative figures.]

## B The Health Measure

### B.1 Health Variables

- Item coding, impairment groups and the COND merge rule; 7.1% of person-waves at the best answer on all items.

Table B.1: Unconditional distribution by health variable

| | $k$ | share of person-waves, best category first (%) | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| GH | 5 | 13.29 | 33.33 | 31.71 | 15.58 | 6.10 | | | | |
| PF | 5 | 65.13 | 11.66 | 11.82 | 4.60 | 6.79 | | | | |
| RP | 9 | 52.18 | 6.85 | 12.77 | 4.55 | 11.57 | 2.68 | 4.76 | 1.20 | 3.44 |
| BP | 5 | 53.72 | 25.24 | 9.14 | 8.26 | 3.64 | | | | |
| LIM_FL | 5 | 83.05 | 6.66 | 5.30 | 3.24 | 1.76 | | | | |
| LIM_SELF | 3 | 94.15 | 4.74 | 1.12 | | | | | | |
| LIM_SENS | 3 | 95.62 | 3.82 | 0.56 | | | | | | |
| COND | 7 | 48.28 | 28.88 | 12.90 | 5.59 | 2.44 | 1.05 | 0.86 | | |

### B.2 The Graded Response Model

Table B.2: Graded response model item parameters.

| Item | $k$ | $a$ | thresholds | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| GH | 5 | 2.03 | −2.10 | −0.98 | 0.16 | 1.45 | | | | |
| PF | 5 | 3.52 | −1.73 | −1.38 | −0.80 | −0.40 | | | | |
| RP | 9 | 3.09 | −2.20 | −2.03 | −1.56 | −1.37 | −0.81 | −0.64 | −0.23 | −0.03 |
| BP | 5 | 2.31 | −2.35 | −1.50 | −0.99 | −0.09 | | | | |
| LIM_FL | 5 | 3.38 | −2.49 | −1.94 | −1.46 | −1.07 | | | | |
| LIM_SELF | 3 | 2.43 | −2.95 | −1.99 | | | | | | |
| LIM_SENS | 3 | 1.36 | −4.49 | −2.85 | | | | | | |
| COND | 7 | 1.18 | −4.61 | −3.90 | −3.12 | −2.26 | −1.25 | 0.11 | | |

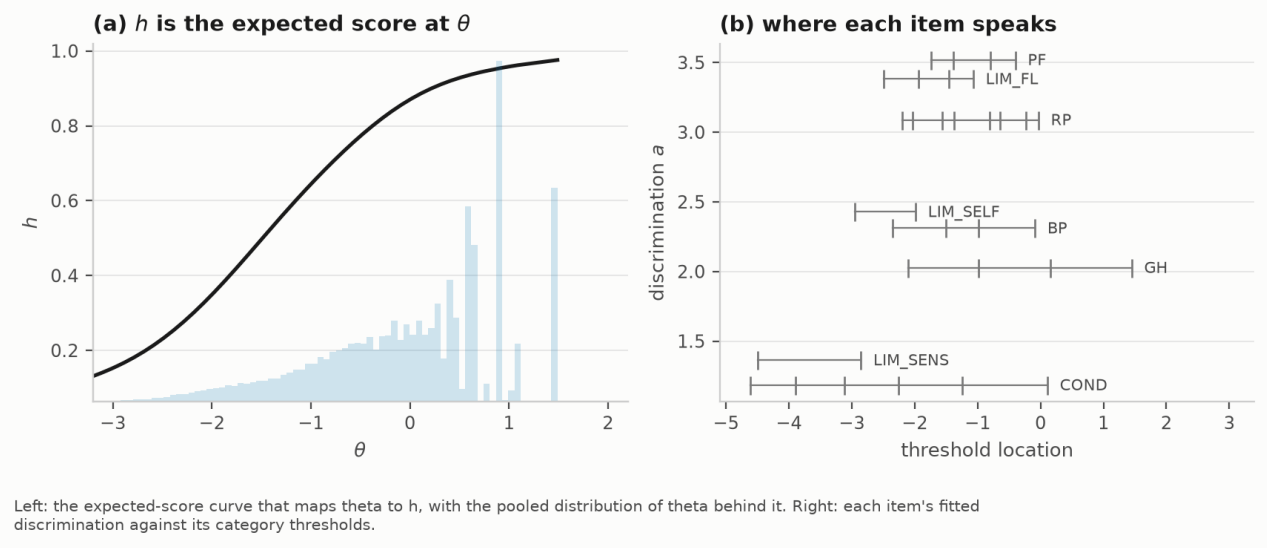



Figure B.1: (a) The map from $\theta$ to $h$; (b) item discriminations against thresholds.

| | $\theta$ | $h$ | frailty index (negated) | log frailty (negated) |
|---|---|---|---|---|
| UKHLS PCS | 0.903 | 0.935 | 0.859 | 0.837 |
| UKHLS MCS | 0.274 | 0.266 | 0.232 | 0.228 |
| chronic count (negated) | 0.576 | 0.587 | 0.785 | 0.763 |
| Spearman, vs $\theta$ | 1.000 | 1.000 | 0.920 | 0.920 |

Table B.3: Correlations of the four measures with comparison metrics.

### B.3 The Frailty Indices

- 31 equal-weight deficits (SF-12 physical items, eight impairment areas, 17 conditions); differs from Hosseini et al. (2022) in deficits, self-rated health and treatment of zeros.

## C The Cost Proxy

### C.1 Inputs

Table C.1: UKHLS variables behind the cost proxy, waves 7–15.

| Quantity | Source | Coverage |
|---|---|---|
| Any admission | `hosp` | complete for respondents |
| Nights in hospital | `hospd` | all admitted; median 3, mean 7.5, max 300 |
| Maternity flag | `hospch` | 12.7% of admissions; left out |
| GP visits, banded | `hl2gp` | 0 / 1–2 / 3–5 / 6–10 / more than 10 |
| Out-patient attendances, banded | `hl2hop` | same bands |

## C.2 Sensitivity of the Convexity

- All twelve costing variants are convex ($t$ 17–36); long-stay pricing moves the sickest-to-healthiest ratio most.

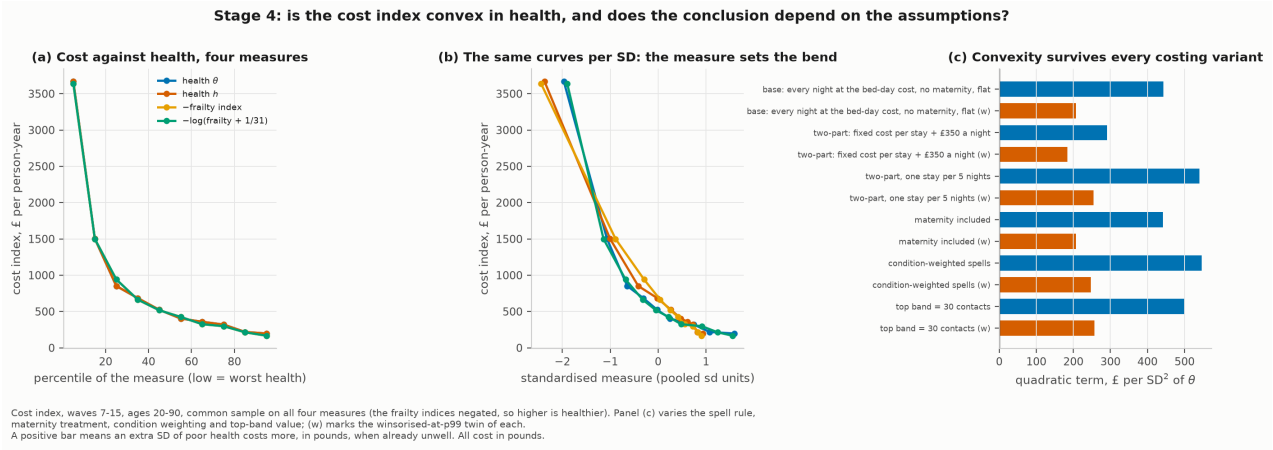



Figure C.1: Cost against the four measures, and sensitivity of the convexity to each costing assumption.

## C.3 Validation against Published Profiles

- Concentration passes; level undershoots as expected; age profile too flat (3× vs 7×), so not for cost-by-age projections. The health gradient is strong.

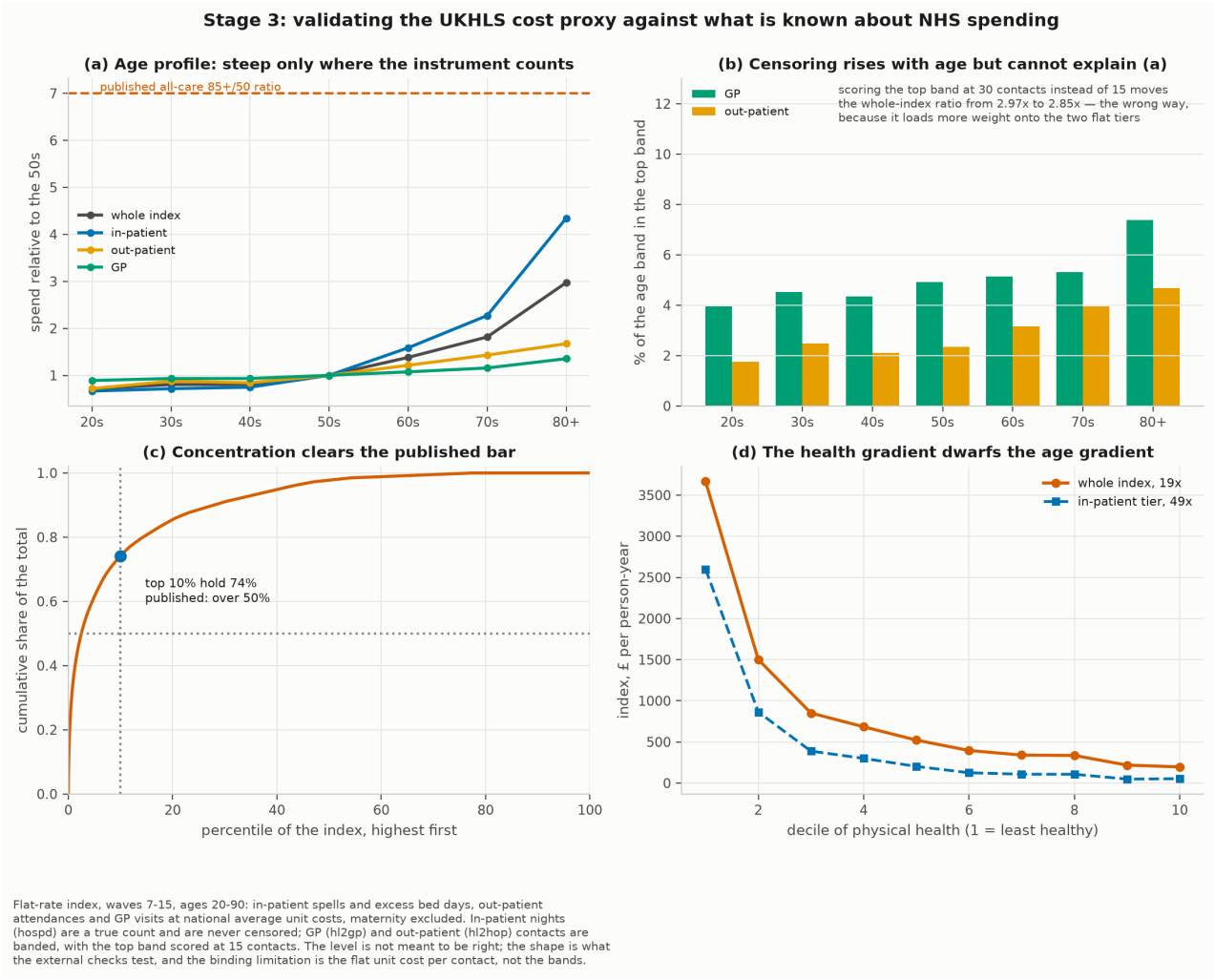



Figure C.2: Validation: age profile, top-coding, concentration, health gradient.

# D Additional Descriptive Results

## D.1 Robustness of the Moments

- Balanced rows sit lower but keep the first-year drop and the slope signs.
- Cost itself is almost entirely a one-period draw: the in-the-moment part of the bill that prevention cannot reach.

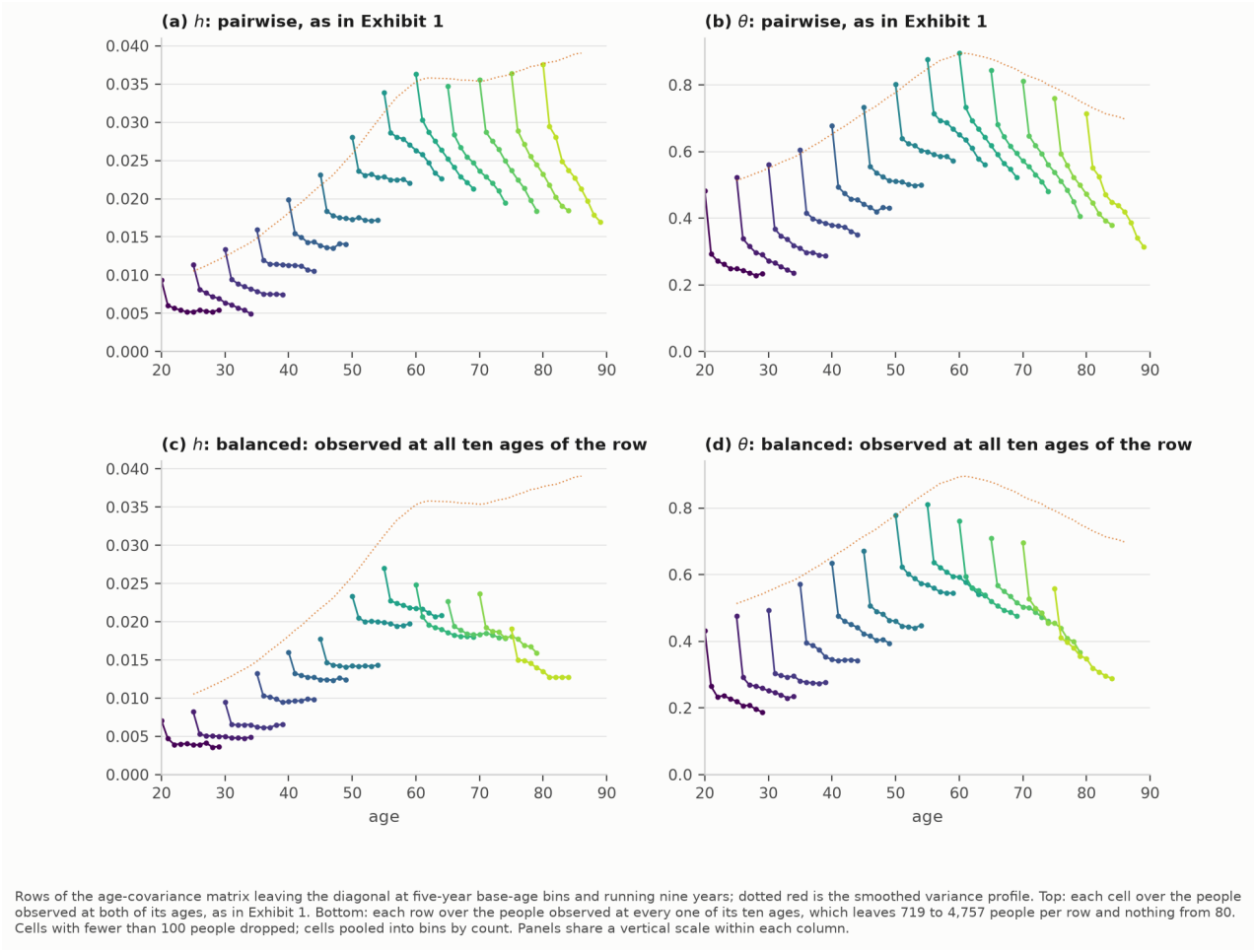



Figure D.1: Covariance rows on pairwise (top) and balanced (bottom) samples.

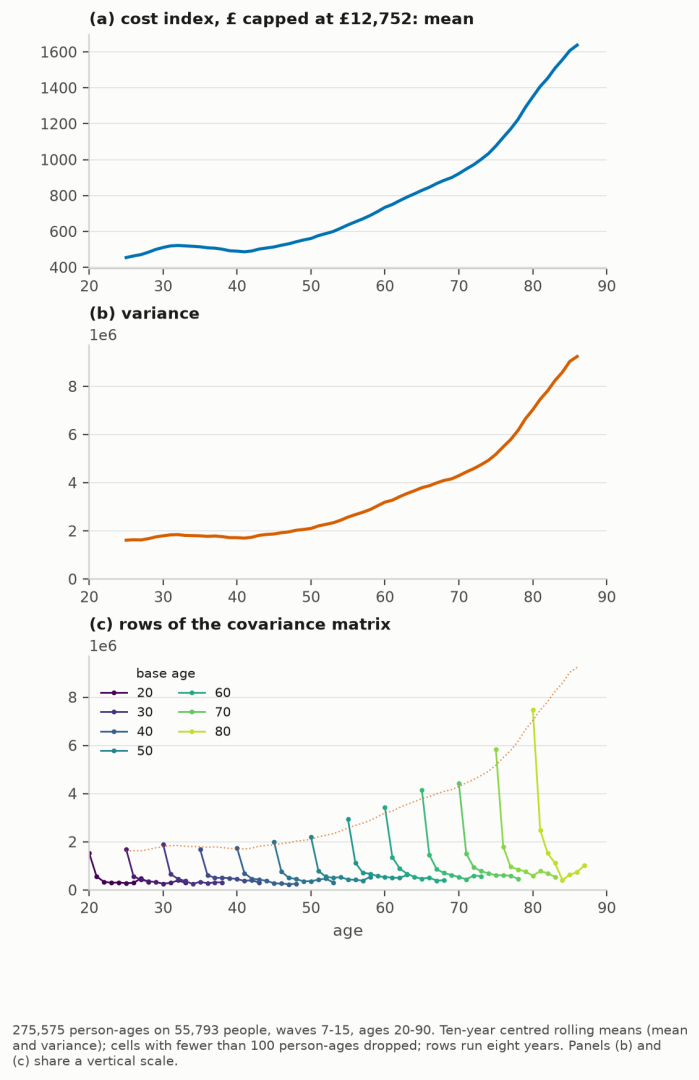



Figure D.2: The cost index through the moments of Figure 2.

## D.2 The Linear Approximation




Case 1 (transitory only) on theta, h, the frailty index and log(frailty + 1/31). Four ages four years apart, pairwise moments pooled over base ages within each band; a base age enters when every cell rests on at least 50 people observed at both of its ages. Properly constrained fit: V_H, V_d and noise variances non-negative and |Corr(H,d)| <= 1 imposed in the estimation, rho profiled. Where the unconstrained fit was inadmissible the title gives it and the fit cost of the constraint in squared error. Where V_H falls below 5% of the first variance the level, and so Corr(H,d), is not identified. Bars stack the fitted layers; black ticks are the observed moments. V(k): the variance at the k-th age; C(k,l): the covariance between ages k and l.

Figure D.3: Figure 3 under Case 1 (transitory only), on all four measures.

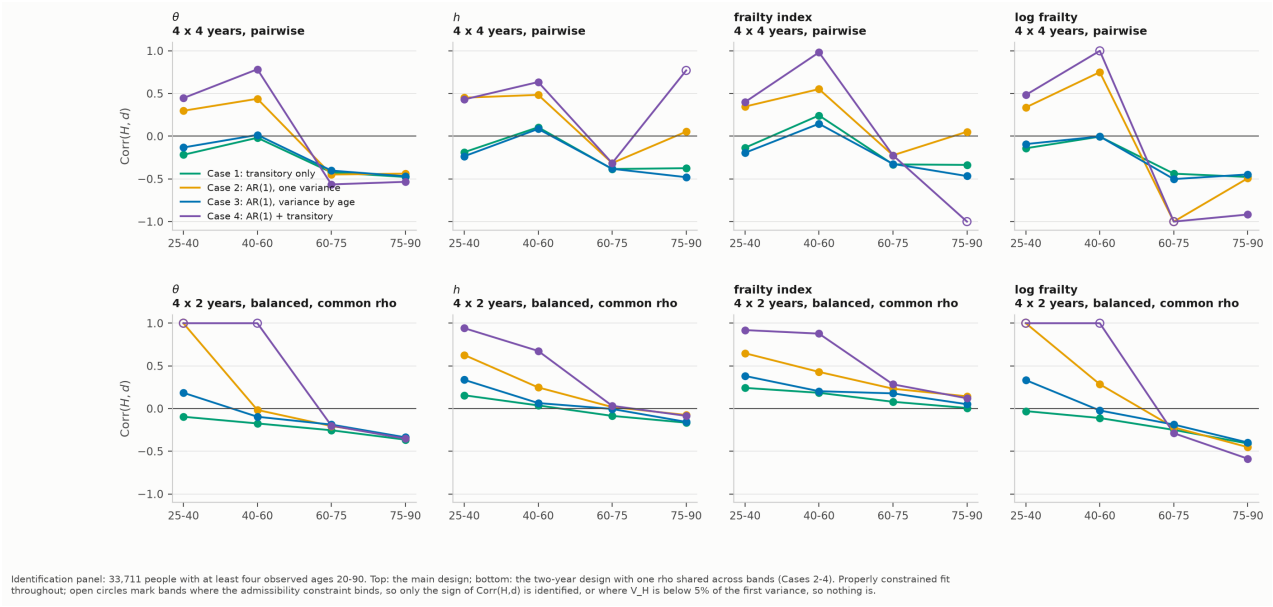



Figure D.4: Implied Corr$(H, d)$ by age band under Cases 1–4, all measures, two designs.




Four ages two years apart, balanced moments (people observed at all four ages), one rho shared by the four bands in Case 4. Properly constrained fit: V_H, V_d and noise variances non-negative and |Corr(H,d)| <= 1 imposed in the estimation, rho profiled. Where the unconstrained fit was inadmissible the title gives it and the fit cost of the constraint in squared error. Where V_H falls below 5% of the first variance the level, and so Corr(H,d), is not identified. Bars stack the fitted layers; black ticks are the observed moments. V(k): the variance at the k-th age; C(k,l): the covariance between ages k and l.

Figure D.5: Figure 3 on the two-year balanced design.

# E Additional K-Means Results

## E.1 The Types on Predicted Cost

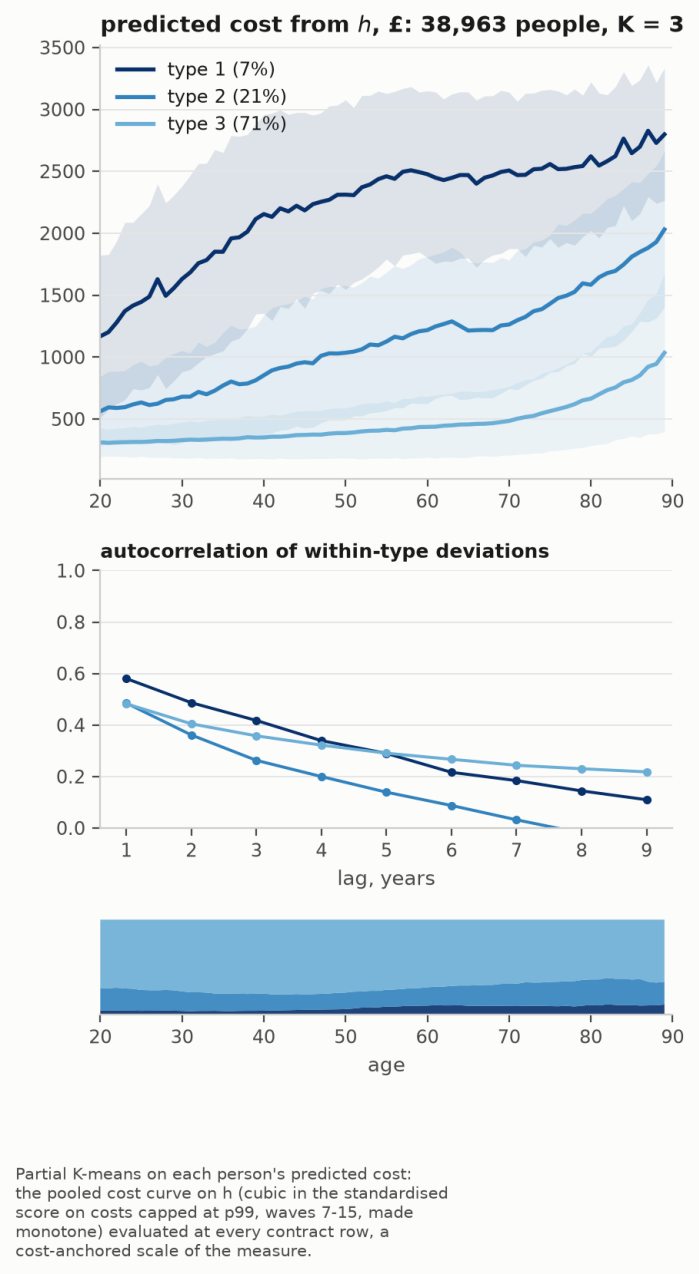
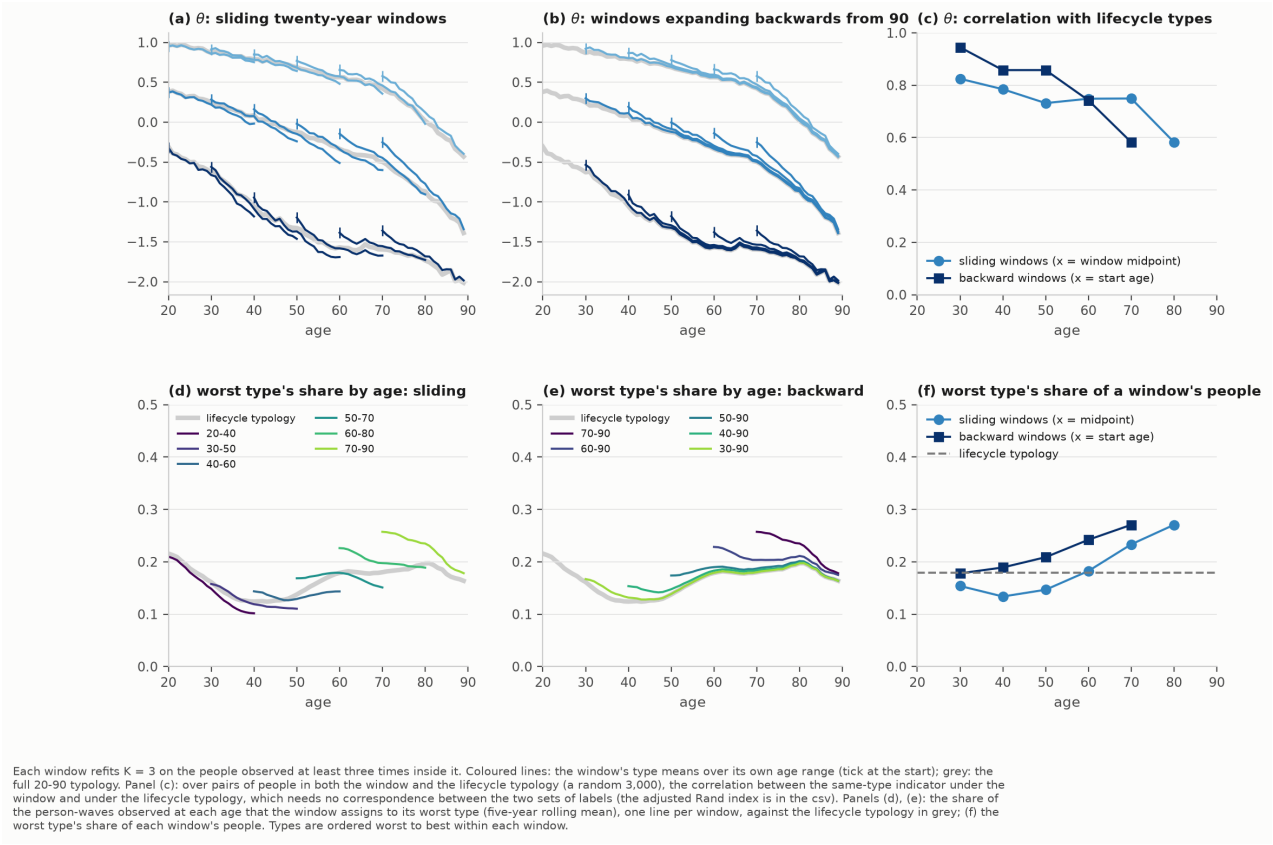



Figure E.1: Partial K-means on predicted cost; lands with the $h$ types (ARI 0.65).

## E.2 Types over Age Windows

- K-means is more stable across windows than the mixture, but only because it is insensitive to where the decline happens.



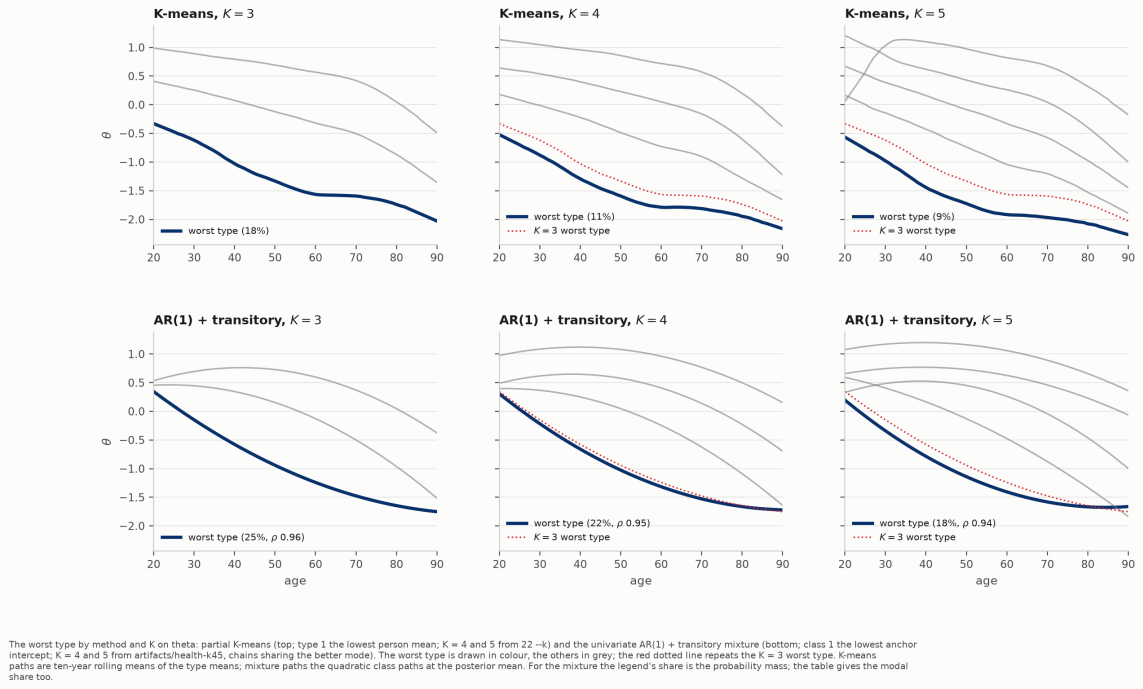

Each window refits K = 3 on the people observed at least three times inside it. Coloured lines: the window's type means over its own age range (tick at the start); grey: the full 20-90 typology. Panel (c): over pairs of people in both the window and the lifecycle typology (a random 3,000), the correlation between the same-type indicator under the window and under the lifecycle typology, which needs no correspondence between the two sets of labels (the adjusted Rand index is in the csv). Panels (d), (e): the share of the person-waves observed at each age that the window assigns to its worst type (five-year rolling mean), one line per window, against the lifecycle typology in grey; (f) the worst type's share of each window's people. Types are ordered worst to best within each window.

Figure E.2: As Figure 8, for partial K-means.

# F Additional Bayesian Results

## F.1 The Worst Type by Method and Number of Classes

- The error structure changes the worst type's path, not its membership; more classes tighten it without changing its shape.




The worst type by method and K on theta: partial K-means (top; type 1 the lowest person mean; K = 4 and 5 from 22 --k) and the univariate AR(1) + transitory mixture (bottom; class 1 the lowest anchor intercept; K = 4 and 5 from artifacts/health-k45, chains sharing the better mode). The worst type is drawn in colour, the others in grey; the red dotted line repeats the K = 3 worst type. K-means paths are ten-year rolling means of the type means; mixture paths the quadratic class paths at the posterior mean. For the mixture the legend's share is the probability mass; the table gives the modal share too.

Figure F.1: The worst type under K-means and the persistent–transitory mixture, $K = 3, 4, 5$.

| method | $K$ | share | $\theta_{20}$ | $\theta_{50}$ | $\theta_{80}$ | slope to 50 | sd at 50 | $\rho$ | of $K$=3 worst | worst under other |
|---|---|---|---|---|---|---|---|---|---|---|
| K-means | 3 | 18% | -0.33 | -1.33 | -1.74 | -0.033 | 0.66 | | 100% | 85% |
| | 4 | 11% | -0.52 | -1.59 | -1.95 | -0.036 | 0.59 | | 63% | 92% |
| | 5 | 9% | -0.57 | -1.72 | -2.08 | -0.039 | 0.55 | | 48% | 94% |
| AR(1) + transitory | 3 | 18% (25%) | 0.35 | -0.94 | -1.65 | -0.043 | 0.81 | 0.96 | 100% | 83% |
| | 4 | 15% (22%) | 0.31 | -1.03 | -1.66 | -0.044 | 0.80 | 0.95 | 84% | 67% |
| | 5 | 14% (18%) | 0.20 | -1.14 | -1.67 | -0.045 | 0.79 | 0.94 | 76% | 57% |

Table F.1: The worst type by method and $K$.

## F.2 Which Type Drives the Level–Slope Reversal?

- The worst type's own convergence carries nearly all the negative covariance after 60, also with within-person slopes.

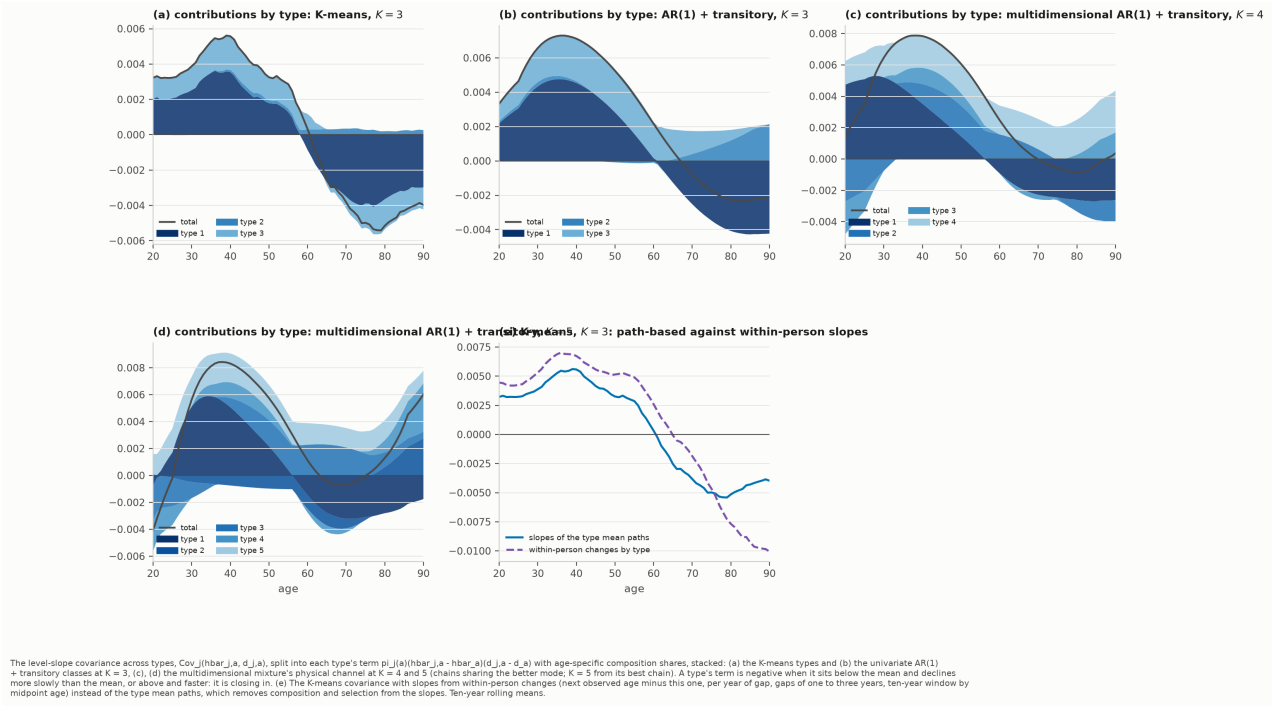
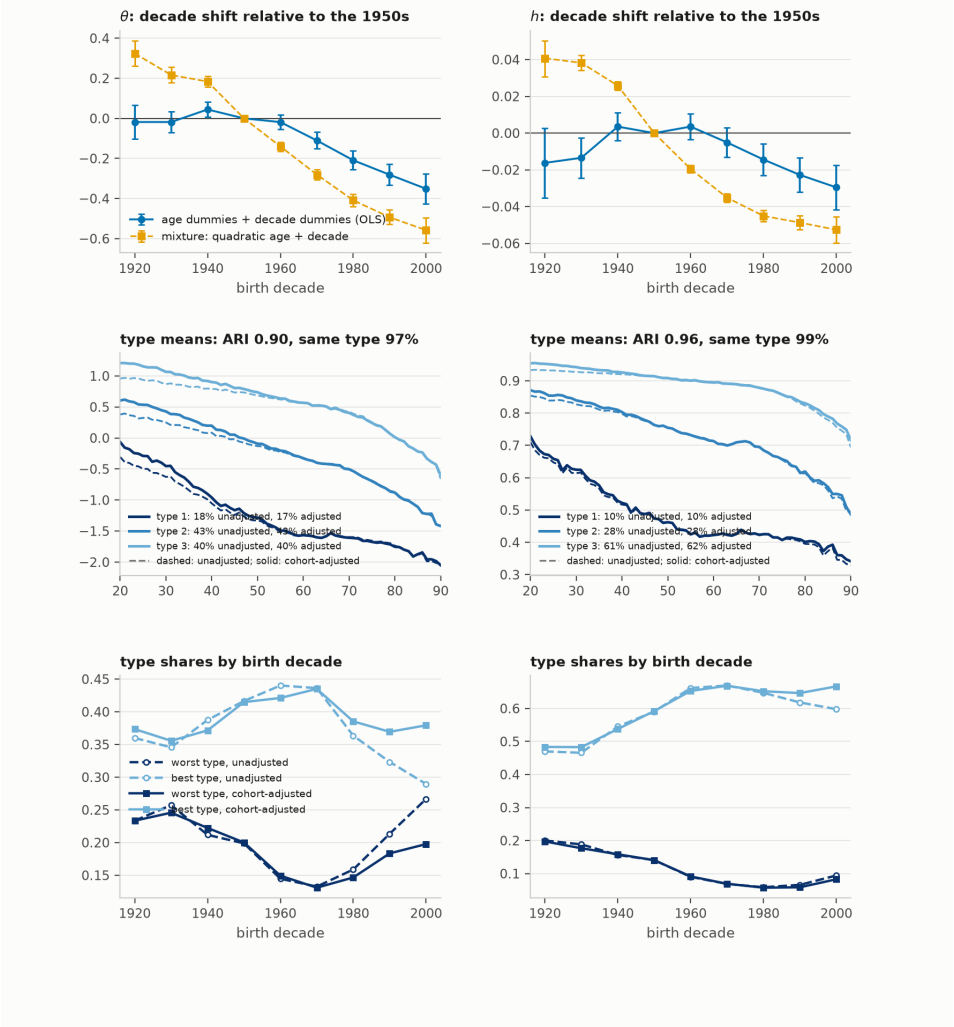



Figure F.2: The level–slope covariance split into each type's term, by typology.

## F.3 Birth-Decade Cohort Effects

- Neither K-means with flexible age nor the mixtures with decade shifts change the typology; believe the flexible version's shifts (zero before 1970).




Health contract people. Top: decade level shifts from OLS on single-year age dummies plus decade dummies (95% person-clustered intervals), against the mixture's quadratic-plus-decade estimates. Middle: K-means type means, unadjusted scores (dashed) and scores net of the OLS shifts re-clustered (solid). Bottom: share of people in the worst and best type by birth decade under both.

Figure F.3: Cohort adjustment within partial K-means.

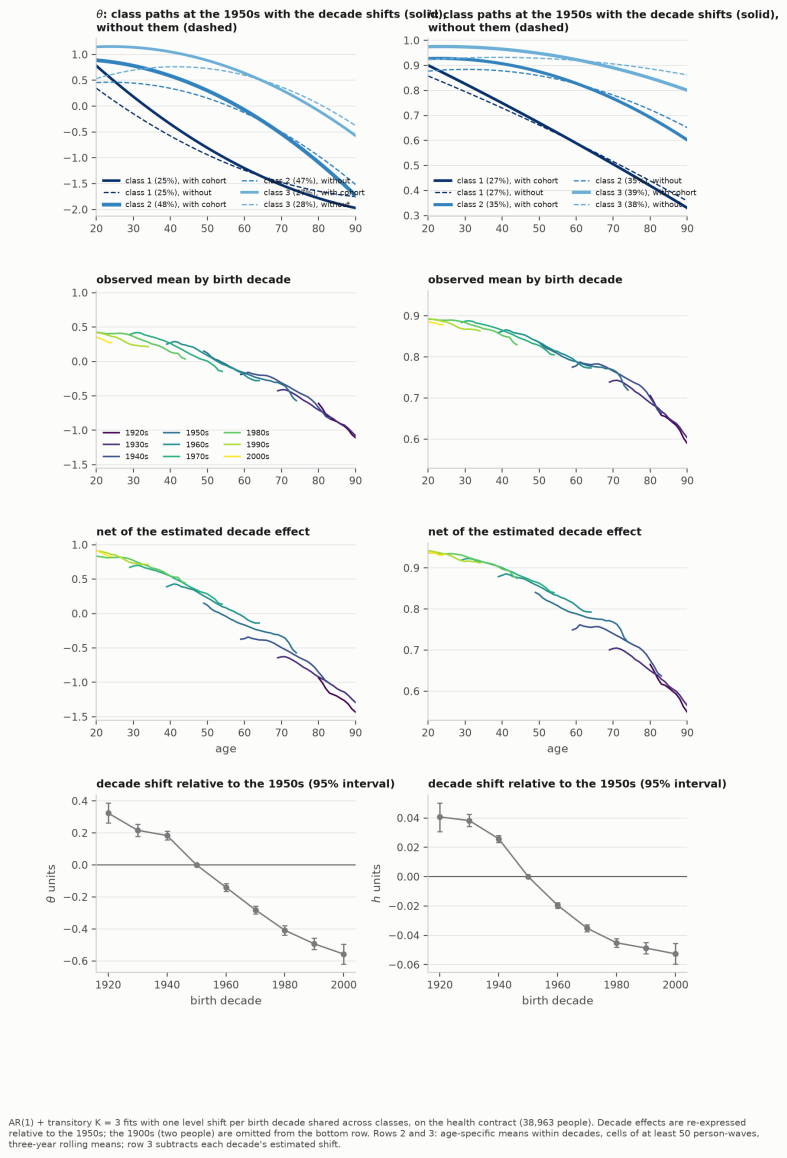
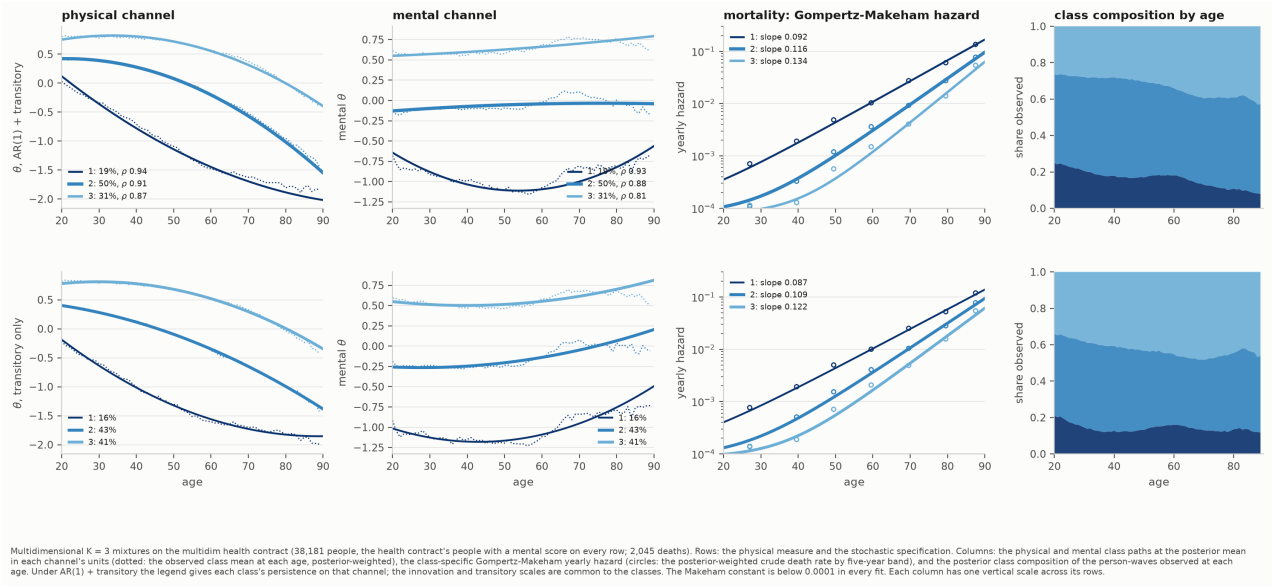
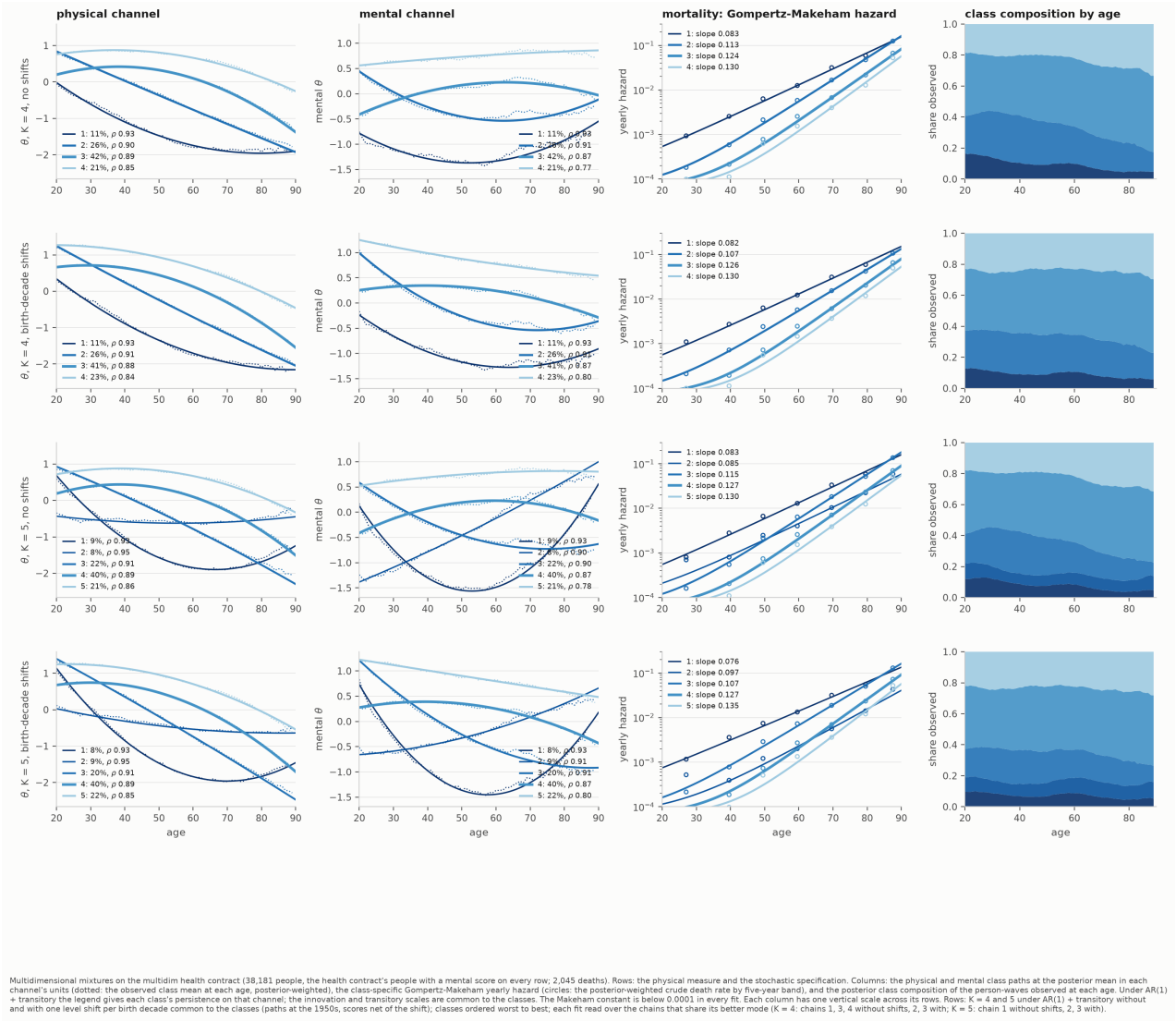



Figure F.4: Single-channel mixtures with birth-decade shifts.




Multidimensional K = 3 mixtures on the multidim health contract (38,181 people, the health contract's people with a mental score on every row; 2,045 deaths). Rows: the physical measure and the stochastic specification. Columns: the physical and mental class paths at the posterior mean in each channel's units (dotted: the observed class mean at each age, posterior-weighted), the class-specific Gompertz-Makeham yearly hazard (circles: the posterior-weighted crude death rate by five-year band), and the posterior class composition of the person-waves observed at each age. Under AR(1) + transitory the legend gives each class's persistence on that channel; the innovation and transitory scales are common to the classes. The Makeham constant is below 0.0001 in every fit. Each column has one vertical scale across its rows.

Figure F.5: Multidimensional $K = 3$ mixtures without the decade shifts.

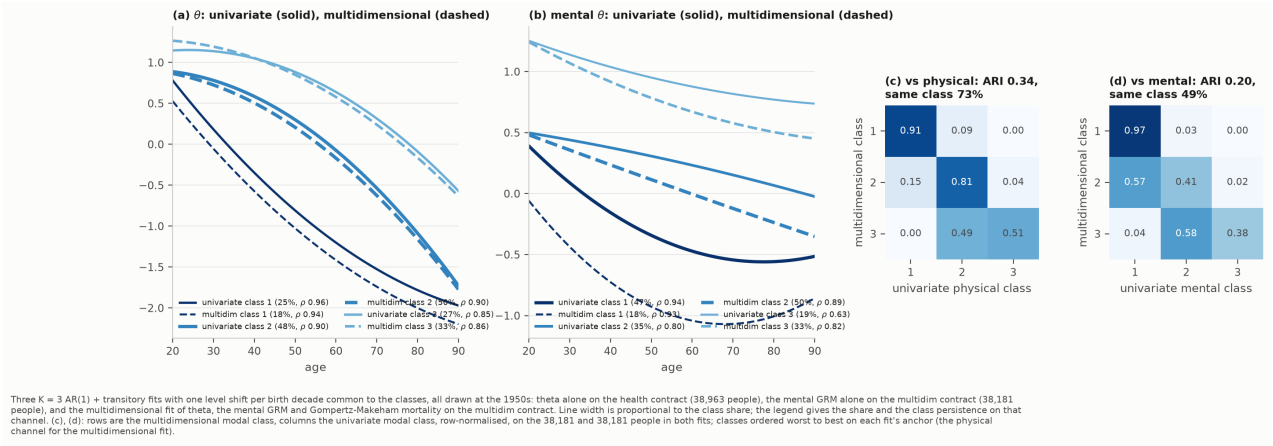



Multidimensional mixtures on the multidim health contract (38,181 people, the health contract's people with a mental score on every row; 2,045 deaths). Rows: the physical measure and the stochastic specification. Columns: the physical and mental class paths at the posterior mean in each channel's units (dotted: the observed class mean at each age, posterior-weighted), the class-specific Gompertz-Makeham yearly hazard (circles: the posterior-weighted crude death rate by five-year band), and the posterior class composition of the person-waves observed at each age. Under AR(1) + transitory the legend gives each class's persistence on that channel; the innovation and transitory scales are common to the classes. The Makeham constant is below 0.0001 in every fit. Each column has one vertical scale across its rows. Rows: K = 4 and 5 under AR(1) + transitory without and with one level shift per birth decade common to the classes (paths at the 1950s, scores net of the shift); classes ordered worst to best; each fit read over the chains that share its better mode (K = 4: chains 1, 3, 4 without shifts, 2, 3 with; K = 5: chain 1 without shifts, 2, 3 with).

Figure F.6: Multidimensional $K = 4$ and $K = 5$ fits with and without decade shifts.

## F.4 Multidimensional against Univariate Classes

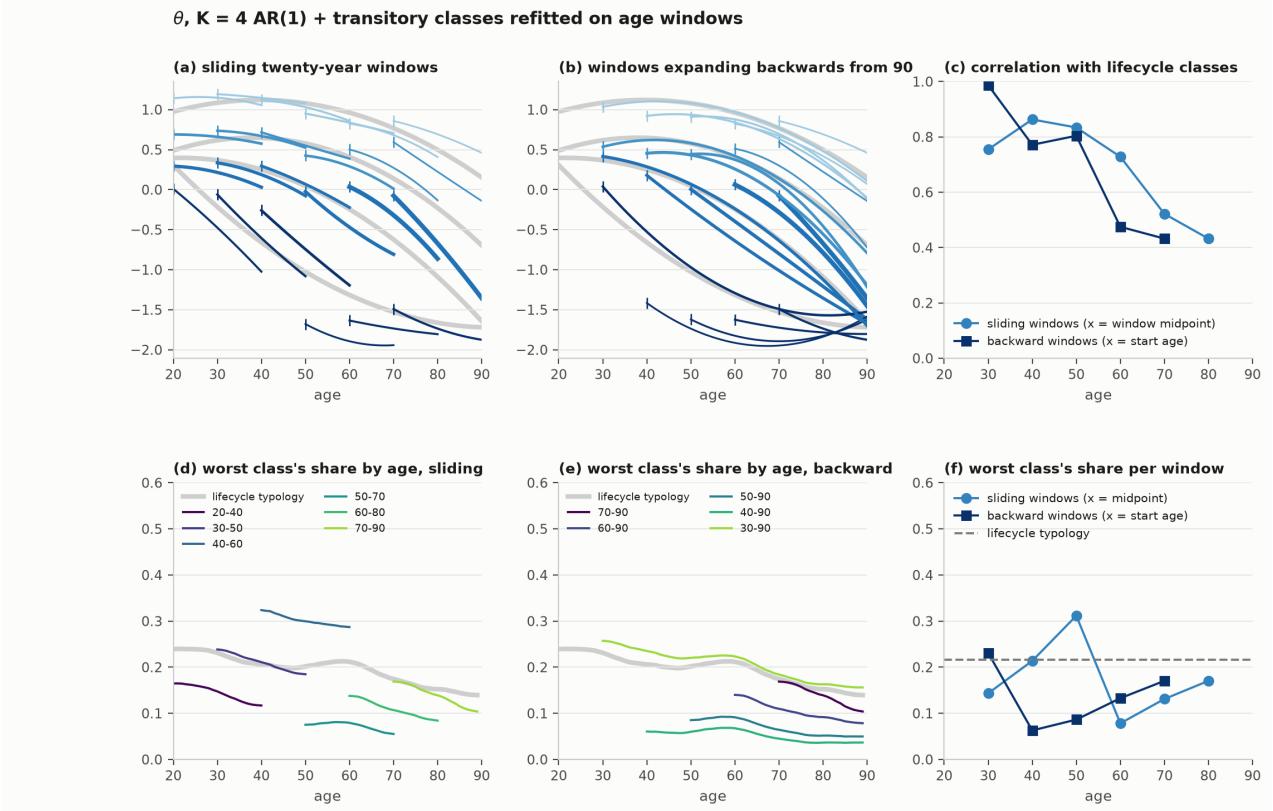
- The multidimensional classes are the physical classes with a mental path attached; the mental score alone gives a different partition.




Three K = 3 AR(1) + transitory fits with one level shift per birth decade common to the classes, all drawn at the 1950s: theta alone on the health contract (38,963 people), the mental GRM alone on the multidim contract (38,181 people), and the multidimensional fit of theta, the mental GRM and Gompertz-Makeham mortality on the multidim contract. Line width is proportional to the class share; the legend gives the share and the class persistence on that channel. (c), (d): rows are the multidimensional modal class, columns the univariate modal class, row-normalised, on the 38,181 and 38,181 people in both fits; classes ordered worst to best on each fit's anchor (the physical channel for the multidimensional fit).

Figure F.7: Multidimensional against univariate physical and mental classes.

## F.5 The Windows at Four Classes




Each window refits the K = 4 AR(1) + transitory mixture on theta on the people observed at least three times inside it (its own contract and fit). Coloured lines: the window's class paths at the posterior mean over its own age range (tick at the start; line width proportional to the class share); grey: the lifecycle 20-90 fit. Panel (c): over pairs of people in both fits (a random 3,000), the correlation between the probability that the window fit puts the pair in the same class and the probability that the lifecycle fit does, sum_k p_ik p_jk under each; it needs no correspondence between the two sets of labels and is 1 whenever the two posteriors coincide (the adjusted Rand index of the modal classes is in the csv). Panels (d), (e): the worst class's posterior share of the person-waves observed at each age (five-year rolling mean), one line per window, against the lifecycle fit in grey; (f) the worst class's share of each window's people. Classes ordered worst to best by the class path's mean over the window. Windows whose chains split into modes are read over the chains that share the better mode: 70-90 over chains 1,4; 60-90 over chains 1,2,3.

Figure F.8: As Figure 8, at $K = 4$.

## F.6 Predicting Death

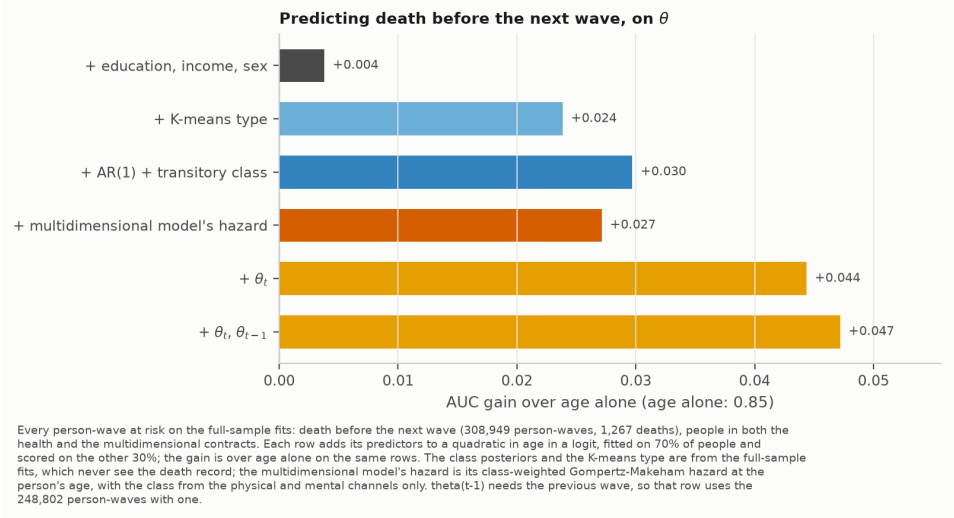



Figure F.9: AUC gain over age in predicting death before the next wave, on $\theta$.

# G Additional Results on the Returns to Prevention

## G.1 The Estimated Cost Curve

- Close to age-invariant below 75; the 75–90 band sits 32% above, so late-life cost is understated.

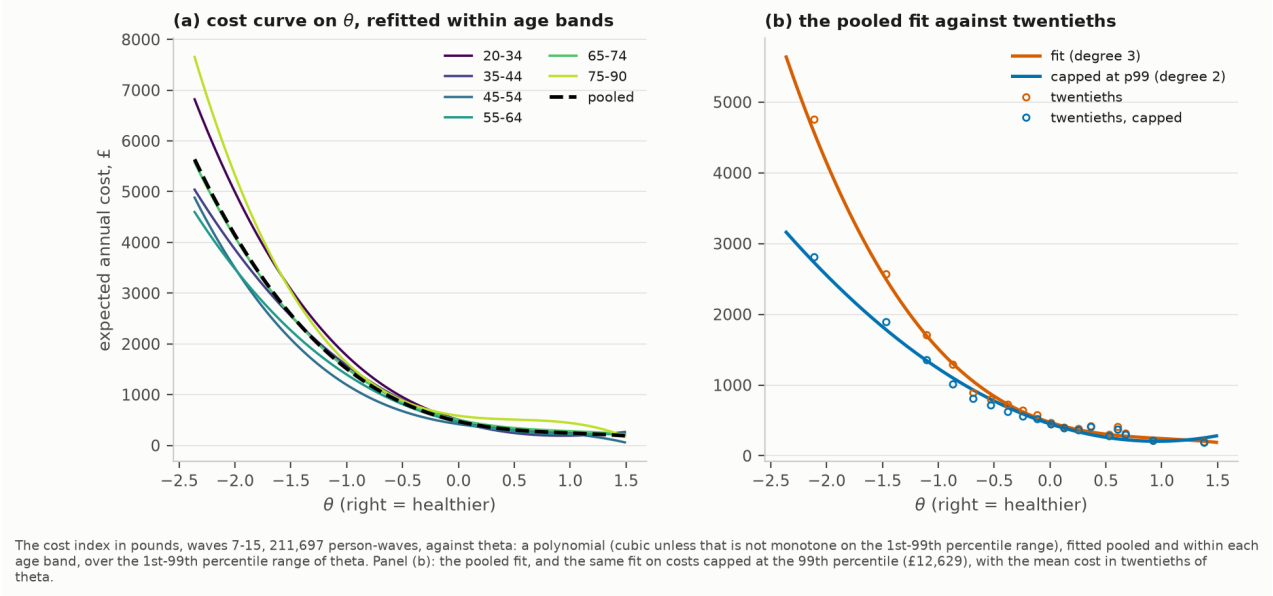
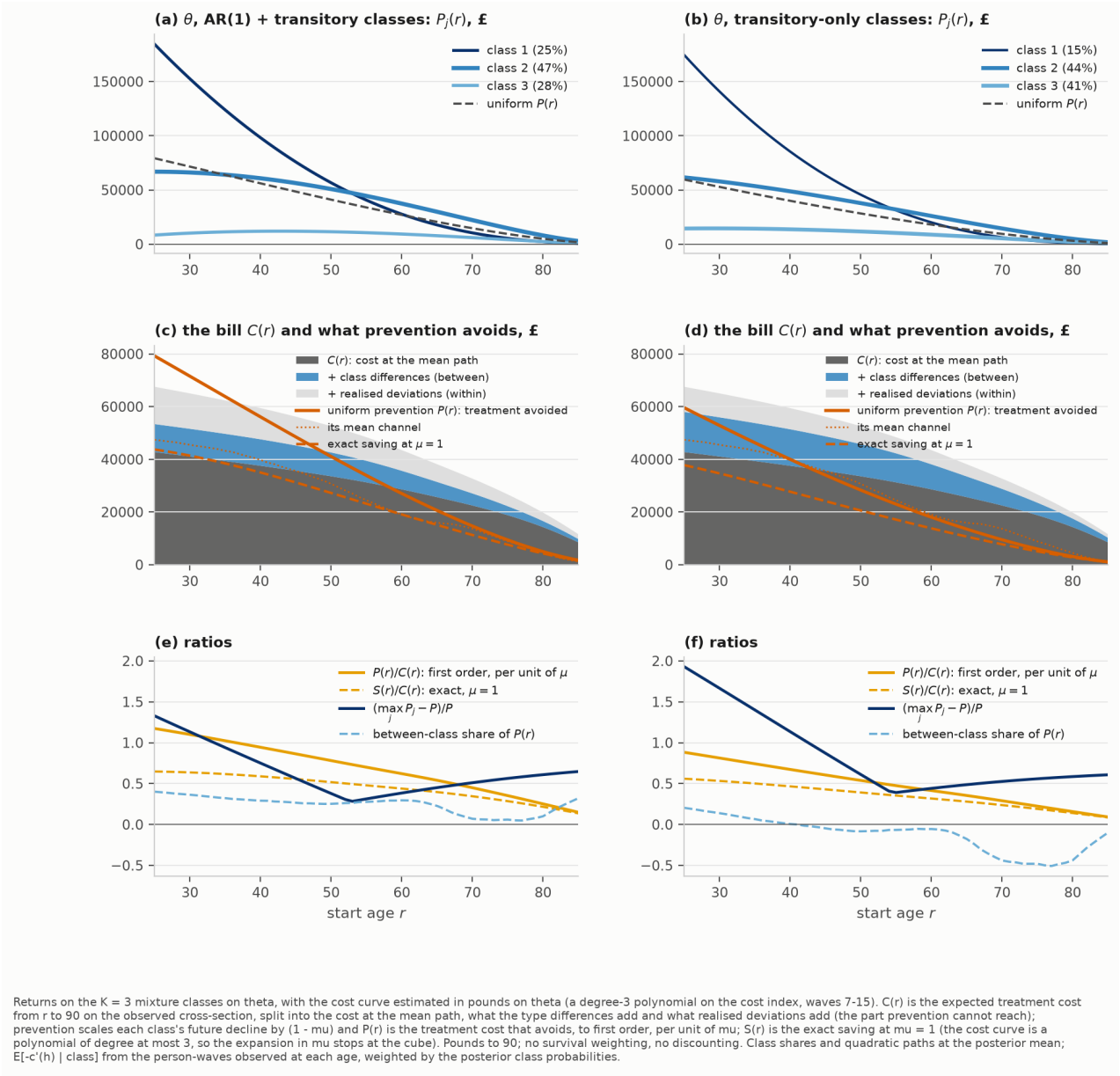


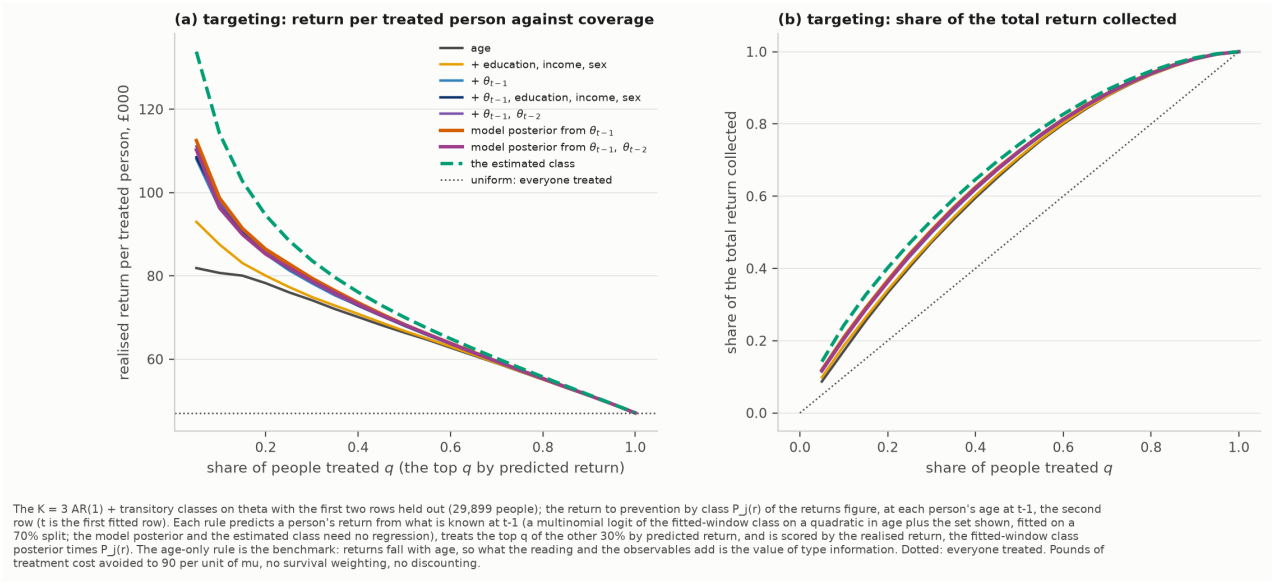

Figure G.1: The cubic cost curve: by age band (a) and capped vs uncapped (b).

## G.2 The Detailed Returns and the Targeting Rules



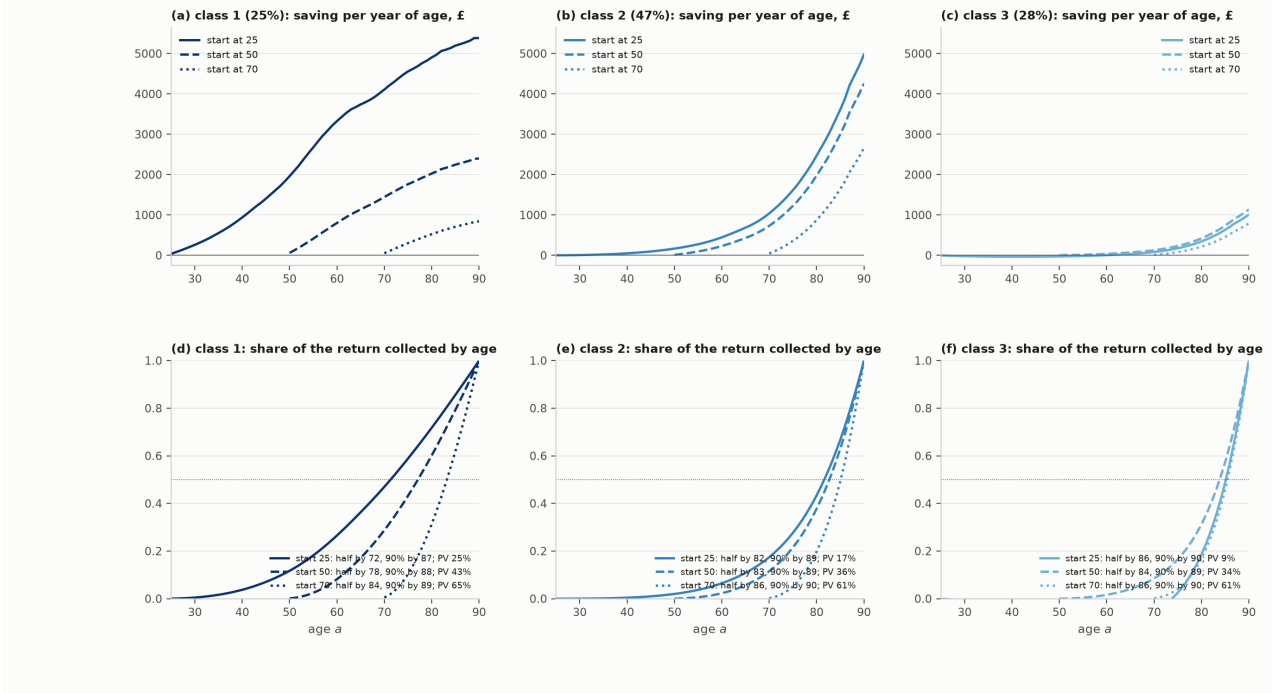

Returns on the K = 3 mixture classes on theta, with the cost curve estimated in pounds on theta (a degree-3 polynomial on the cost index, waves 7-15). C(r) is the expected treatment cost from r to 90 on the observed cross-section, split into the cost at the mean path, what the type differences add and what realised deviations add (the part prevention cannot reach); prevention scales each class's future decline by (1 - mu) and P(r) is the treatment cost that avoids, to first order, per unit of mu; S(r) is the exact saving at mu = 1 (the cost curve is a polynomial of degree at most 3, so the expansion in mu stops at the cube). Pounds to 90; no survival weighting, no discounting. Class shares and quadratic paths at the posterior mean; E[-c'(h) | class] from the person-waves observed at each age, weighted by the posterior class probabilities.

Figure G.2: Bill, $P_j(r)$, $P(r)$, $S(r)$ and ratios by start age; persistent–transitory (left) and transitory only (right).




The K = 3 AR(1) + transitory classes on theta with the first two rows held out (29,899 people); the return to prevention by class P_j(r) of the returns figure, at each person's age at t-1, the second row (t is the first fitted row). Each rule predicts a person's return from what is known at t-1 (a multinomial logit of the fitted-window class on a quadratic in age plus the set shown, fitted on a 70% split; the model posterior and the estimated class need no regression), treats the top q of the other 30% by predicted return, and is scored by the realised return, the fitted-window class posterior times P_j(r). The age-only rule is the benchmark: returns fall with age, so what the reading and the observables add is the value of type information. Dotted: everyone treated. Pounds of treatment cost avoided to 90 per unit of mu, no survival weighting, no discounting.

Figure G.3: Targeting rules when the type is not observed: return per treated person and share collected.

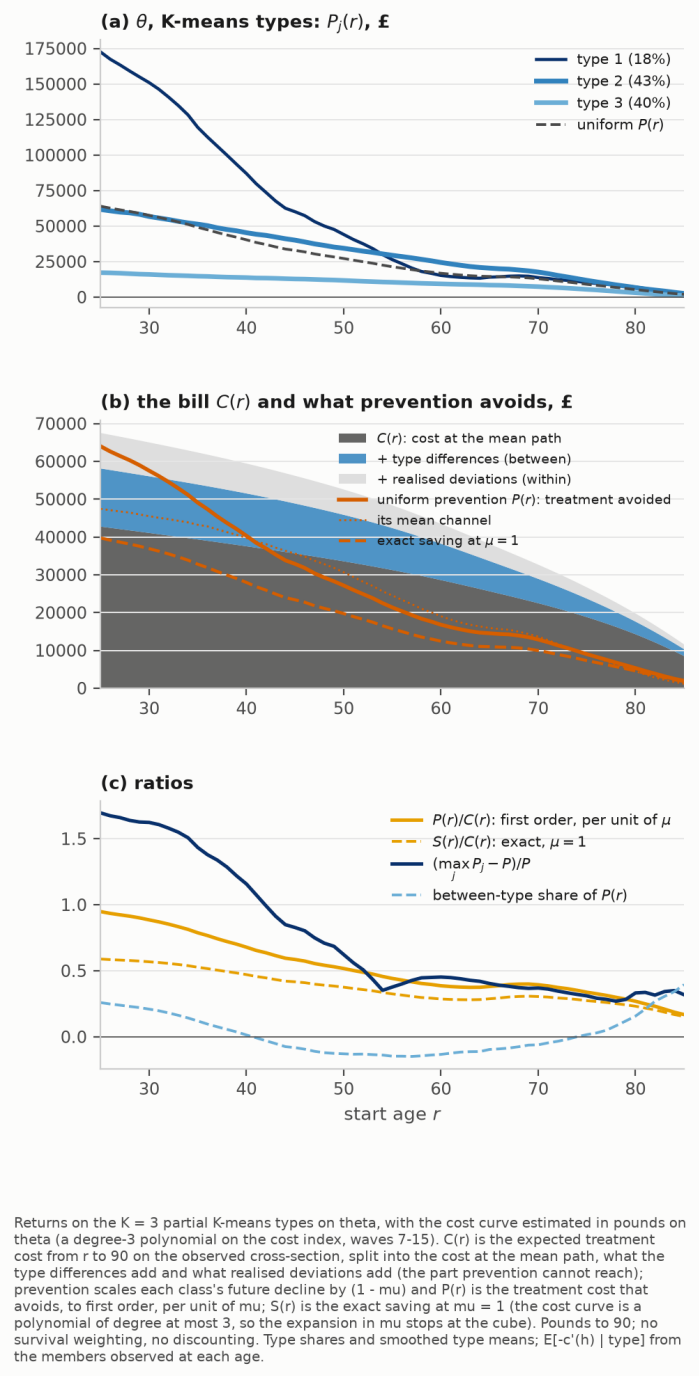



The K = 3 AR(1) + transitory classes on theta with the cost curve in pounds. Top: the saving in each year of age a from prevention started at r, s_j(a | r) = E[-c'(h_a) | j] (hbar_{j,r-1} - hbar_{j,a}), the per-age terms whose sum over a >= r is the class return P_j(r), for starts at 25, 50 and 70. Bottom: the cumulative share of P_j(r) collected by age a, with the ages by which half and 90% has arrived and the present value at 3.5% a year (the Green Book rate) as a share of the undiscounted return. Pounds per unit of mu; no survival weighting.

Figure G.4: When the saving arrives by class, with half-way ages and present values at 3.5%.

## G.3 K-Means Types under the Estimated Cost Curve

- Lower $P/C$ than the mixture, and the between-type channel turns negative from about 40.

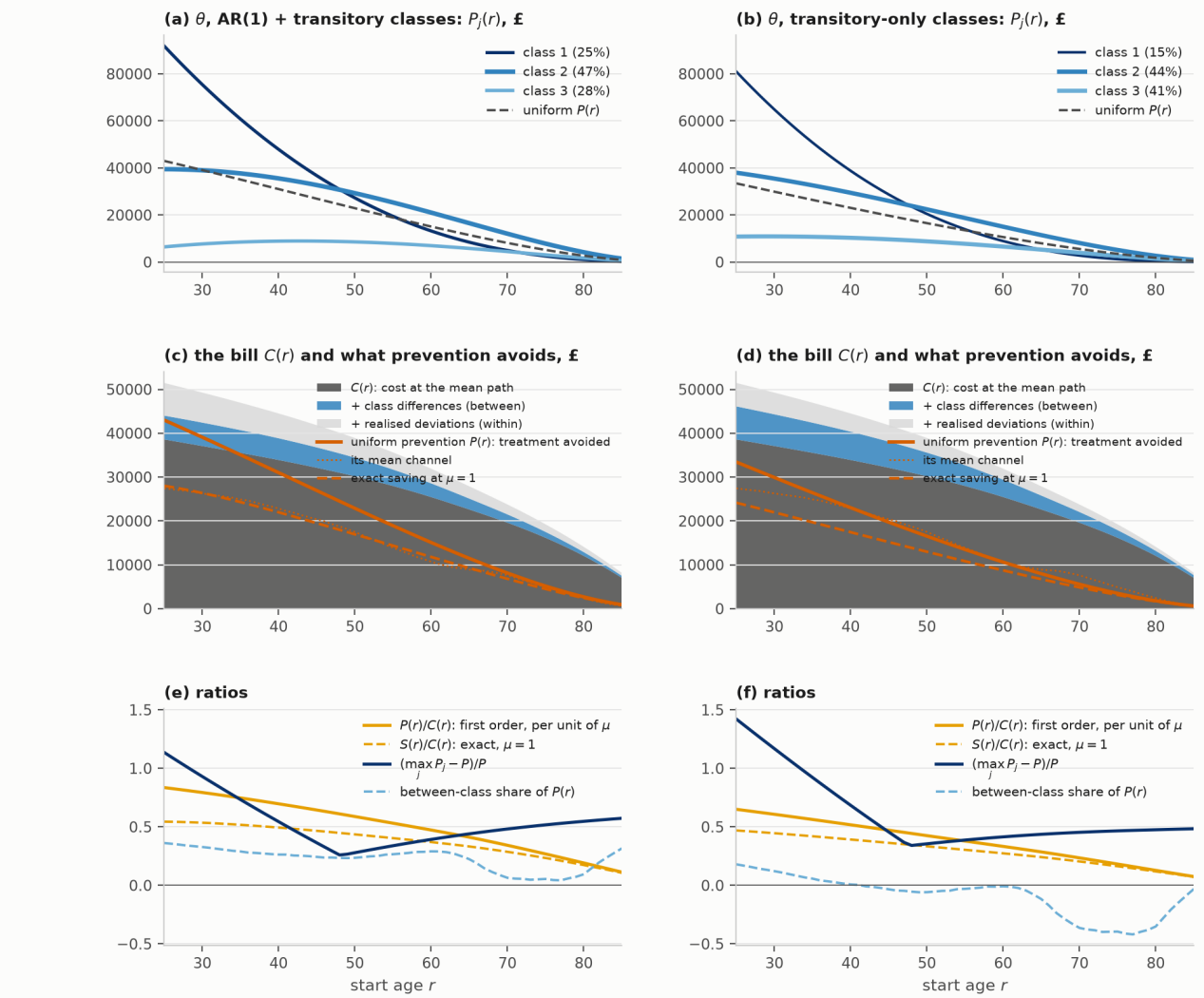



Returns on the K = 3 partial K-means types on theta, with the cost curve estimated in pounds on theta (a degree-3 polynomial on the cost index, waves 7-15). C(r) is the expected treatment cost from r to 90 on the observed cross-section, split into the cost at the mean path, what the type differences add and what realised deviations add (the part prevention cannot reach); prevention scales each class's future decline by (1 - mu) and P(r) is the treatment cost that avoids, to first order, per unit of mu; S(r) is the exact saving at mu = 1 (the cost curve is a polynomial of degree at most 3, so the expansion in mu stops at the cube). Pounds to 90; no survival weighting, no discounting. Type shares and smoothed type means; E[-c'(h) | type] from the members observed at each age.

Figure G.5: As Figure G.2, on the K-means types.

## G.4 The Capped Cost Curve

- Capping roughly halves first-order returns; the ranking survives.




Returns on the K = 3 mixture classes on theta, with the cost curve estimated in pounds on theta (a degree-2 polynomial on the cost index capped at the 99th percentile, waves 7-15). C(r) is the expected treatment cost from r to 90 on the observed cross-section, split into the cost at the mean path, what the type differences add and what realised deviations add (the part prevention cannot reach); prevention scales each class's future decline by (1 - mu) and P(r) is the treatment cost that avoids, to first order, per unit of mu; S(r) is the exact saving at mu = 1 (the cost curve is a polynomial of degree at most 3, so the expansion in mu stops at the cube). Pounds to 90; no survival weighting, no discounting. Class shares and quadratic paths at the posterior mean; E[-c'(h) | class] from the person-waves observed at each age, weighted by the posterior class probabilities.

Figure G.6: As Figure G.2, with costs capped at the 99th percentile.

## G.5 The Return by Class under Other Typologies and Measures

- Same shape under every typology; decade shifts raise returns by about a quarter; the worst class carries the return on every measure; adding classes splits the return, it does not create it.

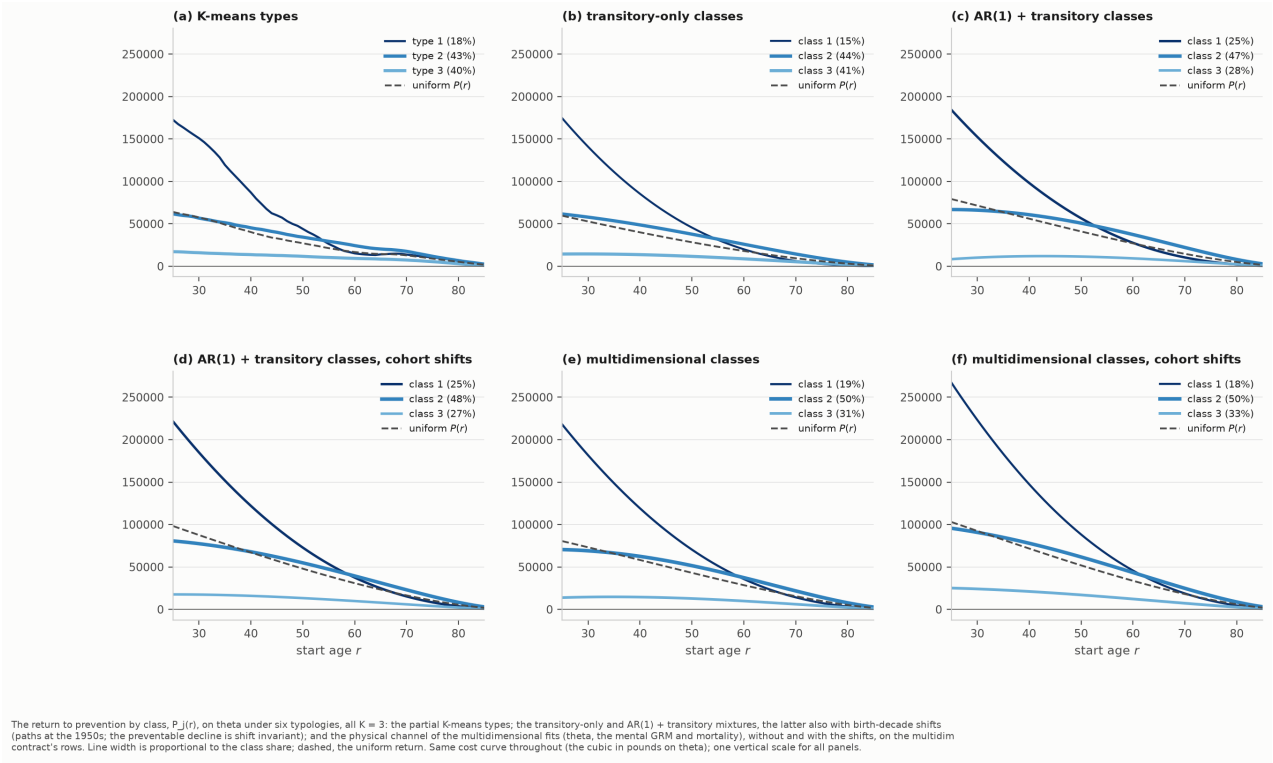



Figure G.7: $P_j(r)$ under six typologies on $\theta$.

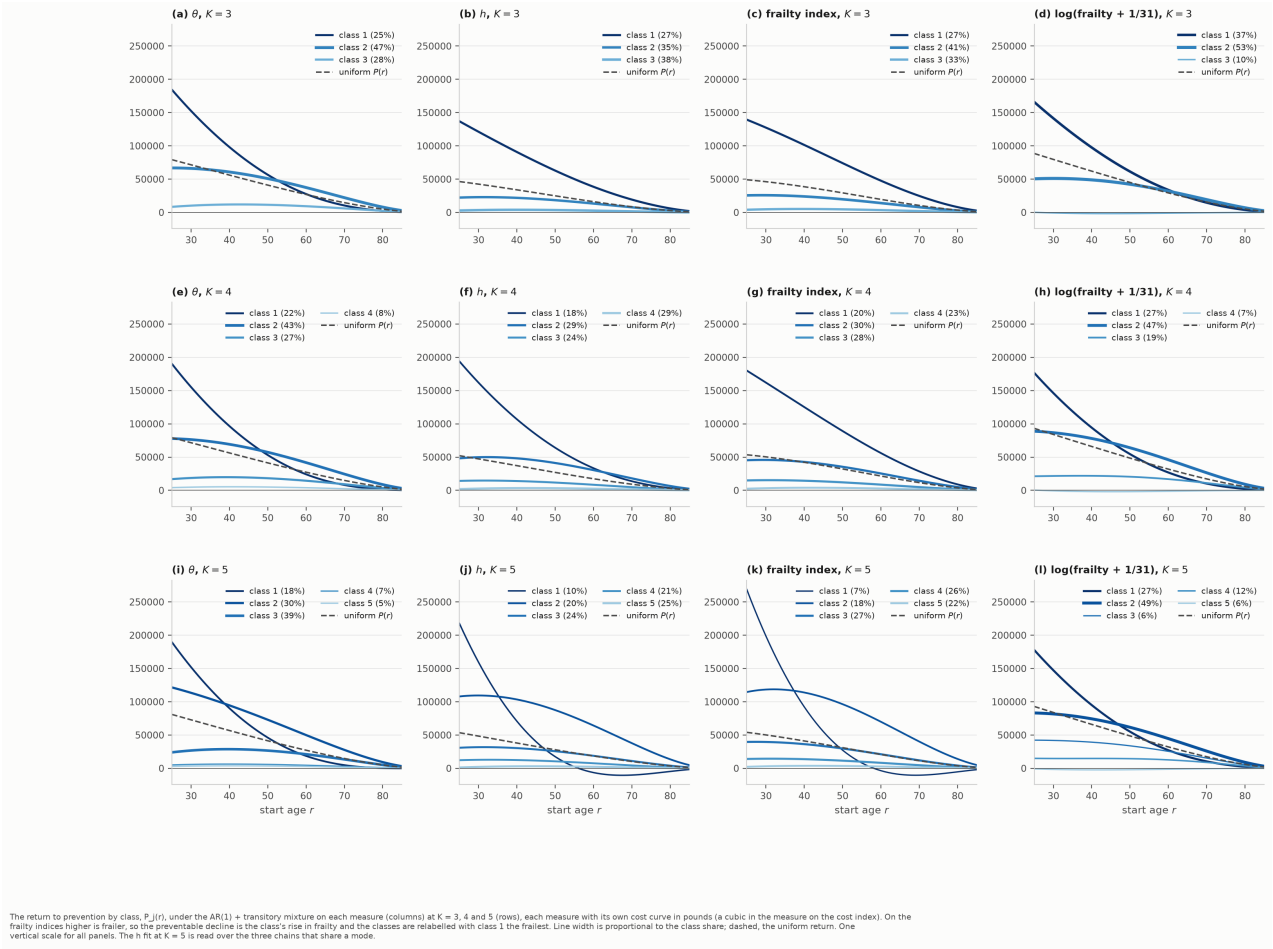
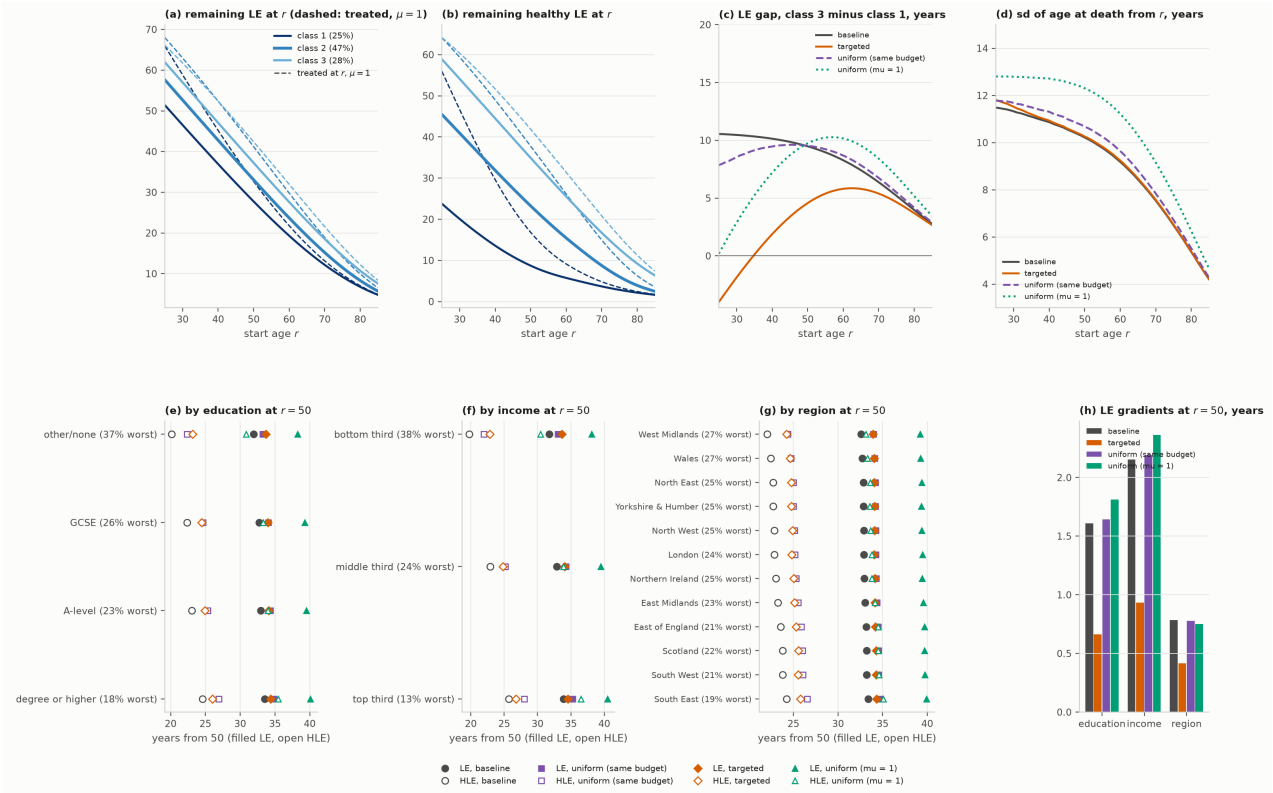



Figure G.8: $P_j(r)$ on each of the four measures at $K = 3, 4, 5$.

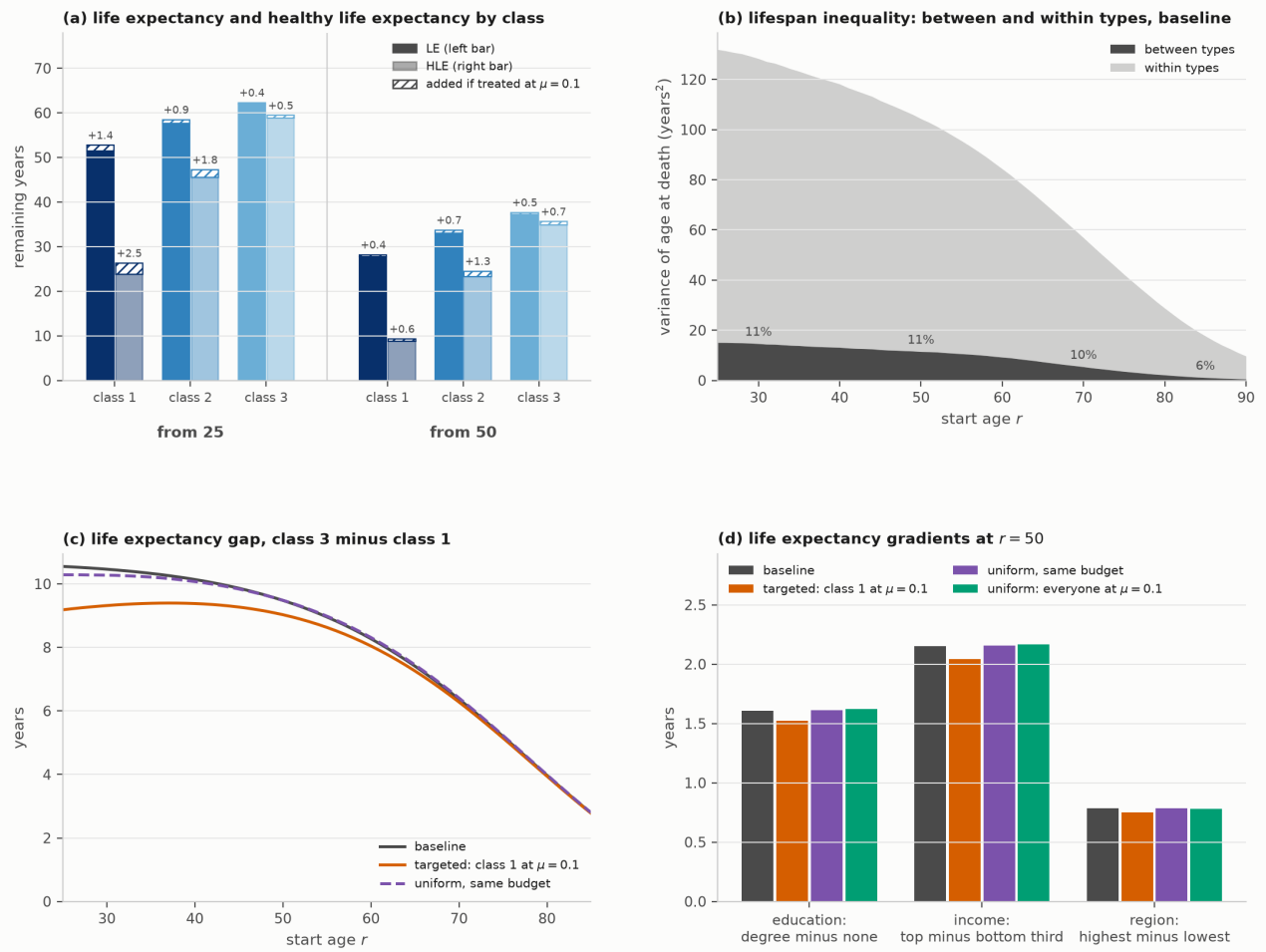
## G.6 Life Years

- Multidimensional classes give wider gaps and the same policy ranking; on the other measures targeting closes the gaps almost entirely.



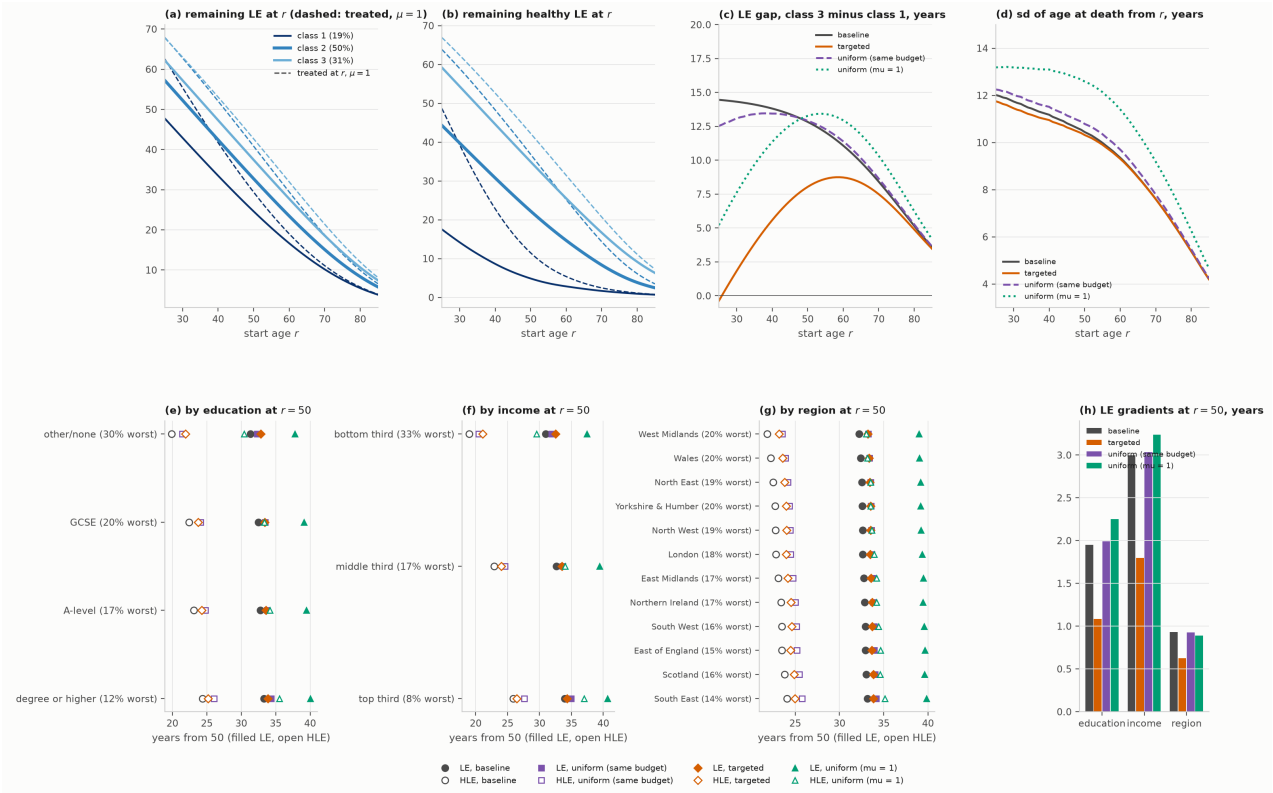

The univariate K = 3 AR(1) + transitory classes on theta (38,963 people), class 1 worst health, with Gompertz-Makeham class hazards estimated by posterior-weighted maximum likelihood on the multidim contract's death intervals (2,045 deaths) and calibrated to the ONS life tables (one common age-varying factor on every class hazard, matching the population LE at 65 and 90). Prevention from r at intensity mu holds back a share mu of the class's mean change in theta after r−1 (the paper's instrument; the gain is held at its value at 90 beyond). Mortality responds through the within-age gradient of death before the next wave in theta (a logit on theta, a quadratic in age and theta x age over 363,033 person-waves at risk, 1,825 deaths: log-odds -0.99 per unit of theta at 55); the healthy share (general health good or better) through the same design fitted to the healthy indicator, applied to each class's person-waves at each age. Policies: targeted, class 1 at mu = 1 (its mean health held at the level of r − 1); uniform with the same budget, everyone at mu = pi_1(r) (class 1's share at r, 0.23 at 50); uniform at mu = 1. (a), (b) dashed: each class treated on its own at mu = 1. (c) class 3 minus class 1. (d) the standard deviation of age at death over the composition-weighted mixture. (e)-(g) each group's LE and HLE at 50 is its posterior class composition among the person-waves within five years of 50 times the class values, so the gradients are the part the types carry (within-class differences by group are not modelled); the label gives the group's worst-class share; regions sorted by baseline LE. (h) LE of the top minus the bottom group: degree or higher minus other/none, top minus bottom income third, and the range across the twelve regions. The within-age mortality gradient is at or below the one the estimated class hazards imply between types before 55 and above it after (-0.9 against -0.7 per unit at 70), so the late-life years are if anything generous.

Figure G.9: Detailed version of Figure 12.



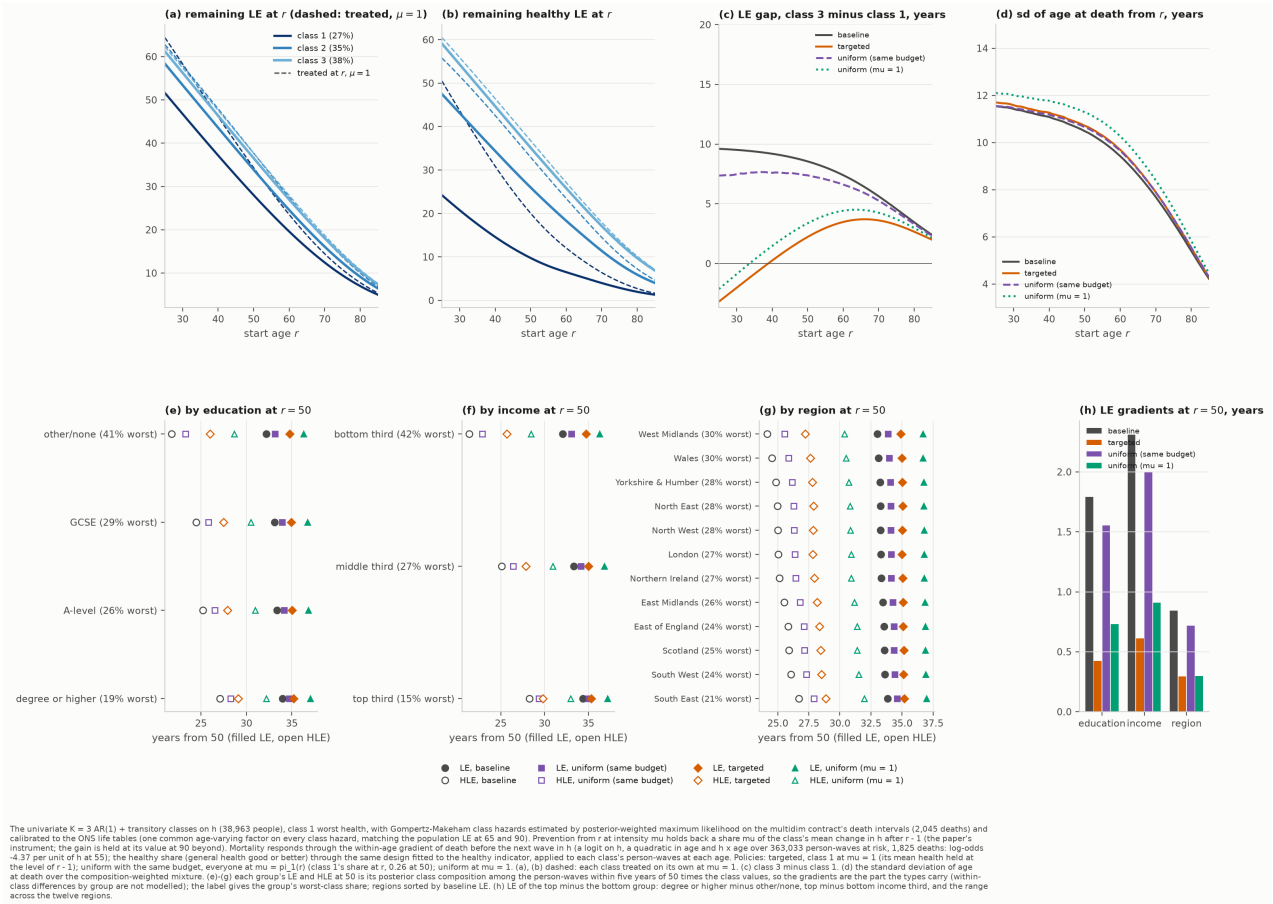

The univariate K = 3 AR(1) + transitory classes on theta (38,963 people), class 1 worst health, with Gompertz-Makeham class hazards estimated by posterior-weighted maximum likelihood on the multidim contract's death intervals (2,045 deaths) and calibrated to the ONS life tables (one common age-varying factor on every class hazard, matching the population LE at 65 and 90). Prevention from r at intensity mu holds back a share mu of the class's mean change in theta after r - 1. Mortality responds through the within-age gradient of death before the next wave in theta (a logit on theta, a quadratic in age and theta x age over 363,033 person-waves at risk, 1,825 deaths: log-odds -0.99 per unit of theta at 55); the healthy share (general health good, very good or excellent) through the same design fitted to the healthy indicator. HLE by the Sullivan method: survival times each class's healthy share at each age. Policies: targeted, class 1 at mu = 0.1; uniform with the same budget, everyone at mu = 0.1 pi_1(r) (class 1's share at r, 0.23 at 50); uniform, everyone at mu = 0.1. (a) solid, the baseline; hatched, what the class gains if treated on its own from r at mu = 0.1. (b) the variance of age at death from r at baseline over the composition-weighted mixture, split by the law of total variance into the part between types, sum_k pi_k (m_k - m)^2 with m_k class k's mean age at death, and the part within them; the labels give the between share. (d) each group's LE at 50 is its class composition among the person-waves within five years of 50 times the class LE, so the gradients are the part the types carry; income thirds of the person's mean percentile rank of equivalised household income within wave, over all adults.

Figure G.10: As Figure 12, at $\mu = 0.1$.



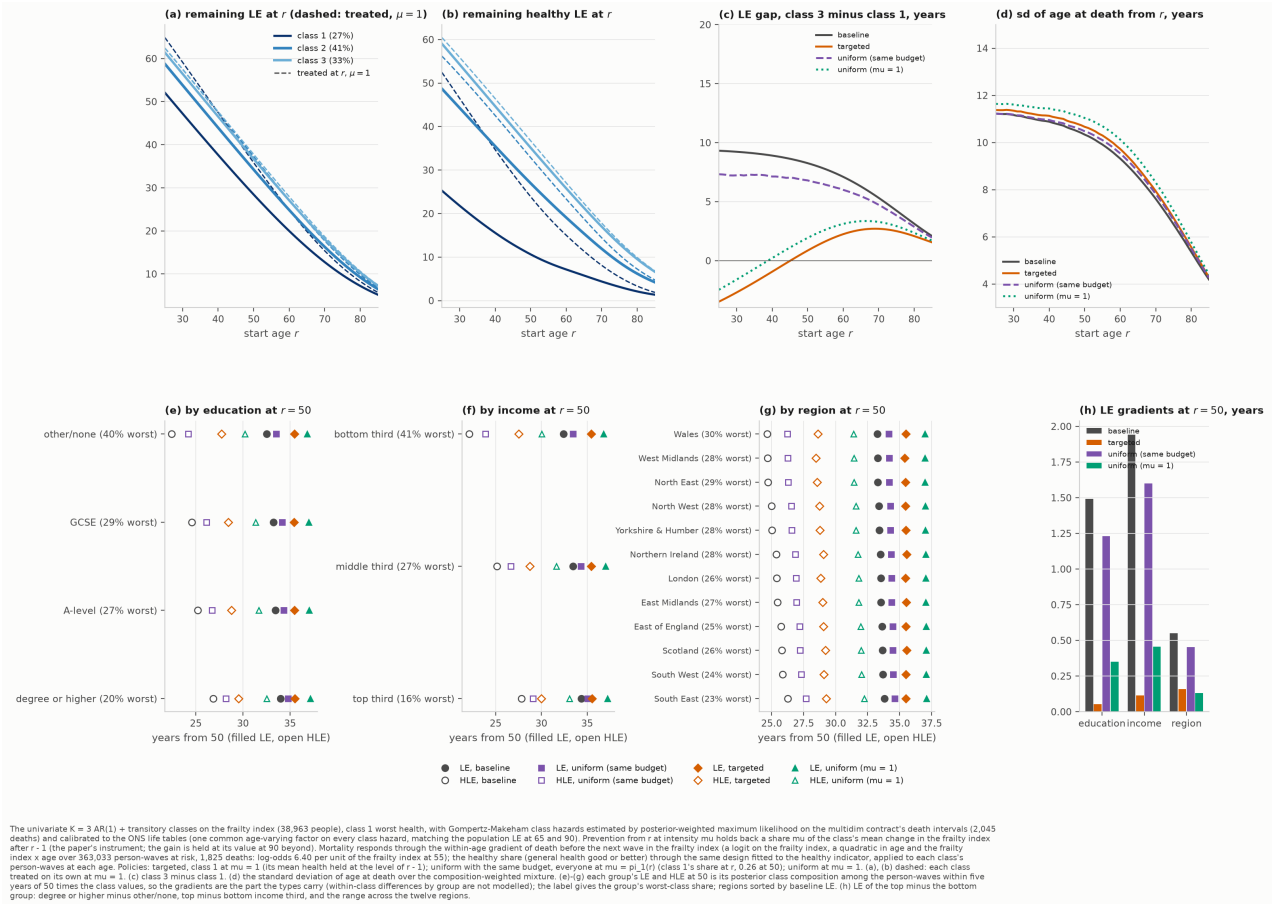

The K = 3 AR(1) + transitory classes of the multidimensional fit (theta, the mental GRM and Gompertz-Makeham mortality; 38,181 people), class 1 worst health; hazards calibrated to the ONS life tables (one common age-varying factor on every class hazard, matching the population LE at 65 and 90). Prevention from r at intensity mu holds back a share mu of the class's mean change in theta after r - 1 (the paper's instrument; the gain is held at its value at 90 beyond). Mortality responds through the within-age gradient of death before the next wave in theta (a logit on theta, a quadratic in age and theta x age over 363,033 person-waves at risk, 1,825 deaths: log-odds -0.99 per unit of theta at 55); the healthy share (general health good or better) through the same design fitted to the healthy indicator, applied to each class's person-waves at each age. Policies: targeted, class 1 at mu = 1 (its mean health held at the level of r - 1); uniform with the same budget, everyone at mu = pi_1(r) (class 1's share at r, 0.17 at 50); uniform at mu = 1. (a), (b) dashed: each class treated on its own at mu = 1. (c) class 3 minus class 1. (d) the standard deviation of age at death over the composition-weighted mixture. (e)-(g) each group's LE and HLE at 50 is its posterior class composition among the person-waves within five years of 50 times the class values, so the gradients are the part the types carry (within-class differences by group are not modelled); the label gives the group's worst-class share; regions sorted by baseline LE. (h) LE of the top minus the bottom group: degree or higher minus other/none, top minus bottom income third, and the range across the twelve regions. The within-age mortality gradient is at or below the one the fit's own class hazards imply between types at every age.

Figure G.11: As Figure G.9, on the multidimensional classes.



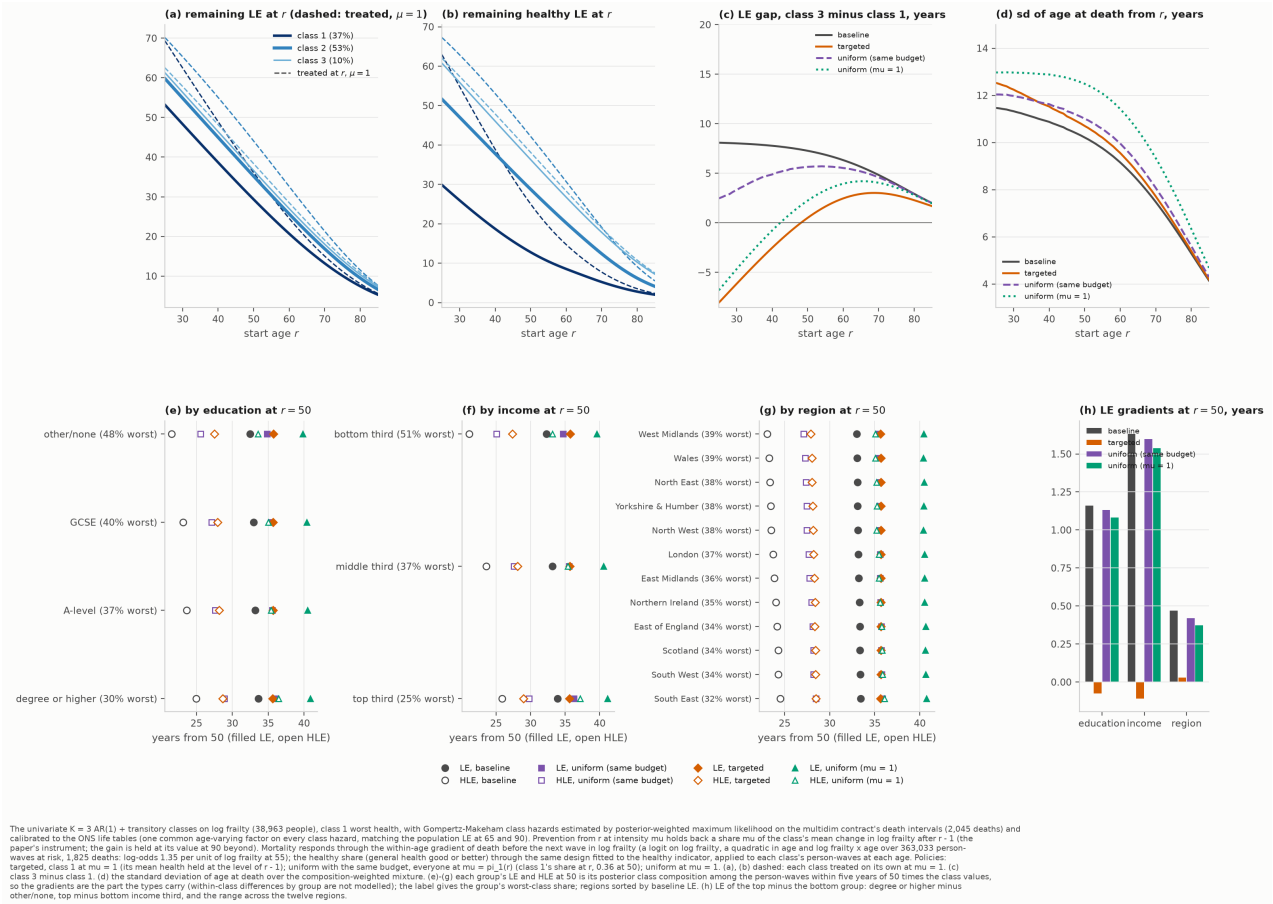

The univariate K = 3 AR(1) + transitory classes on h (38,963 people), class 1 worst health, with Gompertz-Makeham class hazards estimated by posterior-weighted maximum likelihood on the multidim contract's death intervals (2,045 deaths) and calibrated to the ONS life tables (one common age-varying factor on every class hazard, matching the population LE at 65 and 90). Prevention from r at intensity mu holds back a share mu of the class's mean change in h after r - 1 (the paper's instrument; the gain is held at its value at 90 beyond). Mortality responds through the within-age gradient of death before the next wave in h (a logit on h, a quadratic in age and h x age over 363,033 person-waves at risk, 1,825 deaths: log-odds -4.37 per unit of h at 55); the healthy share (general health good or better) through the same design fitted to the healthy indicator, applied to each class's person-waves at each age. Policies: targeted, class 1 at mu = 1 (its mean health held at the level of r - 1); uniform with the same budget, everyone at mu = pi_1(r) (class 1's share at r, 0.26 at 50); uniform at mu = 1. (a), (b) dashed: each class treated on its own at mu = 1. (c) class 3 minus class 1. (d) the standard deviation of age at death over the composition-weighted mixture. (e)-(g) each group's LE and HLE at 50 is its posterior class composition among the person-waves within five years of 50 times the class values, so the gradients are the part the types carry (within-class differences by group are not modelled); the label gives the group's worst-class share; regions sorted by baseline LE. (h) LE of the top minus the bottom group: degree or higher minus other/none, top minus bottom income third, and the range across the twelve regions.

Figure G.12: As Figure G.9, on $h$.




The univariate K = 3 AR(1) + transitory classes on the frailty index (38,963 people), class 1 worst health, with Gompertz-Makeham class hazards estimated by posterior-weighted maximum likelihood on the multidim contract's death intervals (2,045 deaths) and calibrated to the ONS life tables (one common age-varying factor on every class hazard, matching the population LE at 65 and 90). Prevention from r at intensity mu holds back a share mu of the class's mean change in the frailty index after r - 1 (the paper's instrument; the gain is held at its value at 90 beyond). Mortality responds through the within-age gradient of death before the next wave in the frailty index (a logit on the frailty index, a quadratic in age and the frailty index x age over 363,033 person-waves at risk, 1,825 deaths: log-odds 6.40 per unit of the frailty index at 55); the healthy share (general health good or better) through the same design fitted to the healthy indicator, applied to each class's person-waves at each age. Policies: targeted, class 1 at mu = 1 (its mean health held at the level of r - 1); uniform with the same budget, everyone at mu = pi_1(r) (class 1's share at r, 0.26 at 50); uniform at mu = 1. (a), (b) dashed: each class treated on its own at mu = 1. (c) class 3 minus class 1. (d) the standard deviation of age at death over the composition-weighted mixture. (e)-(g) each group's LE and HLE at 50 is its posterior class composition among the person-waves within five years of 50 times the class values, so the gradients are the part the types carry (within-class differences by group are not modelled); the label gives the group's worst-class share; regions sorted by baseline LE. (h) LE of the top minus the bottom group: degree or higher minus other/none, top minus bottom income third, and the range across the twelve regions.

Figure G.13: As Figure G.9, on the frailty index.




The univariate K = 3 AR(1) + transitory classes on log frailty (38,963 people), class 1 worst health, with Gompertz-Makeham class hazards estimated by posterior-weighted maximum likelihood on the multidim contract's death intervals (2,045 deaths) and calibrated to the ONS life tables (one common age-varying factor on every class hazard, matching the population LE at 65 and 90). Prevention from r at intensity mu holds back a share mu of the class's mean change in log frailty after r - 1 (the paper's instrument; the gain is held at its value at 90 beyond). Mortality responds through the within-age gradient of death before the next wave in log frailty (a logit on log frailty, a quadratic in age and log frailty x age over 363,033 person-waves at risk; 1,825 deaths: log-odds 1.35 per unit of log frailty at 55); the healthy share (general health good or better) through the same design fitted to the healthy indicator, applied to each class's person-waves at each age. Policies: targeted, class 1 at mu = 1 (its mean health held at the level of r - 1); uniform with the same budget, everyone at mu = pi_1(r) (class 1's share at r, 0.36 at 50); uniform at mu = 1. (a), (b) dashed: each class treated on its own at mu = 1. (c) class 3 minus class 1. (d) the standard deviation of age at death over the composition-weighted mixture. (e)-(g) each group's LE and HLE at 50 is its posterior class composition among the person-waves within five years of 50 times the class values, so the gradients are the part the types carry (within-class differences by group are not modelled); the label gives the group's worst-class share; regions sorted by baseline LE. (h) LE of the top minus the bottom group: degree or higher minus other/none, top minus bottom income third, and the range across the twelve regions.

Figure G.14: As Figure G.9, on log frailty.

# H Mental Health

- Half the variance is one-period noise; the mixtures then sort by persistence, not level, so mental health cannot carry its own level typology.
- Cost and mortality gradients are weaker than on $\theta$.

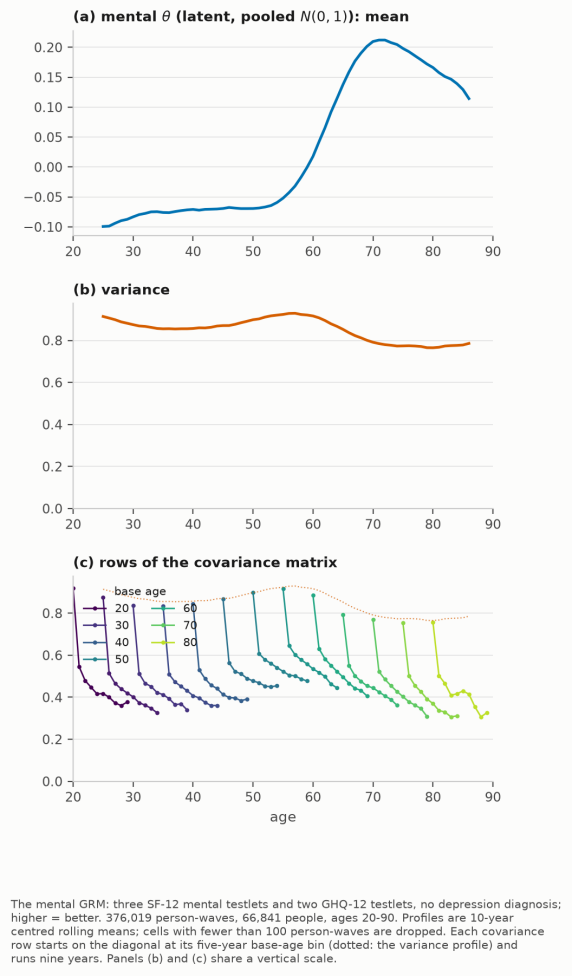
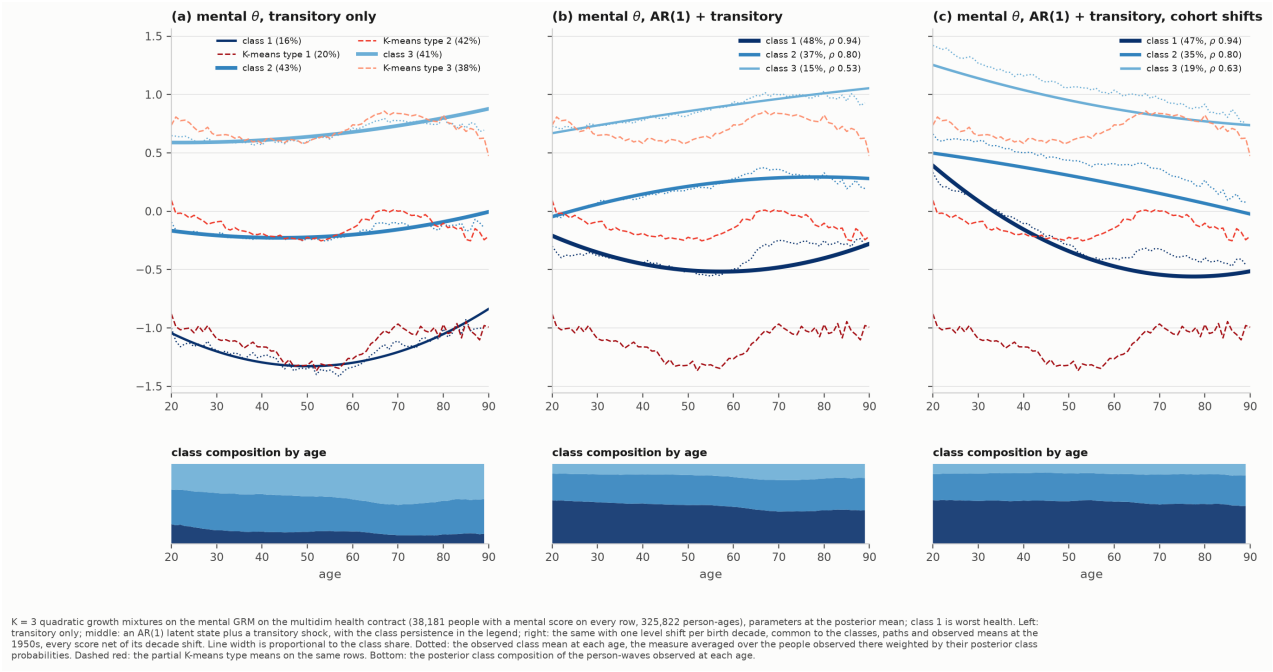


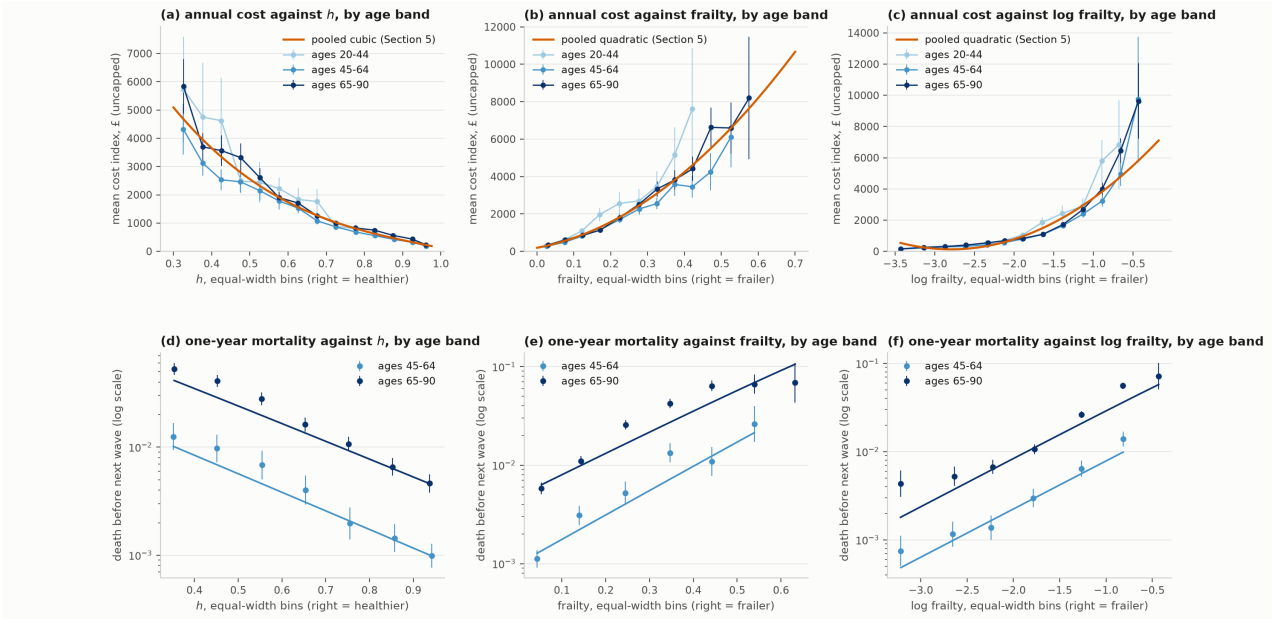

Figure H.1: As Figure 2, on the mental GRM.




K = 3 quadratic growth mixtures on the mental GRM on the multidim health contract (38,181 people with a mental score on every row, 325,822 person-ages), parameters at the posterior mean; class 1 is worst health. Left: transitory only; middle: an AR(1) latent state plus a transitory shock, with the class persistence in the legend; right: the same with one level shift per birth decade, common to the classes, paths and observed means at the 1950s, every score net of its decade shift. Line width is proportional to the class share. Dotted: the observed class mean at each age, the measure averaged over the people observed there weighted by their posterior class probabilities. Dashed red: the partial K-means type means on the same rows. Bottom: the posterior class composition of the person-waves observed at each age.

Figure H.2: $K = 3$ mixtures on the mental GRM.

# I The Main Figures on the Other Measures




UKHLS, h (left), the 31-deficit frailty index (middle) and log(frailty + 1/31) (right), ages 20-90. Cost: the cost index (in-patient, out-patient and GP), uncapped, waves 7-15, 211,697 person-waves; bins with at least 150 person-waves, 95% intervals. Mortality: death before the next wave (one year), waves 1-14, 363,033 person-waves at risk, 1,825 deaths; bins with at least 10 deaths, Wilson 95% intervals; lines are the within-band logit, linear in the measure and quadratic in age, drawn at the band's mean age. Measures standardised on the pooled sample, so a bin means the same health at every age.

Figure I.1: As Figure 1, on $h$, frailty and log frailty.

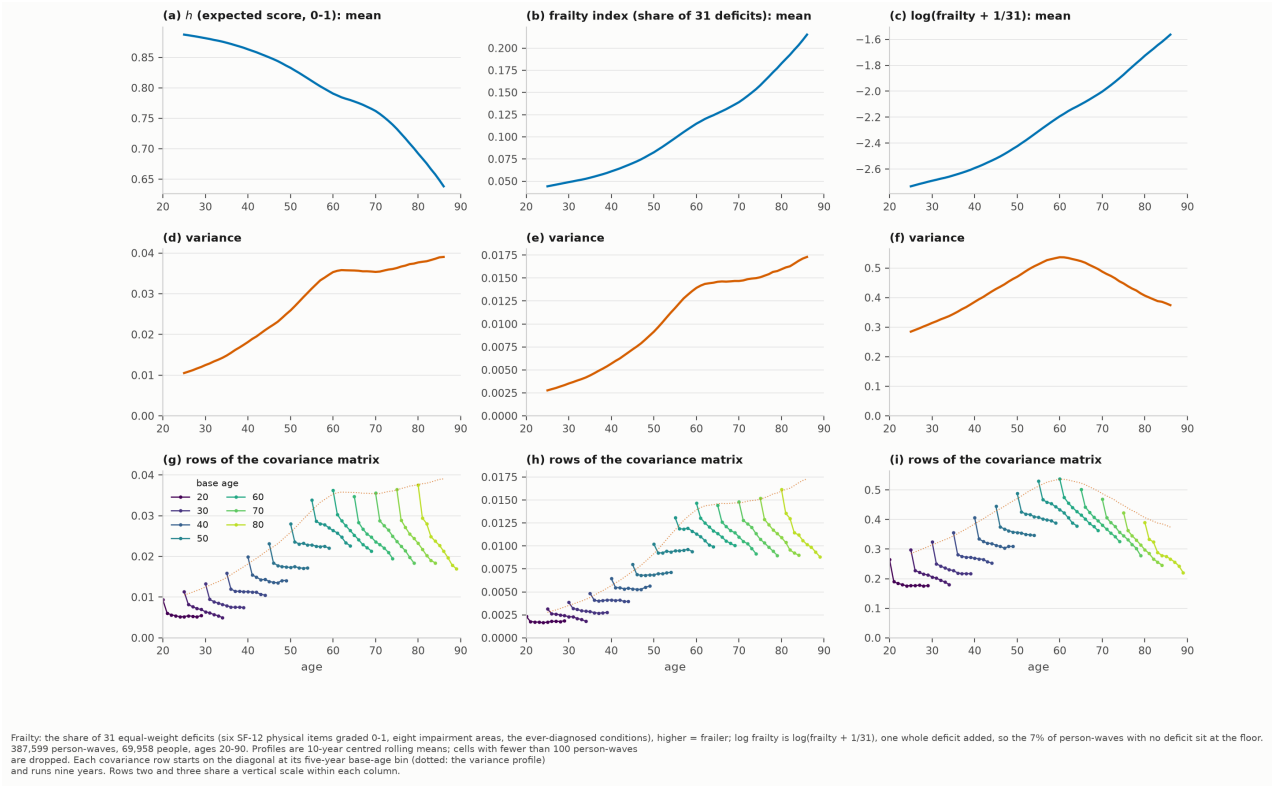
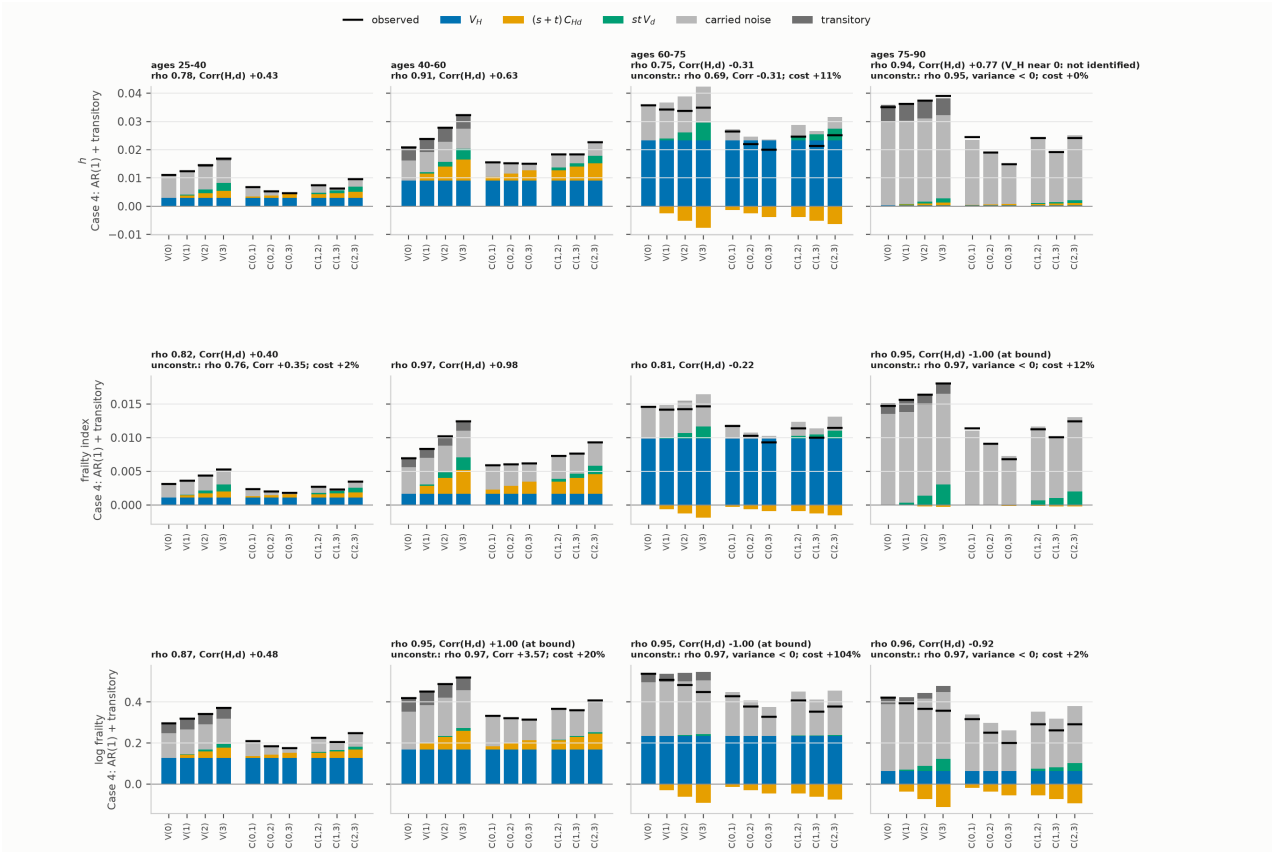
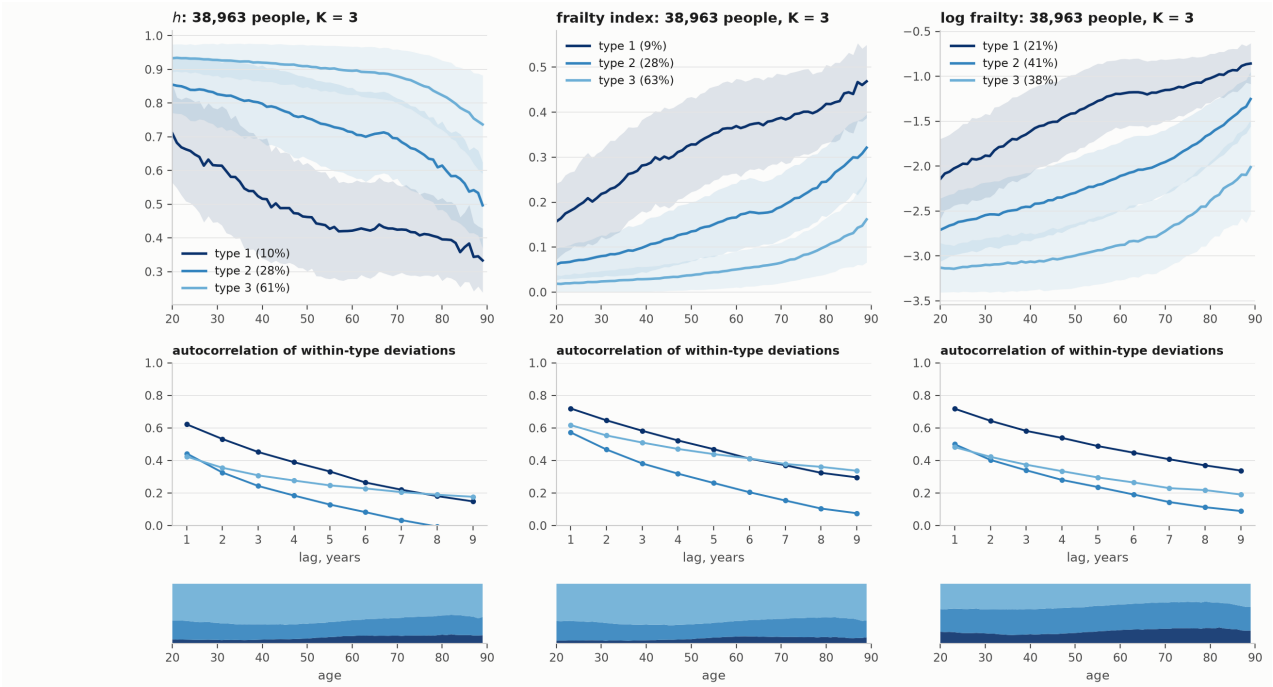



Figure I.2: As Figure 2, on $h$, frailty and log frailty.



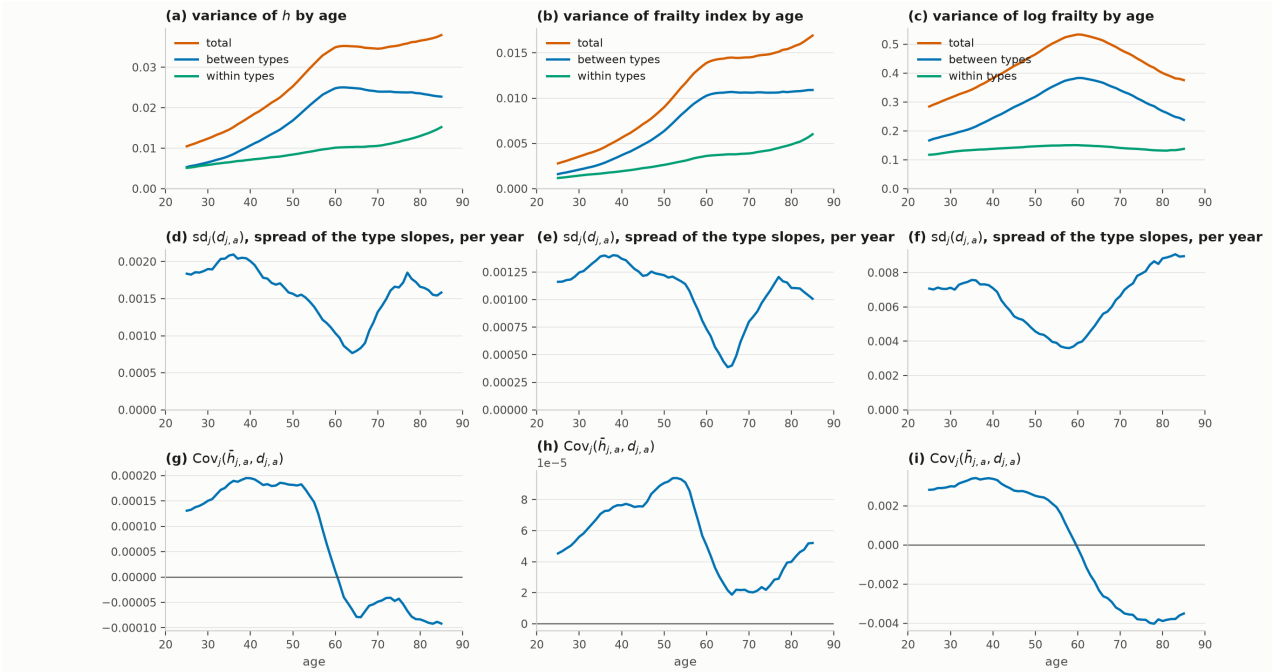

Case 4 (AR(1) + transitory) on h, the frailty index and log(frailty + 1/31). Four ages four years apart, pairwise moments pooled over base ages within each band; a base age enters when every cell rests on at least 50 people observed at both of its ages. Properly constrained fit: V_H, V_d and noise variances non-negative and |Corr(H,d)| <= 1 imposed in the estimation, rho profiled. Where the unconstrained fit was inadmissible the title gives it and the fit cost of the constraint in squared error. Where V_H falls below 5% of the first variance the level, and so Corr(H,d), is not identified. Bars stack the fitted layers; black ticks are the observed moments. V(k): the variance at the k-th age; C(k,l): the covariance between ages k and l.

Figure I.3: As Figure 3, on $h$, frailty and log frailty.



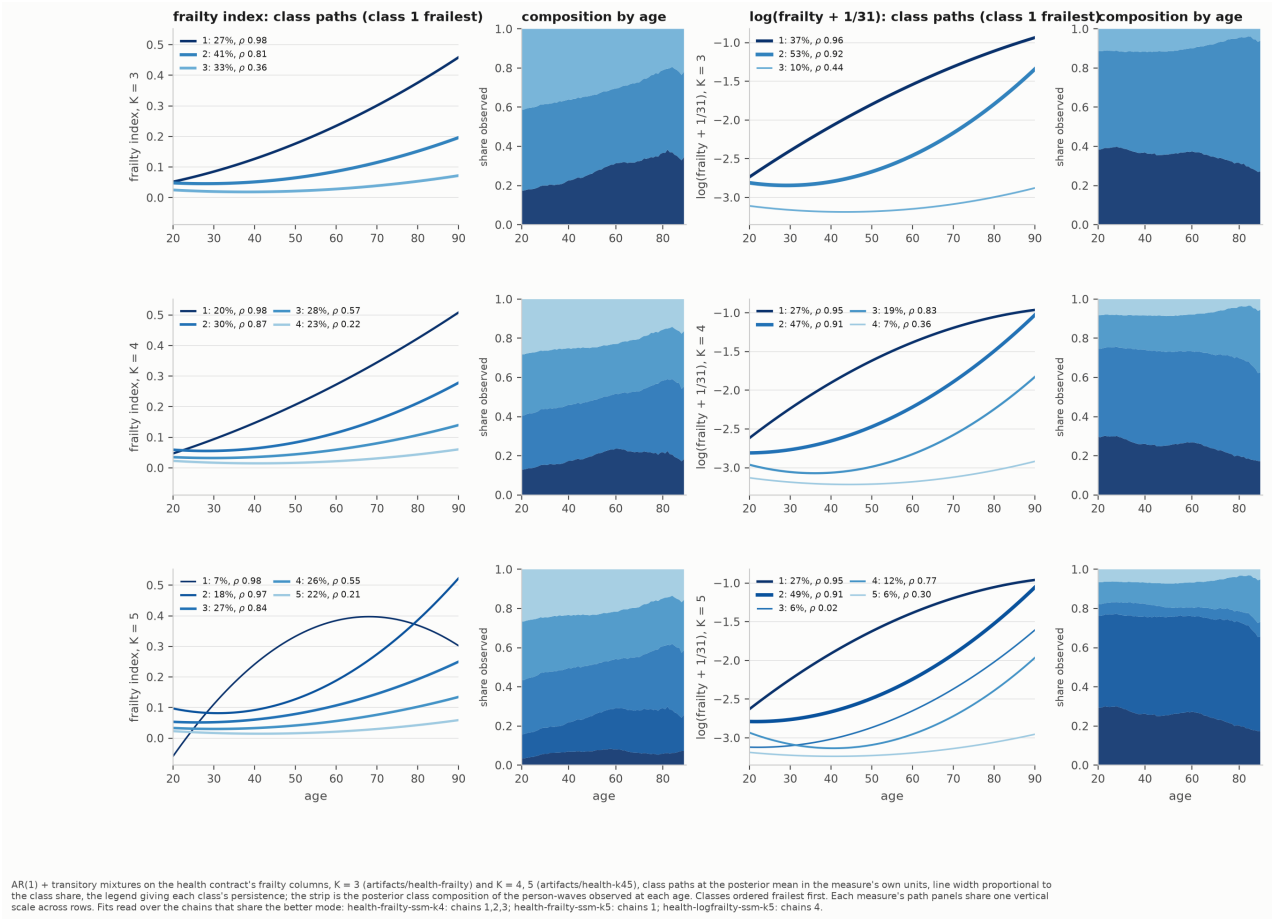

The health contract's people, on h (left), the 31-deficit frailty index (middle) and log(frailty + 1/31) (right): partial K-means with 50 seeded starts, type 1 worst health (frailest on the frailty indices). Top: type means, +/- 1 within-type sd. Middle: autocorrelation of within-type deviations. Bottom: type composition by age. Log frailty is log(frailty + 1/31), so person-ages with no deficit sit at the floor.

Figure I.4: As Figure 4, on $h$, frailty and log frailty.



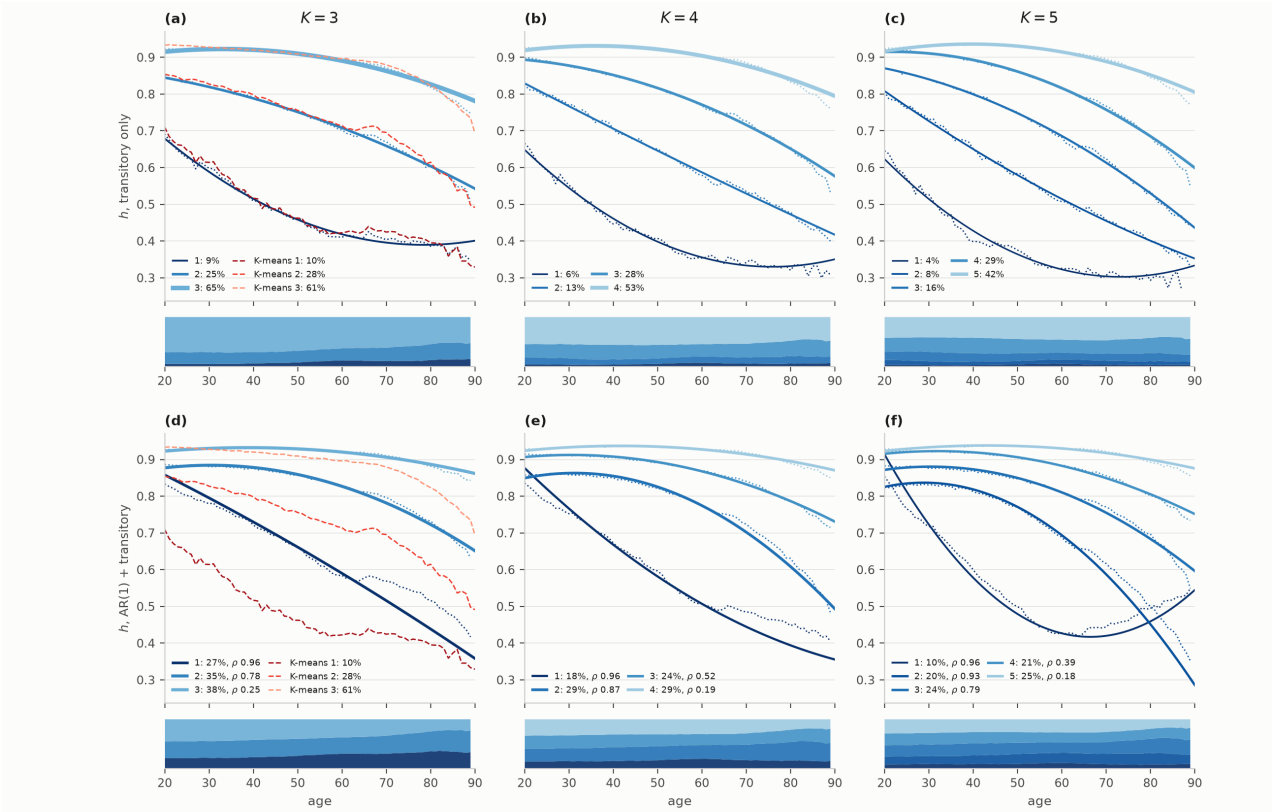

As the main-text decomposition figure, on h (left), the 31-deficit frailty index (middle) and log(frailty + 1/31) (right): 10-year rolling means, age-specific composition shares.

Figure I.5: As Figure 5, on $h$, frailty and log frailty.



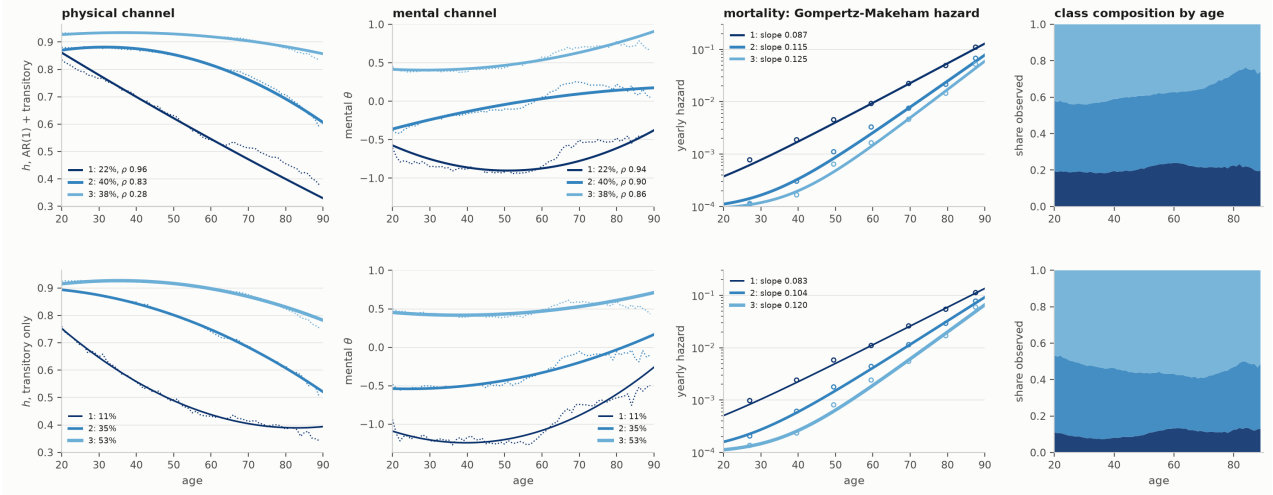

AR(1) + transitory mixtures on the health contract's frailty columns, K = 3 (artifacts/health-frailty) and K = 4, 5 (artifacts/health-k45), class paths at the posterior mean in the measure's own units, line width proportional to the class share, the legend giving each class's persistence; the strip is the posterior class composition of the person-waves observed at each age. Classes ordered frailest first. Each measure's path panels share one vertical scale across rows. Fits read over the chains that share the better mode: health-frailty-ssm-k4: chains 1,2,3; health-frailty-ssm-k5: chains 1; health-logfrailty-ssm-k5: chains 4.

Figure I.6: Frailty and log frailty under the persistent–transitory specification at $K = 3, 4, 5$.



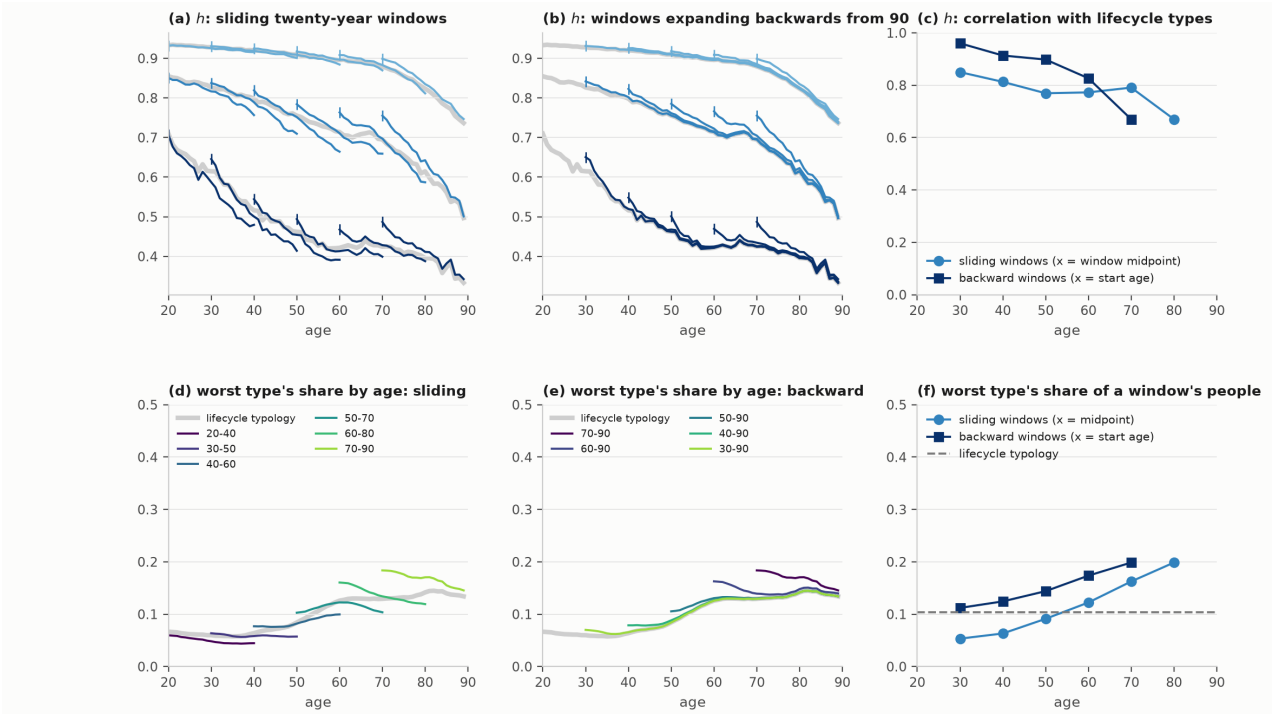

Quadratic growth mixtures on the health contract (38,963 people, 334,194 person-ages), parameters at the posterior mean, in each variant's own units; class 1 is worst health. Line width is proportional to the class share; the legend gives the share and, under AR(1), the class persistence. Dotted: the observed class mean at each age, the measure averaged over the people observed at that age weighted by their posterior class probabilities (where the summed weight is at least 25). Dashed red ($K = 3$ only): the partial K-means type means on the same rows, with the type shares in the legend. The strip beneath each panel is the posterior class composition of the person-waves observed at each age. Rows: the stochastic specification, transitory only or AR(1) + transitory; columns: the number of classes. All six panels share one vertical scale. Posterior mean over the chains that share a mode for health-h-ssm-k5: chains 1, 2, 4 of 4 (the other chain sat in a worse mode).

Figure I.7: As Figure 6, on $h$.



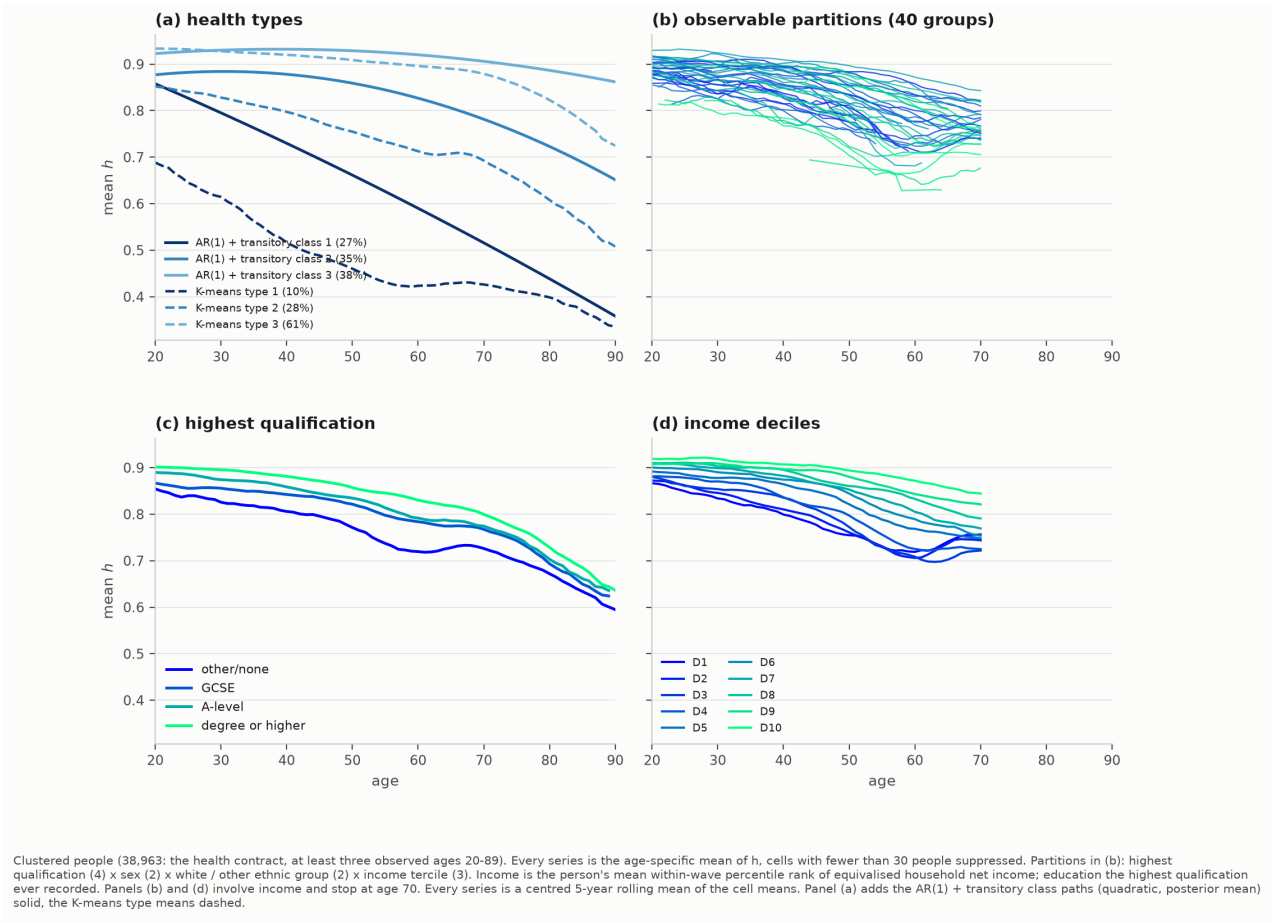

Multidimensional K = 3 mixtures on the multidim health contract (38,181 people, the health contract's people with a mental score on every row; 2,045 deaths). Rows: the physical measure and the stochastic specification. Columns: the physical and mental class paths at the posterior mean in each channel's units (dotted: the observed class mean at each age, posterior-weighted), the class-specific Gompertz-Makeham yearly hazard (circles: the posterior-weighted crude death rate by five-year band), and the posterior class composition of the person-waves observed at each age. Under AR(1) + transitory the legend gives each class's persistence on that channel; the innovation and transitory scales are common to the classes. The Makeham constant is below 0.0001 in every fit. Each column has one vertical scale across its rows. h + AR(1) + transitory is read over the three chains that share a mode; the fourth sat in the mode with the class 2 and 3 intercepts pressed together on the ordering constraint, about 2,950 log-posterior units lower.

Figure I.8: As Figure 7, on $h$.




Each window refits K = 3 on the people observed at least three times inside it. Coloured lines: the window's type means over its own age range (tick at the start); grey: the full 20-90 typology. Panel (c): over pairs of people in both the window and the lifecycle typology (a random 3,000), the correlation between the same-type indicator under the window and under the lifecycle typology, which needs no correspondence between the two sets of labels (the adjusted Rand index is in the csv). Panels (d), (e): the share of the person-waves observed at each age that the window assigns to its worst type (five-year rolling mean), one line per window, against the lifecycle typology in grey; (f) the worst type's share of each window's people. Types are ordered worst to best within each window.

Figure I.9: As Figure E.2, on $h$.




Clustered people (38,963: the health contract, at least three observed ages 20-89). Every series is the age-specific mean of h, cells with fewer than 30 people suppressed. Partitions in (b): highest qualification (4) x sex (2) x white / other ethnic group (2) x income tercile (3). Income is the person's mean within-wave percentile rank of equivalised household net income; education the highest qualification ever recorded. Panels (b) and (d) involve income and stop at age 70. Every series is a centred 5-year rolling mean of the cell means. Panel (a) adds the AR(1) + transitory class paths (quadratic, posterior mean) solid, the K-means type means dashed.

Figure I.10: As Figure 9, on $h$.

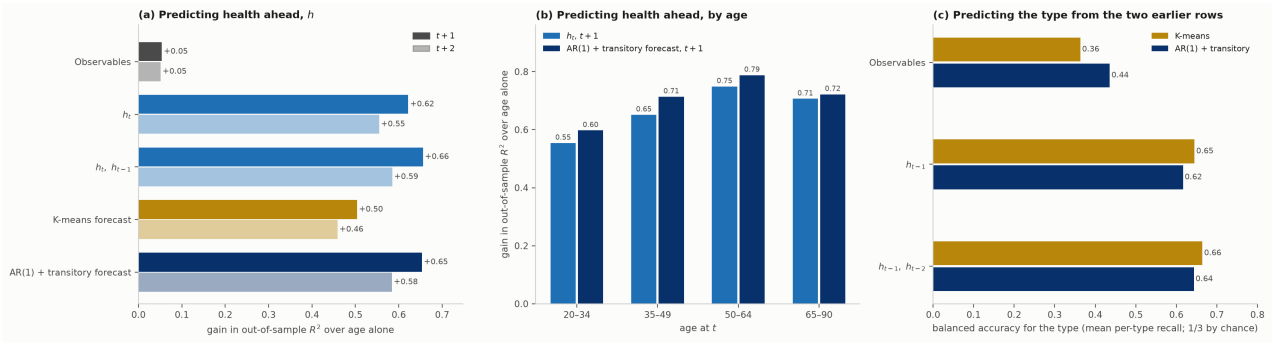



Figure I.11: As Figure 10, on $h$.

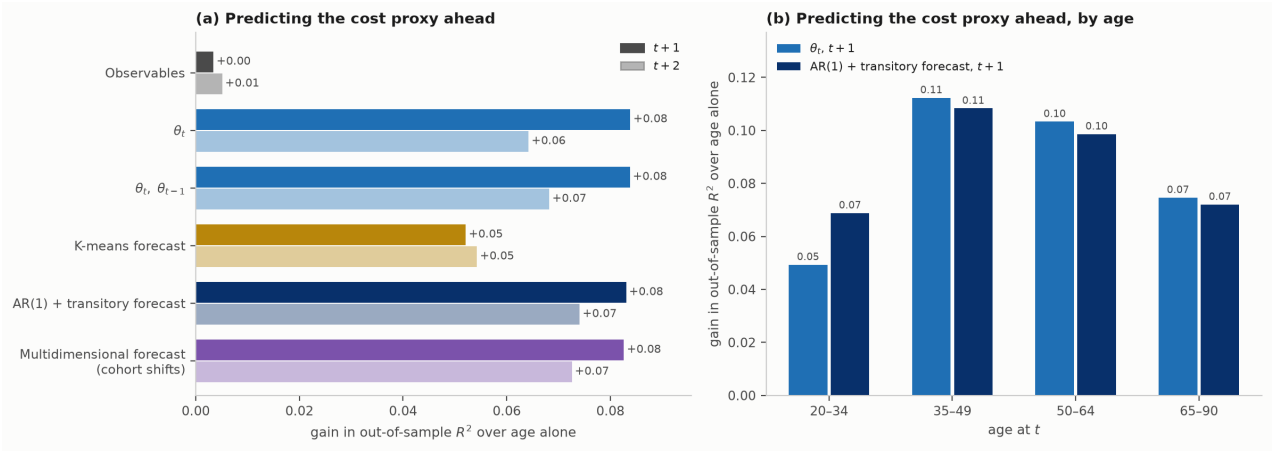



Figure I.12: As Figure 10(a), (c), predicting the cost proxy.

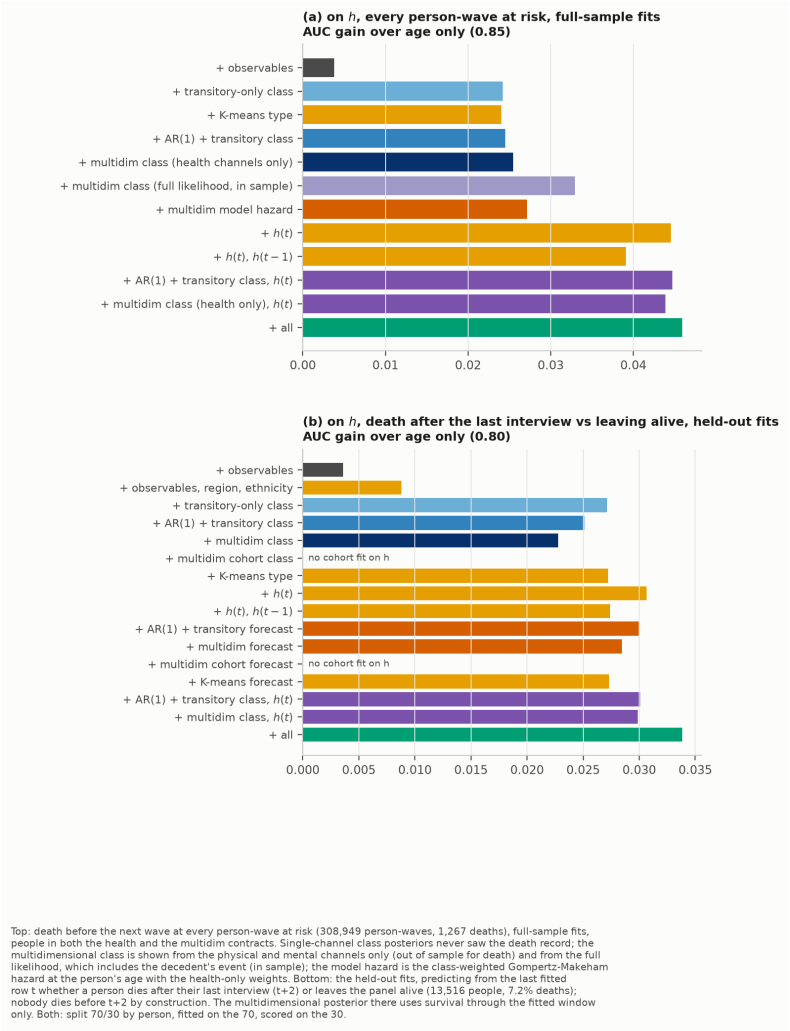
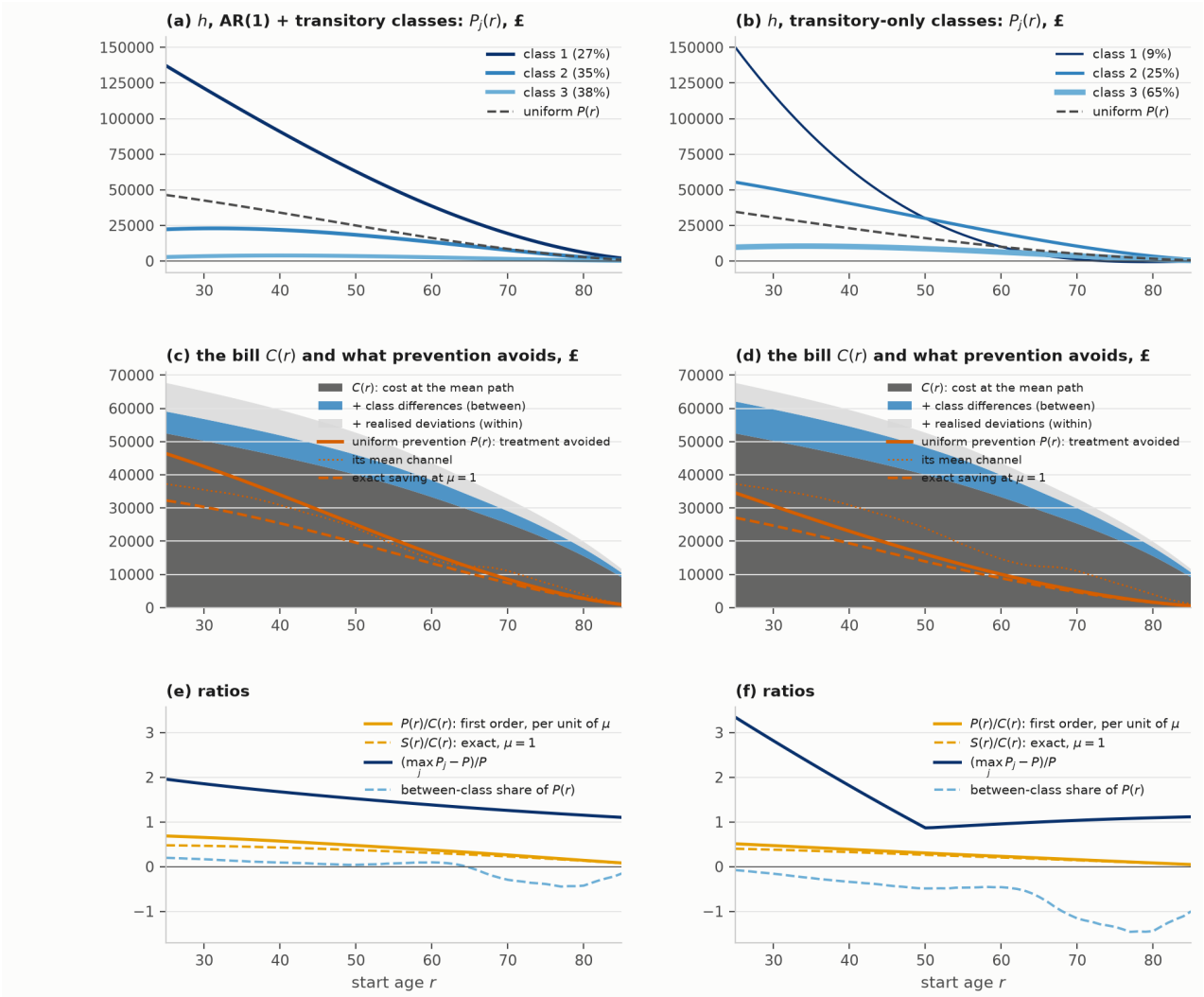
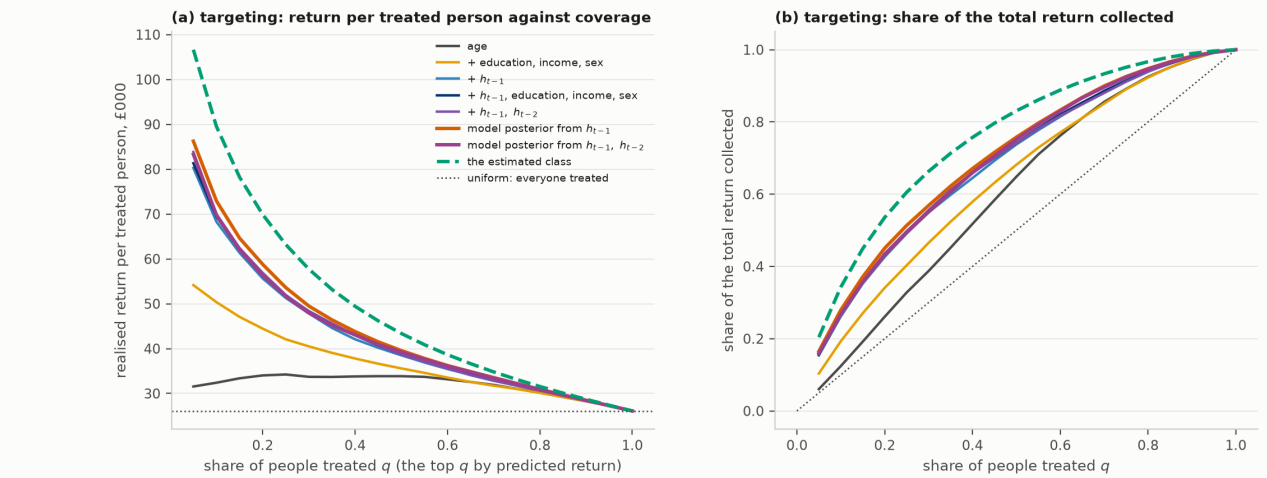



Figure I.13: Predicting death on $h$.




Returns on the K = 3 mixture classes on h, with the cost curve estimated in pounds on h (a degree-3 polynomial on the cost index, waves 7-15). C(r) is the expected treatment cost from r to 90 on the observed cross-section, split into the cost at the mean path, what the type differences add and what realised deviations add (the part prevention cannot reach); prevention scales each class's future decline by (1 - mu) and P(r) is the treatment cost that avoids, to first order, per unit of mu; S(r) is the exact saving at mu = 1 (the cost curve is a polynomial of degree at most 3, so the expansion in mu stops at the cube). Pounds to 90; no survival weighting, no discounting. Class shares and quadratic paths at the posterior mean; E[-c'(h) | class] from the person-waves observed at each age, weighted by the posterior class probabilities.

Figure I.14: As Figure G.2, on $h$.




The K = 3 AR(1) + transitory classes on h with the first two rows held out (29,899 people); the return to prevention by class P_j(r) of the returns figure, at each person's age at t-1, the second row (t is the first fitted row). Each rule predicts a person's return from what is known at t-1 (a multinomial logit of the fitted-window class on a quadratic in age plus the set shown, fitted on a 70% split; the model posterior and the estimated class need no regression), treats the top q of the other 30% by predicted return, and is scored by the realised return, the fitted-window class posterior times P_j(r). The age-only rule is the benchmark: returns fall with age, so what the reading and the observables add is the value of type information. Dotted: everyone treated. Pounds of treatment cost avoided to 90 per unit of mu, no survival weighting, no discounting.

Figure I.15: As Figure G.3, on $h$.